

Dirty Wisdom

a novel

Pumulo Sikaneta

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PASSAGE 0

Preface

Oil is not what most people think it is. That is the first claim in this book and the most important.

The substance in the hold of a tanker is not, in any meaningful sense, the fuel for cars and airplanes. It is that too, but that is the smallest part of what it is. Oil is the foundation of almost every physical object a modern person touches in the course of an ordinary day. The furniture, the packaging, the clothing, the appliances, the medicines, the fertilizer in the food, the coatings on the walls, the asphalt on the road, even the components of the solar panels and the electric vehicles that are supposed to replace oil. When we speak about oil dependency as a transport problem, we are using a word that covers about a fifth of the actual dependency and leaves the other four-fifths unexamined. This book is the other four-fifths.

Once you see the dependency at that scale, the question changes. It is no longer whether we should use less oil in our cars. It is whether we have mapped the dependencies we do not think about, stockpiled the products we cannot manufacture without oil, and built alternatives for the sectors where oil cannot be swapped out quickly. The short answer to all three questions, for almost every country on earth, is no. The longer answer, which is the argument of this book, is that we have instead built a global system that is efficient in ways that are impressive and fragile in ways we do not notice until something stops moving through one of a small number of straits.

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Along with the facts, I wanted to put a human face on them.

I have watched very good books about very important subjects fail to reach the readers who most needed them, because the facts arrived as statistics and went nowhere useful. When we read about eight-day jet fuel reserves, or ninety-nine percent of the raw materials in our medicines, or ninety-five percent Bangladeshi

energy imports, or twenty-two days of Indonesian fuel reserves, we skim past. The numbers are large. They sound serious. Then they are gone. We do not know what to do with them because we do not know who they happen to.

The ship in this book is fictional. The crew is fictional. The captain is fictional. The voyage, the anchorage, and the satellite phone calls home after the crisis, all of this is invented. The closure of the strait is real. The cascade that moves through the global system is real. The solution models named in the appendices are real. The philosophers are real. The wisdom that a crewman's mother shares at the end of the book, in her own language, is a real Chewa proverb carried by real women across generations in Eastern Zambia and Malawi. I invented the characters to carry the facts, because the facts needed to arrive through people. Numbers dehumanize. Statistics turn neighbours into abstractions. The job of a book that is asking the reader to care about people they have never met is to refuse that abstraction.

The crew of the *Noor Salaam* are not Africans, or Asians, or Europeans. They are all of those things, and they are specific. A Syrian captain from Damascus who went to sea because the country he grew up in became a country people left. A Filipino chief engineer whose daughter keeps his letter under her pillow and reads it twenty times. A Zambian crewman whose mother plants maize on a small plot of land north of Chipata, and a Bangladeshi cook whose mother has used a kerosene stove and then a gas stove and still knows how to cook with wood if she has to. A Ukrainian officer who knows what a blackout schedule does to a country, because he has lived inside one, and a Greek helmsman who has been helping the captain think for fourteen years and does not need to be told when to pay closer attention. A Thai deck foreman who remembers the names of every crewman he has sailed with. They are people. The reader is asked to know them, because the reader already knows people like them, even if they have not met these particular ones.

When we know who our neighbours are, not only in our own countries but across the world, we get the empathy that solving these problems requires. That, more than anything, is what I hoped this story could do.

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There is a line early in the book, spoken by one of the younger men on the ship, that became a principle for me while I was writing it, and that I would now offer to you as a principle for reading it. *The learning is more important than the lesson.*

He was talking to his captain, who had just worked out something significant about the cargo in the hold and was wondering whether to share it with the crew as a finding

or to invite them to arrive at it with him. The young man, barely out of his twenties, told him that the finding would land better if they were all figuring it out together, in real time, with the captain admitting that he did not yet have all the answers.

He was right, and the captain listened, and the book unfolded the way it did because of that moment. I would like to offer you the same arrangement. I do not want to hand you a set of conclusions. I want to walk you through what the ship's crew was learning, in the order they learned it, so that by the end you have something more useful than a lesson. You have the habit of looking. Once we know how to see the connections between a closed strait and a bag of fertilizer, between the chips in our phones and a gas plant in Qatar, between the statistic we skimmed this morning and a specific person waiting for something that is not coming, we become different kinds of citizens. We ask different questions of our governments. We buy things a little differently. We vote a little differently. We talk to our neighbours a little differently, because we have noticed that they are actually our neighbours.

That is what the book is trying to give you. Not a conclusion. A way of seeing.

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I have my own philosophies about what is happening in the world. I have tried to keep most of them out of the book, because a reader who comes to a book to find out what the author thinks is often making a poor exchange for their time. What I have done instead is draw on eight thinkers from eight traditions, spaced across twenty-five centuries and six cultures, whose words map onto this crisis in ways I did not have to force.

Ibn Khaldun on the cohesion that holds civilisations together. Lao Tzu on the water that carves the stone. Sun Tzu on the terrain a general has to know before the fighting starts. Confucius on the absence of courage. Seneca on preparation and negligence. Taleb on the difference between fragile, robust, and antifragile systems. Donella Meadows on the twelve places where a person can intervene to change a system, and why almost everybody works on the weakest of them. Ubuntu, the southern African concept that opens the book, which the captain reaches for on night one of the anchorage, and which closes the book through a woman in Chipata who says the same thing in a different language.

These are not new problems. The patterns have been described by thoughtful people for two and a half millennia. The wisdom is available. The question has always been whether we listen to it.

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I have not tried to be apolitical in these pages, and I have not tried to be merely factual either. There is always a human aspect to the choices I make when I look at what is happening in the world, and those choices show up in what I wrote here. I have tried to be careful and honest. I have not tried to be neutral, because I do not think neutrality is what this moment asks for.

If I have done nothing else, even if you disagree with what I have said, I hope you will think about the people and the problems in this book differently than you did before you picked it up. You may not agree with my conclusions. My hope is that you will agree it was important to say this, and important to start the conversation.

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If I have done this right, you will finish the book and look at your own room and see, for the first time, what the objects in it actually are. You will understand that the fertilizer in Chipata, the dialysis machine in Thane, the cooking gas in Dhaka, the blackout schedule in Kyiv, and the jet fuel at Heathrow are connected by the same waterway, and that the waterway was not designed to be the single passage for a fifth of what modern life is built from. We built this, over many decades of small decisions each of which seemed rational at the time. We can unbuild it. Not quickly, and not easily, but the models exist and the technologies exist and the wisdom exists. What has been missing is the will to act on any of it before the next crisis forces the question again.

That is the argument. I have tried to make it through the lives of people who are not real but whose problems are real, in a voice that tries to respect your time, with facts that have been checked against primary sources, and with the benefit of new tools that did not exist when I began this work. The methodology is declared in detail in Appendix E, for readers who want to understand how a book like this is made in 2026. The short version is that I wrote it, and I used artificial intelligence as a research accelerator and a drafting partner, and every sentence in the book was reviewed by me, and the facts are all mine to defend.

I am grateful to you for picking this up. My hope is that you finish it with a clearer picture of the world you are standing in, and with some empathy for the people you have never met who are standing in it with you, and with a little more of the courage it takes to look down at the floor instead of away from it.

We are all on the same ship.

Pumulo Sikaneta

April 2026

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Part One — We Are Connected

Ubuntu. "I am because we are."

PASSAGE 1

The Ship

The strait is narrow enough to see both shores on a clear day.

Captain Sami Kassar stood on the bridge of the *Noor Salaam* and watched the water ahead. He had made this transit more times than he could count, always in the early hours, always with the same quiet focus he brought to anything that mattered. The water here moves differently than open ocean. It funnels, tightens, presses the current between Iran to the north and Oman to the south until the passage is barely thirty kilometers across at its narrowest point. A fifth of the world's petroleum passes through this gap every day. Sami had carried a fair share of it himself over three decades.

The bridge was quiet. At the helm, Petros, the Greek helmsman, held course with both hands on the wheel and his eyes on the water, the way he always did when they entered a busy channel. Petros was fifty-one and had been sailing almost as long as Sami. He did not need to be told when to pay closer attention. He could feel it in the traffic pattern, the way the VHF radio, the short-range channel that ships and port authorities use to talk to each other, picked up more voices as the strait narrowed. A dozen conversations overlapping in different accents, all saying variations of the same thing: I am here, I am heading this direction, please confirm you see me.

Sami poured two cups of coffee from the thermos he kept on the chart table and brought one to the helm. "Coffee, Petros?"

"Please."

Petros took it without looking away from the water. Neither of them commented on the gesture. It had been happening every morning for eleven weeks.

Below, the engine room hummed. That low vibration was so constant that you stopped noticing it after the first few days at sea. It became part of your body's rhythm, like a second heartbeat. You only knew it was there when it went away.

The *Noor Salaam* was loaded to her marks, nearly two million barrels of crude in her tanks. She was a VLCC, a Very Large Crude Carrier, which was the shipping

industry's way of saying she was longer than most city blocks and held enough oil to power a small country for a day. Loaded this heavy, she moved through the water with a particular steadiness, a reluctance to change direction quickly, a weight that you could feel in the way the deck responded to your footsteps. Sami had always liked that feeling. A loaded ship was a purposeful ship.

He ran through his checks. Radar, electronic chart, VHF, AIS. The automatic identification system broadcast the ship's name, position, speed, and heading to every vessel within range. He had been running through this sequence since he was a cadet. The instruments had changed every decade. The captain still needed to check.

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His name was Sami Kassar. He was fifty-seven, Syrian by birth, a citizen of the sea by circumstance and by choice. He had studied history at the University of Damascus before the country he grew up in became a country people left. He went to sea at twenty-three, originally because it paid, eventually because the ship was the only place where the world made sense to him. From the water, you could see how things connected: trade routes, port economies, the invisible threads between a loading dock in Kuwait and a gas station in Manila. On land, everyone saw their own country. From the bridge of a tanker, you could see the system.

He was a curious man. Not in the restless way that exhausts the people around you, but in the quiet way that makes them feel noticed. He read constantly: history, philosophy, science, whatever the ship's wifi could deliver to his laptop during the slow satellite hours. He followed threads the way some people follow sports: obsessively, happily, for no practical reason other than the pleasure of understanding how one thing led to another. His crew had learned that if you mentioned something interesting in passing, Sami would bring it up three days later having read everything he could find about it.

He also listened. This was the thing his crew would have mentioned first if someone had asked them to describe their captain, though most of them would not have used that word. They would have said he pays attention, or he notices things, or he remembers what you said. What they meant was that Sami had a way of hearing not just the words but the situation behind them. When Jun, the chief engineer, said the auxiliary cooling pump was showing early signs of wear, Sami heard that Jun was worried about being responsible for a mechanical failure on a ship that could not get to a repair yard. When Tendai, the youngest deckhand, said his mother was fine, Sami heard that Tendai did not know whether his mother was fine and did not have a way to

find out that felt honest.

This quality made him a good captain. It also made him a tired one. Carrying twenty-three people's situations in your mind alongside your own was not something they taught at maritime academy.

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The crew of the *Noor Salaam* numbered twenty-three. They came from seven countries, spoke four languages well and three others in fragments, and had been together on this rotation for eleven weeks.

Sami met Tendai on the foredeck during the morning inspection, coiling a line that had come loose overnight. Tendai was twenty-four, Zambian, broad-shouldered, and still new enough to the sea that he took his duties with a seriousness that the older hands found endearing. He had taken this job because it was the first opportunity he had found that let him support his family without a degree he did not have. His mother farmed maize on a small plot outside Chipata, and the cost of fertilizer made the difference between a season that fed the family and one that did not.

"Morning, Tendai. Everything secure?"

"All good, Captain. One line needed re-coiling. The rest is fine."

Sami nodded and leaned against the rail for a moment, looking out at the water. The Iranian coast was visible to port, low and brown in the early light. Tendai followed his gaze.

"Is that Iran?"

"It is."

"It looks peaceful."

"Most places do, from a distance."

Tendai went back to his line. Then, after a moment, he said something that stayed with Sami for the rest of the day. "Captain, what exactly are we carrying? I know it is crude oil. But what does that actually become?"

It was an honest question. Tendai had been on tankers for less than a year. He knew the cargo was crude oil the way a mail carrier knows the bag contains letters, by category rather than by content.

"It becomes almost everything," Sami said. He was not exaggerating, and something in his tone made Tendai look up from the line. "When this oil reaches the refinery, they break it apart into thousands of different things. The fuel in cars and planes, yes. But also the fertilizer your mother uses on her farm. The plastic in your phone case. The asphalt on roads. The chemicals in medicine. The material in your

shoes." He gestured at the deck. "The paint on this ship."

Tendai looked at the deck. Then at his shoes. Then at the hold covers, behind which two million barrels of crude oil waited for a refinery that was still two days of sailing away.

"It's just oil," he said. But the way he said it was different now. It was a statement turning into a question.

"It's the thing underneath everything else," Sami said. "Take it away and nothing above it works. Not the food, not the medicine, not the planes, not the plastic your phone case is made of. It's the floor everything stands on, except nobody looks down."

Tendai thought about this. "Even the fertilizer?"

"Especially the fertilizer."

He looked at the deck beneath his feet again, as if the steel plate had become transparent and he could see the oil below. Then he finished the line and went to find the bosun, the senior deckhand who ran the working day on every ship, for his next task. Sami watched him go and felt the particular satisfaction of a conversation that had changed how someone saw something without either person making a speech about it.

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He followed Tendai at a distance and found Somchai on the starboard walkway, checking the mooring winches with the methodical patience of a man who had done this job on twelve ships over twenty-three years. Somchai was Thai, fifty-one, small and quiet and missing nothing. Crews learned quickly that if the bosun did not say anything when you came on watch, everything was in order, and if he said something, you had done something that needed to be discussed. He had been on the *Noor Salaam* for six years, which was two years longer than Sami, and he was the member of the crew Sami asked first when he wanted to know something the official channels would not tell him.

"Morning, Captain."

"Morning, Somchai. Tendai on his way?"

"He is. I have him on the port windlass today. He asks good questions. He asked me last week why the windlass motor runs hotter on the port side than the starboard side. I had to look it up. I have been avoiding the question for fifteen years because nobody had ever asked it. Now I know the answer."

"What is the answer?"

"Slight misalignment in the bearing housing from a re-fit in 2019. Not critical. Worth watching." Somchai paused with his hand on the winch drum. "He will be a good bosun one day. If he chooses to stay."

"You would take him?"

"I would train him. Whether I take him on permanently is another matter."

Sami nodded. This was how Somchai spoke. A Thai bosun trained in the British merchant tradition, careful with his words the way some people were careful with their tools. What Somchai did not say was usually as important as what he did, and Sami had learned over the years to pay attention to both.

"Somchai."

"Yes."

"The transit this morning. Anything in the traffic that feels off to you?"

"No, Captain. The usual density. The usual voices on the VHF. I have been listening since four."

"Good."

"Captain."

"Yes."

"I will tell you if anything changes."

"I know you will."

* * *

He continued his inspection. In the galley, Rafiq, the Bangladeshi cook, was preparing lunch. The smell of spices reached the corridor before Sami reached the door. Rafiq was thirty-eight, meticulous about his kitchen, and quietly proud of the fact that his food was the one thing on the ship that everyone agreed on regardless of nationality. He cooked for twenty-three people three times a day and treated each meal as if it mattered, because on a ship where the days blend into each other, it did.

"Rafiq. Smells good."

"Dal and rice today, Captain. Fish curry for dinner." He paused. "How long are we in the strait?"

"We should be through by mid-morning. Usual transit."

"Good. I want to provision at Jamnagar if we can. The onions are not lasting."

Sami made a mental note. Provisioning at Jamnagar was routine. They would discharge the cargo, load ballast water, resupply the galley, and head back for another load. That was the plan. That was always the plan.

He passed through the crew quarters on his way back to the bridge. The doors were thin, steel frames with minimal insulation, designed for function rather than comfort. You could hear televisions, phone calls, music. The ship had wifi, slow but functional, and during off-watch hours the corridors sounded like a small apartment building where everyone was home. Someone was watching a Filipino game show. Someone else was on a video call, speaking fast in Hindi. A phone rang behind a closed door and was answered in Ukrainian.

Twenty-three people from seven countries, sharing a steel building that moved across oceans. On good days, it felt like a village. On long days, it felt like a village with no way to leave.

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The *Noor Salaam* was headed for Jamnagar, on the northwest coast of India, where the world's largest refinery complex would break her cargo into a hundred different products. Naphtha for plastics. Diesel for trucks and generators. Kerosene for cooking stoves. Jet fuel for airlines. Lubricants, solvents, asphalt, paraffin wax. Even the residue had a destination. Petroleum coke, processed into synthetic graphite, shaped into the anode of an electric vehicle battery. The green technology of the future, built from the substance it was supposed to replace.

If the refinery did its work, pieces of the *Noor Salaam's* cargo would end up in a bag of fertilizer in Zambia, a blister pack of aspirin in Germany, a fuel tank at Heathrow, a plastics factory in Vietnam, and a Tesla battery in California. The people at each end of those journeys would never think about each other. They would never think about the ship. They would think about the price, if they thought about anything at all.

Sami thought about these connections more than most captains. It was the historian in him, the part that could not stop tracing how one thing led to another. A ship was not just a ship. It was a thread in something so large that no single person could see its edges. He had spent thirty years carrying the thread, and what surprised him was how few people on shore knew it existed.

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He was back on the bridge when it started.

The morning of February 28, 2026 was clear. Visibility was good. He had checked the weather at four and found nothing worth worrying about. Traffic in the strait was the usual dense procession, all following the separation lanes that divided inbound

from outbound like a highway median painted on the ocean.

Less than twenty-four hours ago, he had reviewed the latest diplomatic reporting and concluded that the negotiations between the warring parties were progressing. He had checked carefully before committing to this transit, knowing that his route took him through the one place where Iran held a card that had always served as a deterrent. Control the strait and you control a fifth of the world's oil. That threat had kept a fragile peace for decades. Or so he had believed.

The first sign was a VHF transmission from a vessel ahead. Her master reported that he had been warned to reverse course by an Iranian naval patrol. Then another vessel, same message. Petros looked at Sami. Sami picked up the handset and listened. A third vessel now, this one reporting that the warning included a threat: any ship attempting to transit would be considered hostile.

Sami's phone buzzed on the chart table. A news alert. He picked it up and read it, then read it again.

The United States and Israel had launched an air war against Iran.

Neither Sami nor Petros spoke. The VHF was alive with voices, captains calling each other, calling port authorities, calling their companies. The professional calm of maritime communication was cracking under the weight of what was happening.

Within the hour, the Islamic Revolutionary Guard Corps issued a formal warning: passage through the Strait of Hormuz was forbidden. Any ship that attempted to transit would be treated as hostile.

"Reduce speed," Sami said.

Petros pulled back the engine telegraph.

"All stop."

The telegraph moved again.

"Drop anchor."

The chain rattled through the hawsepipe with a sound like industrial thunder, and the *Noor Salaam* caught and held.

Then the engines wound down.

The vibration that had been part of every waking and sleeping moment for eleven weeks faded from Sami's body like a held breath being released. The bridge, the corridors, the cabins, the engine room, the galley. Every space on the ship changed in that moment. Not visually. Acoustically. The ship went from a humming, living machine to something closer to a building. A very large, very heavy building, floating in a strait between two countries, one of which had just gone to war.

Petros said, quietly, "How long?"

Sami did not answer because he did not know, and he had made it a rule early in his career never to guess when people needed facts.

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The news moved through the ship the way news moves through every small community: fast, distorted, and accompanied by speculation. By lunch, everyone knew the basics. By the afternoon, the speculation had expanded to fill every gap in the facts. Some crew members thought the war would last days. Others thought weeks. Arjun, the Indian second officer, said his wife had called to ask if he was safe, and he had told her he was fine, which was true, and that it would be over soon, which he did not know.

The mood was uncertain but not frightened. This was a professional crew, and most of them had sailed through difficult situations before. Jun had been on a ship that was approached by pirates off the coast of Nigeria in 2019. Petros had transited the strait during a previous period of Iranian threats. Even Tendai, the youngest, had been briefed during his maritime training on what to do in a crisis. The consensus at dinner was cautious optimism. These things happened. They were disruptive and uncomfortable. They ended.

"Remember the Houthis?" Jun said, picking up a piece of fish with his fork. "Everybody panicked. It went on for a while. Then it settled."

"This is bigger than the Houthis," Arjun said.

"Everything is bigger when it is happening to you," Jun replied. A few people laughed.

Sami listened from his end of the table. He did not correct the optimism and he did not add to the worry. He ate Rafiq's fish curry, which was excellent, and he watched his crew handle the first hours of uncertainty the way good crews do: with food, with humor, and with the quiet reassurance of being around people who knew what they were doing, even if none of them knew what was going to happen next.

After dinner, a few people drifted toward the recreation room. Someone put on a movie. The sound of it carried through the thin walls, muffled but recognizable, an action film, lots of explosions, which under the circumstances seemed like either a poor choice or a perfect one. Sami heard laughter from the room and decided it was a good sign.

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He went to his cabin. The space was not large: a bunk, a desk, a chair, a small sofa, a porthole that looked out over the water. The desk held his laptop, a stack of books, and his personal logbook. The walls were the same thin steel as everywhere else on the ship. He could hear the movie from the recreation room two doors down. He could hear someone's phone ringing in the cabin next to his. He could hear the water against the hull, a gentle, rhythmic sound that was the only thing that had not changed since the engines stopped.

He opened his logbook. He had kept this personal log for thirty years, separate from the official ship's log that recorded positions and weather and mechanical data. His history professors at Damascus had insisted that observation without documentation was tourism. The habit had stayed with him long after the university was gone.

He wrote:

28 February. Anchored 14nm south of Bandar Abbas. Engines stopped 0847 local. Crew informed. No injuries, no mechanical issues. The ship is fine. The world she serves is not.

I have carried oil through this strait perhaps three hundred times. I have never been asked to stop. I am not sure anyone who built the system that depends on this passage ever considered the possibility that someone might say no.

The crew is steady. Jun made a joke about the Houthis at dinner. Arjun said this was bigger. They are both right, and neither of them knows by how much. I checked the diplomatic reporting yesterday and thought the negotiations were holding. What changed in twenty-four hours? What was I not seeing?

He closed the logbook and opened his laptop. The news was a flood: missile strikes, diplomatic statements, oil prices surging. But what caught his attention was a smaller detail, buried beneath the military coverage. A shipping analyst estimated that 170 million barrels of oil were now trapped in the Persian Gulf. He tried to picture the number and could not. He could picture two million, because that was beneath his feet. Eighty-five ships like his, fully loaded, going nowhere.

He thought about Tendai's question from this morning. "What does it actually become?" The question had stayed with him all day, and now, with the engines stopped and the ship going nowhere, it felt different. It felt like the first question of a much longer investigation. What was in the hold beneath him, really? Not crude oil. That was just the beginning. What did it become, and who was waiting for it, and what would happen to them now that it was not coming?

He thought about something he had first encountered as a student, years before Damascus became a place people fled. A concept from southern Africa, something he had come across again in port cities along the east coast of the continent. The word meant something like "I am because we are." He had liked it then as an idea. Sitting on a still ship with a war starting on the shore he could see from his window, it felt less like an idea and more like a description of what was actually happening. The farmer in Zambia. The pharmacist in Frankfurt. The airline passenger in Sydney. None of them knew the *Noor Salaam* existed. All of their lives were about to change because she had stopped.

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Through the wall, the movie had ended. The ship was settling into its first night at anchor. Sami could hear footsteps in the corridor, someone going to the bridge for the night watch. He could hear Jun's door close, and then the faint blue light of a phone screen through the gap under the door, and the silence of a man reading news he wished he was not reading.

Tendai's mother was probably asleep in Chipata. She did not know about the strait. She would know about the price of fertilizer when planting season came, because the price was the part of the system that reached her, the only part she could see. The rest, the ship and the war and the oil and the ammonia plants and the natural gas that would not flow, was out of sight. Nobody in Chipata had ever been asked to think about any of it. Nobody there would be warned before it reached them either.

Sami turned off the light. The ship rocked gently at anchor, a rhythm he had known his entire career but that felt different tonight. Slower. Heavier. As if the *Noor Salaam* understood, in whatever way a ship understands, that something fundamental had changed.

I am because we are.

Sami lay in the dark and listened to the water and the silence where the engines should have been, and the question that would carry the rest of this story was already forming, though he did not yet know how far it would take him:

What was she carrying, and who was waiting for it?

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PASSAGE 2

The Manifest

Sami could not sleep.

This was not unusual for the first night at anchor. The body adjusts to the rhythm of engines the way it adjusts to a heartbeat, and when the rhythm stops, something in you stays alert, waiting for it to come back. Sami had learned over the years not to fight it. He made coffee in the small electric kettle on the shelf above his desk, took it to his chair, and opened his laptop.

He was not the only one awake. Through the thin wall to his left, he could hear Jun moving around. A drawer opening, closing. The tap running briefly. Jun's cabin was next to his. Captain and chief engineer housed on the same deck, close enough to reach each other quickly if something went wrong. On normal nights, this proximity was invisible. Both men slept through each other's routines without noticing. Tonight, with the engines off and the ship's usual background noise stripped away, every sound was specific and close. Sami could hear the creak of Jun's bunk when he shifted. He could hear the muffled chime of a phone notification.

Sami considered knocking on Jun's door. Then he decided to let the man have his silence. There were times when being next to someone who understood was better than being with them.

The news on his laptop was a wall of noise. Missile trajectories, diplomatic statements, oil futures climbing in real time. Sami read for a while, then closed the browser. The news would be there tomorrow. It would be louder and no more useful. What he wanted to understand was not what was happening in the war. He wanted to understand what was happening because of the ship.

He opened a file he had not looked at in weeks: the cargo manifest.

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Every tanker carries one. It describes, in technical language and precise quantities, exactly what is in the hold. For the *Noor Salaam*, the manifest was straightforward.

Grade: Arab Heavy crude. Volume: approximately 1.98 million barrels. Origin: Ras Tanura terminal, Saudi Arabia. Destination: Reliance Jamnagar refinery, Gujarat, India.

Sami had read hundreds of these documents. They were as routine as weather reports. But tonight, on a ship that was going nowhere, with 170 million barrels trapped around him in the Gulf, he read this one differently. Not as a shipping document. As an inventory of everything the world was about to run short of.

He had always known, in general terms, that crude oil became more than fuel. He had told Tendai as much that morning: the fertilizer, the medicine, the plastic. But general terms were not the same as understanding. He had picked up the broad idea from years of being in the industry, from conversations in port, from a podcast he had half-listened to on a long watch. He had never actually traced the specifics. He had never opened a manifest and followed each line to its end. Tonight, with the ship going nowhere and his curiosity fully awake, he wanted to know exactly how many things depended on what was beneath his feet. He suspected the answer was large. He did not suspect how large.

Sami opened a new tab and started looking.

* * *

A barrel of Arab Heavy crude, once it reaches a refinery like Jamnagar, does not become one thing. It becomes dozens of things, separated by heat in a process called fractional distillation. Sami found a technical article that explained it and worked through it slowly.

The crude is heated in a furnace until it vaporizes, then fed into a distillation column, a tower as tall as a ten-story building, where different components condense at different heights depending on their boiling points. The lightest gases rise to the top. The heaviest residues sink to the bottom. Everything in between is captured and sent to secondary processing units that refine it further into finished products.

Sami found a diagram and studied it carefully. He knew the general picture: oil becomes fuel, plastics, chemicals. What he wanted was the detail. Which specific products came from which fraction of the barrel, and where each one ended up. Knowing those specific chains might help him understand why this strait mattered enough to fight a war over.

A soft knock on the door. Jun, in a T-shirt and shorts, holding an empty mug.

"I heard your kettle," Jun said. "Can I borrow some hot water?"

Sami gestured to the kettle. Jun came in, filled his mug, and dropped a teabag into it. Then he noticed the laptop screen.

"What are you reading?"

"I am trying to understand what happens to our cargo after we deliver it. What it actually turns into."

Jun leaned over and looked at the diagram. His engineer's eye caught the structure immediately. "Fractional distillation. Same principle as the fuel purifiers in our engine room. We separate water and sediment from fuel oil by density and temperature. The refinery does it on a much larger scale."

"You understand this better than I do," Sami said.

Jun pulled a chair over. He was not someone who socialized easily, but machinery was his language, and this was a conversation about machinery. He pointed to the diagram. "The column works by temperature gradient. Top is coolest, bottom is hottest. Light fractions come off the top: gases, naphtha. Middle fractions are your kerosene and diesel. Heavy fractions at the bottom: fuel oil, bitumen, residuals." He traced the flow with his finger. "Every fraction gets sent to a different secondary unit. Nothing is wasted."

"What is naphtha?" Sami asked. He had seen the word but wanted to hear Jun explain it.

"Naphtha is the base material for the entire petrochemical industry. Plastics, synthetic fibers, detergents." Jun gestured toward the corridor. "The plastic containers in Rafiq's galley started as naphtha. The nylon in our mooring ropes. The polyester in my shirt. The insulation on every wire I maintain. All naphtha."

"And kerosene?"

"Two things people would never connect. Jet fuel for airlines and cooking fuel for families who burn it in simple stoves. Same fraction of the same barrel. The airline passenger at Heathrow and a family in Bangladesh cooking dinner are using the same product."

Sami wondered whether Rafiq knew that connection. Whether he had ever thought about the kerosene stoves in the Bangladeshi countryside as being linked to the jet fuel at Heathrow. It was the kind of thing Sami wanted to ask him about, when the moment was right.

"What about the bottom?" Sami asked. "The residuals."

Jun leaned forward. This was his territory. "The bottom is where it gets interesting. Bitumen. That is the thick, black, sticky material they make roads from. Every highway, every airport runway, every parking lot is paved with what comes out of the

bottom of a refinery." He sipped his tea. "Then petroleum coke, which is a hard carbon material. It gets used in industrial furnaces, but here is the part that will interest you." He looked at Sami. "Petroleum coke is also the raw material for synthetic graphite. And synthetic graphite is what they shape into the anodes of lithium-ion batteries. The batteries in electric cars."

Sami let that register. "So the batteries in electric vehicles are made from a material that comes out of an oil refinery."

"From the very bottom. The part most people think is waste." Jun finished his tea. "The green revolution, built from the leftovers of the oil industry. I am an engineer. I find this fascinating. Most people would find it disturbing."

"Both," Sami said.

Jun smiled. It was the first genuine smile Sami had seen from him since the engines stopped. "Both," he agreed.

Jun explained paraffin wax, the residue that ends up in food packaging, cosmetics, candles, and the smooth coating on pharmaceutical tablets that makes them easy to swallow. He explained lubricants, the oils that keep every engine on earth from grinding itself apart, without which nothing with moving parts works for long. He explained fuel oil, the heavy petroleum product that powers the largest ships at sea, including tankers like the *Noor Salaam* herself.

"So the ship that carries crude oil burns a refined version of crude oil to carry it," Sami said.

"Welcome to the oil industry," Jun said. He stood, rinsed his mug in Sami's small sink, and paused at the door. "Captain. I called my wife earlier. She was at the petrol station. The queue had doubled in a day and the price had already gone up. She said her sister was in a longer queue in Manila. People are starting to buy what they can."

"I saw that."

"She asked me when I am coming home. I told her soon." He paused. "I do not know if that is true."

"Neither do I," Sami said. "But I will find out as soon as I can and I will tell you honestly."

Jun nodded and went back to his cabin. Through the wall, Sami heard him settle into his bunk. The phone screen glowed for a few more minutes. Then it went dark.

* * *

Sami turned back to his laptop. The conversation with Jun had given him the technical framework, and now the implications were opening faster than he could follow.

He thought about the rooms in his apartment in Limassol, the one he rented between contracts and saw for perhaps two months a year. The kitchen floor was vinyl. Petroleum. The countertops sealed with a petrochemical coating. The refrigerator insulation was polyurethane foam. The food packaging inside was polyethylene. The rubber gasket on the door was synthetic. The cleaning products under the sink were petrochemical. The detergent. The sponge.

The bathroom. The toothbrush, the shampoo bottle, the soap dispenser, the shower curtain, the contact lens solution, the razor handle, the adhesive bandages in the cabinet.

The bedroom. The mattress foam, the pillow filling, the polyester in the sheets, the phone charging cable, the phone case itself.

Sami stopped counting somewhere past sixty and looked up the actual number. Various industry sources put it between 6,000 and 8,000 distinct products derived from petroleum. He picked the conservative lower number and carefully considered the massive impact that still represented. Six thousand products. The ship beneath him was not carrying fuel. She was carrying the material vocabulary of modern life, and almost nobody on shore knew it.

A sound from the corridor. Footsteps, slow and quiet, someone trying not to wake anyone. Then a pause outside his door. Sami looked up. Tendai was standing in the corridor in a T-shirt and tracksuit pants, drawn by the light spilling out of Sami's open doorway. He caught Sami's eye and started to move on.

"Tendai. Come in."

The deckhand hesitated, then stepped inside. He looked at the laptop screen, the notepad covered in Sami's handwriting, the empty coffee cup.

"I could not sleep, Captain."

"Neither could I. Neither could Jun." Sami gestured to the chair Jun had been sitting in. "Sit. Remember what you asked me this morning? What the cargo becomes?"

Tendai nodded.

"I told you it becomes almost everything. I was right, but I did not know by how much." Sami showed him the notepad. "I counted more than sixty petroleum products just in my apartment. The actual number across the whole economy is over six thousand. And the connection to your mother's farm is closer and more complicated than I thought."

He explained it in terms Tendai would understand. Not through chemistry but through the price of things in Chipata.

"Your mother buys urea fertilizer. That fertilizer is made from ammonia, which is made from natural gas. Natural gas is not exactly the same thing as what we are carrying, but they come from the same ground and travel through the same strait. When this waterway closes, the gas stops too. And when the gas stops, the factories that make fertilizer slow down. And when they slow down, there is less fertilizer available. And when there is less available, the price goes up."

"So my mother is paying more because of this," Tendai said, gesturing at the water outside the porthole.

"She will be. If she is not already."

Tendai was quiet for a moment. "She will not know why."

"No. She will know the cost. She will not know the reason behind the cost."

"That seems unfair."

"It is."

* * *

They were quiet together for a minute. Through the thin wall, someone coughed in the next cabin. The water moved against the hull with a sound like slow breathing. Sami turned back to his laptop and pulled up something else he had found while reading about fertilizer. He wanted Tendai to hear this part.

"There is something else about the fertilizer that I think your mother might already know, even if she does not have a word for it."

Tendai looked up.

"Synthetic fertilizer, the kind your mother buys, works. It makes crops grow bigger and faster. But it does something to the soil over time." Sami turned the laptop so Tendai could see the article he had found. "The chemicals feed the plant, but they do not feed the soil. The organisms that live in healthy soil, the bacteria, the fungi, the small creatures that break down organic matter and make nutrients available naturally, they weaken when synthetic fertilizer does their job for them. Over years, the soil becomes less alive. It holds less water. It loses structure. It produces less on its own."

Tendai was listening carefully. "So the soil gets worse."

"The soil gets more dependent. Each season, the natural fertility drops a little, and the farmer needs a little more synthetic fertilizer to get the same yield. The more fertilizer you use, the more you need. And the more you need, the more dependent you are on the natural gas that makes it, and the more dependent you are on the strait that the gas flows through."

Sami watched Tendai process this. The deckhand was not an educated man in the formal sense, but he understood systems that mattered to him. His mother's farm was a system he had been watching his entire life.

"My mother says she uses more every year," Tendai said slowly. "She says the soil is not like it was when her mother farmed."

"She is right. And she is not the only one. The United Nations estimates that topsoil is being lost at ten to forty times the rate it regenerates. It takes five hundred to a thousand years to create one inch of topsoil. We are using it faster than the earth can make it, and we are compensating with fertilizer that makes the problem worse."

"So it is a circle."

"It is a circle that gets tighter every year. The soil degrades, so you add more fertilizer. The fertilizer degrades the soil further, so you need even more. And every bag of fertilizer runs through a supply chain that depends on this strait staying open." Sami gestured at the dark water outside the porthole. "Your mother is at the end of a chain that starts here. And the chain is getting more fragile, not less, because the soil underneath it is getting weaker every season."

Tendai stared at the porthole for a long time. Sami did not rush him. Some things need time to land, and the connection between a bag of urea in Chipata and a closed strait in the Persian Gulf was not a small thing to absorb at one in the morning on a ship that was going nowhere.

"You should tell the others," Tendai said finally. "What you have been finding. Jun knows, but the rest of us do not."

Sami considered this. "Maybe. I am still learning myself."

"That is what would make it interesting," Tendai said. "If you are figuring it out at the same time we are."

He said goodnight and went back to his quarters. Sami watched him go and thought that Tendai had just said something genuinely wise without knowing it. The captain figuring it out in real time, with the crew, was more honest than the captain delivering conclusions from his cabin. The learning was the point, not the lesson.

Sami made a note in his logbook: *Consider sharing the research with the crew. Not a lecture. A question. Let them find the connections themselves.*

* * *

It was nearly two in the morning. The ship was as quiet as a ship gets, which is never fully quiet. Water against the hull. The creak of steel contracting as the night air cooled the deck plates. A generator running somewhere below, providing the

electricity the dead main engine no longer produced.

Sami returned to his laptop. He had told Tendai that oil was the floor everything stood on. He had meant it as a general truth, based on something he had heard on a podcast months ago. They had used the word "substrate," almost as if they did not want the average person to understand how important it was. If they had called it a foundation, people would feel the dependence. If they called it a substrate, most people would not know what that meant and would keep scrolling. Now Sami wanted to know exactly how the floor connected to the crew on his ship. How did it connect to a bag of urea in Zambia? How did it connect to Jun's family in Cavite? His laptop faithfully gave him the answers to his growing questions.

The fertilizer connection was not direct, and Sami wanted to be honest with himself about that. Natural gas, not crude oil, is the primary feedstock for ammonia production. The *Noor Salaam* was not literally carrying Tendai's mother's fertilizer in her hold. But the more he read, the less the distinction mattered in practice. Natural gas and crude oil come from the same geological formations, move through the same pipelines and shipping lanes, and are disrupted by the same events. When the Strait of Hormuz closes, both stop. Close enough to the truth that the point held. He had told Tendai the connection was real, and he did not feel the need to go back and correct himself over a technicality.

The process that made the fertilizer had a name: Haber-Bosch. Sami had heard of it but had never followed the thread, and tonight the still ship gave him the time he needed to find out what it really was.

Fritz Haber was a German chemist who, in 1909, figured out how to pull nitrogen from the air and combine it with hydrogen derived from natural gas to produce ammonia. His colleague Carl Bosch scaled the process to industrial production. The result was synthetic fertilizer, and it changed the species. Before Haber-Bosch, the amount of food the planet could grow was limited by the nitrogen naturally available in the soil. After Haber-Bosch, that limit was gone. Crop yields doubled, then tripled. By most estimates, roughly half the people alive today owe their existence to this single chemical reaction. Four billion people, fed by a process most of them have never heard of.

Sami liked that kind of fact. The ones that most people ignored, but that once you understood them, you could never figure out why everyone was not already aware.

And now Sami understood why Tendai's mother was caught in the circle they had just discussed. Haber-Bosch had removed the ceiling on food production, but it had made the floor dependent on a continuous supply from outside. The world had traded

natural limits for chemical dependency, and the dependency ran through a strait that was now closed.

Fritz Haber himself had followed a similar circle, though a darker one. He did not stop at fertilizer. During World War I, he turned his talents to chemical weapons, developing the chlorine gas attacks used at Ypres, in Belgium, in 1915. His wife, Clara Immerwahr, who was also a chemist, opposed the work. After Haber personally supervised the first successful gas attack, Clara took his military pistol and shot herself in their garden. Haber left for the Eastern Front the next morning.

Sami read this twice. The man who figured out how to feed half the planet also figured out how to poison soldiers in trenches, and the person closest to him chose death over complicity. The substance that gives life and the substance that takes it, developed by the same mind, funded by the same government, driven by the same chemistry. It was not a contradiction. It was a snake biting its own tail. The things we depend on most are the things that carry the most dangerous second-order consequences, and the people who see those consequences earliest are usually the ones with the least power to stop them.

Clara Immerwahr saw it. She could not stop it.

Tendai's mother could see the soil changing. She could not stop that either.

* * *

Sami leaned back and rubbed his eyes. It was past three. Through the wall, Jun's cabin was finally silent. Through the porthole, the lights of other anchored vessels dotted the water in every direction, each one carrying cargo the world had already counted on receiving.

He was thinking about a book he had read years ago, a thin paperback from a used bookshop in Singapore. Sun Tzu, *The Art of War*. He had bought it because the shopkeeper said every ship captain should read it, and Sami had liked the shopkeeper and trusted his judgment.

Most people who mentioned Sun Tzu talked about strategy and competition. Sami had taken something different from it. The book was not about fighting. It was about knowing. Knowing the terrain, knowing your supply lines, knowing what you depended on before someone took it away. The general who understood the ground he stood on could win without fighting. The general who did not understand the ground would lose regardless of his strength.

Twenty-five centuries ago, a Chinese military strategist wrote that an army without provisions had already lost the war before the first battle. He was talking about

soldiers and grain. He could have been talking about the *Noor Salaam* and the civilization waiting on her cargo.

The crew were beginning to understand their terrain because Sami had started asking questions and they had started looking for answers. The civilization on shore had not. The airline passenger did not know that 40% of the world's jet fuel transited this strait. The farmer did not know that the price of her fertilizer was determined in a gas field two continents away. The pharmacist did not know that the blister pack, the gel cap, the syringe, the sterile packaging, and the latex gloves were all petroleum. They did not know the terrain because nobody had ever asked them to look down at the floor they were standing on.

Sun Tzu would have understood immediately. A civilization that depends on a single passage for a fifth of its most critical resource, that does not map the dependency, and does not build alternatives, and does not stockpile the finished products it actually needs, that civilization has already lost. Not because an enemy attacked. Because it never learned what it was standing on.

* * *

1 March. Could not sleep. Read the manifest, then spent four hours performing research on my laptop. Jun came by, could not sleep either. He explained the refinery process better than any article could because he is an engineer and this is how he sees the world. Between the two of us, we traced our cargo from the hold to six thousand finished products. I had not known the number.

Tendai came by later. I told him about the fertilizer, and he told me something I had not found in any article. His mother says she uses more fertilizer every year. She says the soil is not what it used to be. She is right, and the science confirms it. Synthetic fertilizer feeds the plant but starves the soil. The more you use, the more you need. A circle that gets tighter every year, running through a strait that just closed.

Fritz Haber fed half the world and poisoned the trenches. His wife killed herself over it. Tendai's mother can see her soil dying and cannot afford the fertilizer that is killing it. The things we depend on most are the things that carry the most dangerous consequences. I do not know what to do with this yet, but I will not forget it.

Tendai said I should share what I am learning with the crew. He said it would be more interesting if I was figuring it out at the same time they were. He is twenty-four and he just described the only kind of teaching that actually works.

Not a lecture. A question. Let them find the connections themselves.

* * *

Sami turned off the lamp. The ship moved gently at anchor. Through the porthole, the lights of eighty-five stranded tankers moved with the same gentle rhythm, each one a small world of people wondering the same things his crew was wondering, each captain deciding how much to share and how much to carry alone.

He would not carry it alone. Tendai was right. The learning was the point, not the lesson.

Through the wall, silence from Jun's cabin. Below, somewhere in the crew quarters, the faint sound of someone's alarm being set for the morning watch. The ship held them all in her steel body, and the oil held them all in its invisible web, and none of them could do anything about it except try to understand.

Sami closed his eyes and let the water rock him toward something that was not quite sleep but was close enough to rest.

* * *

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PASSAGE 3

The Ground You Never Examined

Sami slept for three hours and woke feeling like he had not slept at all.

The coffee helped. The morning light helped more. He showered, dressed, and went to the bridge for the dawn watch. Petros was already there, binoculars raised, scanning the water out of professional habit. The anchored fleet had grown overnight. More ships stretched to the horizon in every direction.

"Three more vessels since midnight," Petros said without lowering the binoculars. "One is a gas carrier. The VHF has been busy. A vessel near the Omani coast was approached by an Iranian patrol. They asked for her flag state and destination, then told her to stay where she was."

"Allowed to stay. Not allowed to leave."

"Correct."

Sami noted it in the official log and went below to check on Jun. He found him in the engine room with Tendai beside him, working on the fuel centrifuge. Tendai handed Jun tools without being asked, learning the rhythm of the work by watching and adapting. Sami stood in the doorway and appreciated this quietly. A young man who learned by being present was a young man who would be good at whatever he chose.

"Centrifuge today, cooling pump tomorrow," Jun said without looking up. "She is manageable, but every week at anchor will be harder than the one before."

Sami left them to it and went to his cabin to think about the evening.

* * *

Tendai's advice was still with him. *It would be more interesting if you are figuring it out at the same time we are.* Not a lecture. A question.

He asked Rafiq to set up coffee and whatever he could spare. He sent word through Somchai: everyone off watch in the mess hall at eight.

* * *

That evening, they came in ones and twos. Twenty of the twenty-three. Sami pulled a chair into the circle and sat. He did not stand behind the table.

"I am not going to pretend I have information you do not have. The company sent an email. Conserve provisions, follow welfare protocols, wait. That is what I know."

He looked around the room. Not scanning. Looking.

"The best crews I have sailed with were not the ones that never had problems. They were the ones that talked when things got hard."

He turned to Jun. "How is everything below?"

Jun gave a brief update on the engines. Then Sami waited. The first answer was never the real one.

"My wife is worried," Jun said. "Oil prices in the Philippines have already gone up. She says people are starting to queue at the petrol stations. It has only been two days."

"If you need more time on the satellite phone, take it. No paperwork. Just tell me."

Jun's shoulders loosened slightly. Sami moved on.

He spoke to Tendai, to Rafiq, to the room. He did not address each person individually. That would have been a performance. But he looked at each of them, and every person understood they had been seen. He told them about the Evening Question. Coffee, cards, a conversation. Not a lecture.

"I will start now. How many products do you think are made from a single barrel of crude oil?"

The guesses came fast. Ten. Thirty. A hundred. Tendai did not guess. He had heard the number from Sami the night before, and he was watching the room the way Sami had watched him on the deck the morning the question had started.

"Six thousand."

The room went still.

"How is that possible?" Arjun asked.

"That is tomorrow's question."

* * *

The Evening Question took hold faster than Sami expected. The next night, people arrived early. Cards appeared. Rafiq had made biscuits. Sami walked them through the refinery process using objects in the room: the plastic cups, the synthetic cushion fabric, the nylon rope coils. By the end, people were looking around the mess hall the way Tendai had looked at the deck on the first day.

On the third evening, Sami changed the approach.

"I want each of you to find one thing. Use the wifi, use your phone, come ask me. How much of your country's energy comes from the Gulf? How dependent is your home on what passes through this strait? Bring back a number."

* * *

They brought back numbers. And the numbers were worse than most of them had expected.

Arjun came first. He had spent his afternoon watch looking up India's energy profile on his phone. "India imports about 85% of its crude oil. More than 60% of that comes from the Middle East." He let this land. "I did not know these numbers before yesterday. I assumed India was more self-sufficient."

"What does your family use for cooking?" Sami asked.

"LPG. Gas cylinders. Everyone in our building uses them." Arjun paused. "I do not know where the gas comes from. I never thought about it."

"You might want to find out," Sami said. "That could be tomorrow's question for you."

Jun's number was the one that quieted the room. "The Philippines imports 98% of its oil from the Middle East."

Nobody spoke for a few seconds.

"Ninety-eight percent," Jun repeated. He was not being dramatic. He was trying to make the number feel real to himself. "I have been an engineer my whole career. I understand systems. I understand single points of failure. If someone showed me a machine with 98% of its power supply running through one valve, I would tell them the machine was badly designed."

"And the valve just closed," Petros said from the back of the room.

"The valve just closed," Jun agreed.

Dani, the younger Indonesian cadet, had found that Indonesia imports about a third of its oil, with 22 days of fuel reserves. "I thought it was more," he said. "I thought we had months stored somewhere."

"Twenty-two days is not months," Eko, the older cadet, said quietly.

Dani put his phone down on the table and did not say anything else for a while. He was nineteen. He had joined the merchant service six months ago, straight out of the maritime academy at Semarang, and this was only his second ship. His father was a fisherman in a village on the north coast of Java. The family had one boat, a small one, with a diesel engine that his father had bought secondhand in 2019 and had been repairing himself ever since because a proper mechanic was an hour's drive along a

road that flooded in the wet season. Dani sent money home every month. His father used some of it for fuel.

"My father takes the boat out every morning," Dani said to the room, to nobody in particular. "If the diesel is not there, he does not go out. If he does not go out, the family does not eat the way we are used to eating." He looked at his phone as if it might have the answer. "Twenty-two days. I did not know it was twenty-two days."

Eko put his hand on Dani's shoulder. Eko was forty-two, a former fisherman himself, and he had taken Dani under his wing on the first day of the rotation in a way that was not formal and not announced but was clearly understood by everyone on the ship. He did not say anything. He just left his hand there for a few seconds and then took it away.

Tendai had not looked up percentages. He had called his mother. "She says prices have not changed yet in Chipata. But she has heard on the radio that there are problems with fuel in Lusaka. She does not know what problems exactly." He looked at Sami. "She does not know any of this is connected to where I am." He paused. "My sister Nambela was with her. She is fourteen. She wanted to know whether I was safe on the ship. I told her I was, and I told her the ship was fine, and she said she was going to tell her friends at school that her brother had called from a ship in the Persian Gulf. She wanted a specific place name to tell them, and I gave her one. She will look it up."

Rafiq had found that Bangladesh imports roughly 95% of its energy. He said the number and then said nothing else. The number was enough.

Viktor did not share a number. He sat in his chair with his arms crossed and listened to everyone else. When the room was quiet, he said, "These are the numbers before anything has actually happened to any of you. Oil prices went up two days ago. Your families are noticing the first effects. In a week, they will notice more. In a month, they will notice much more." He looked around the room. "I know because I have seen it before."

Nobody asked Viktor to elaborate. His country's experience was not something you asked a person to explain for educational purposes. It was something you respected by not pushing.

But Sami watched him for a moment longer than he might have otherwise. Viktor's hands were resting on his crossed arms, and Sami noticed that they were not still. The fingers of his right hand were pressing gently into the fabric of his left sleeve, a small, repetitive pressure that looked like someone counting something without knowing he was counting. Sami had watched sailors do this before, on other ships, after other

news. Usually the thing being counted was the number of people a person had lost, or the number of days since they had heard from someone they were waiting to hear from. Sami did not know what Viktor was counting. He knew it was not nothing.

When the room broke up, Viktor did not leave with the others. He stayed in his chair. Rafiq brought him a fresh cup of coffee without being asked, set it on the table in front of him, and walked back to the galley without speaking. Viktor drank the coffee slowly. Sami stayed at his own table and finished his. Neither of them said anything. That was a kind of conversation too.

* * *

The next morning, March 4th, the situation escalated.

Sami was on the bridge when the news came through. QatarEnergy, the company that operates the world's largest liquefied natural gas facility at Ras Laffan, had declared force majeure, a legal announcement meaning the company could not fulfill its contracts because circumstances beyond its control had made delivery impossible.

He read the details carefully. Ras Laffan produces roughly 20% of the world's LNG. Most of it goes to Asia: China, Japan, India, South Korea. With that single declaration, a fifth of the global natural gas supply was officially unavailable. LNG spot prices in Asia more than doubled by midday.

This was not just about heating and electricity. Sami had learned enough over the past few nights to know that natural gas was also the feedstock for ammonia, which was the feedstock for urea fertilizer. The connection to Tendai's mother had just become more direct.

He told the crew at that evening's gathering. He explained force majeure the way he would explain any contract term. It means the company is saying the situation is out of their hands, and their customers should not expect delivery. Then he explained what Ras Laffan was, how much of the world's gas it produced, and where that gas went.

"Twenty percent," he said. "Gone with one announcement."

He watched the room absorb this. The crew was starting to understand that the situation was not a temporary disruption that would resolve in a few days. Each evening brought new information that made the picture larger and more serious. The optimism from the first night, Jun's joke about the Houthis and the consensus that these things happen and then they end, was still present but it was getting thinner.

Sami told them about helium that same evening. He had been saving it because it was the connection that surprised him most. Helium, he explained, is not just for

balloons. It is essential for cooling during semiconductor manufacturing. The process of fabricating computer chips requires temperatures so extreme that helium is one of the few substances that can reach them. Qatar supplies a third of the world's helium. The same force majeure that shut down LNG had shut down helium exports.

Jun understood immediately through his own engineering experience. "Same principle as our refrigerant systems. Without the right cooling gas, it does not matter how good the equipment is. The process cannot function."

"So the chips in our phones," Dani said, looking at the device in his hand.

"The chips in your phones. The chips in laptops, in cars, in medical equipment, in military systems. All of them require helium during manufacturing. And a third of the world's supply just became unavailable."

Dani put his phone down on the table as if it had become heavier.

* * *

Two days later, on March 6th, Arjun came to Sami's cabin during the morning watch. He had found the answer to the question Sami had given him about his family's cooking gas.

"LPG. Liquefied petroleum gas. Most of India's supply comes from the Gulf: Qatar, UAE, Saudi Arabia. All through this strait." He sat in the chair that was becoming the unofficial visitor's seat in Sami's cabin. "I called my mother last night. She said the gas price in Mumbai has not changed yet, but there are rumors it will. She wanted to buy an extra cylinder in case it goes up, but the shop would only sell her one."

"Rationing?"

"Not officially. But the shopkeeper said they were limiting purchases. One per household." Arjun looked at the porthole. "My mother asked me what was happening. I tried to explain. I told her the strait was closed and it affected the gas supply. She asked me which strait. I told her the Strait of Hormuz. She had never heard of it."

Sami thought about Tendai's mother and the fertilizer. Arjun's mother and the cooking gas. Both women in different countries on different continents, affected by the same waterway, neither of whom had heard its name.

"I also found out something else," Arjun said. "Yesterday, the United States gave India an emergency waiver to buy Russian oil. Thirty days. Because the normal supply through the Gulf is cut off." He shook his head. "America is telling India it is acceptable to buy oil from Russia, the country America has been sanctioning for years, because there is no other option."

"When the floor falls out," Sami said, "the rules change."

Arjun nodded. "My father has a small manufacturing business in Gujarat. He uses plastic components from a factory in Mumbai. He called me yesterday because his delivery was late and the price had gone up. He asked if I knew why, because I work on ships and maybe I understand these things." Arjun paused. "I did not understand these things a week ago. Now I do, and I am not sure it helps."

* * *

That evening, Sami told the crew about methanol and naphtha. Methanol is a chemical building block used to make resins, coatings, and adhesives. Naphtha is the feedstock for the entire plastics industry. Both move through the strait. Asian petrochemical firms were already reporting supply concerns, and China's methanol port inventories were approaching warning levels.

"If the plastics factory in Mumbai cannot get naphtha," Arjun said to the room, "my father cannot get components. And my father has never heard the word naphtha. He just knows the delivery is late."

Sami told Rafiq about LPG separately, in the galley, while Rafiq was preparing dinner. Liquefied petroleum gas, the cooking fuel that hundreds of millions of families across South Asia and East Africa depend on, flowing through the same strait, subject to the same disruption. Rafiq listened while working and gave a response that was more precise than any analysis. "My mother cooks with gas. If the gas stops, she cooks with wood. If there is no wood, she does not cook."

Three sentences. Sami would think about them for days.

An hour later, Sami went back to the galley. He told himself he was going to ask Rafiq something about the provisioning, but he was not, and when he got there, Rafiq was alone at the counter with the onions and the knife, and Sami pulled up a stool without saying anything and sat.

Rafiq glanced at him and went back to the onions. He was chopping them the way he always did, quickly and precisely, with the kind of blade control that came from twenty years of cooking for people who had somewhere to be. The galley smelled of cumin and heated oil. A pot of dal was simmering on the stove.

"You are here for a reason, Captain."

"I am here because I am thinking about what you said."

"You heard it. You do not need me to say it again."

"I do not need you to say it again. I am here because I want you to know that I heard it."

Rafiq did not stop chopping. But his hands slowed slightly, which was the galley equivalent of looking up.

"My mother has cooked with LPG for sixteen years," he said. "Before that, she cooked with kerosene. Before that, with wood. My grandmother cooked with wood her whole life. My great-grandmother, too. The gas is new. It came with the electricity and the paved road and the mobile phone. Everyone in the village knows the gas is new, and everyone in the village acts like the gas has always been there, because that is what people do when something good happens. You get used to it, and then you forget it was new."

He scraped the onions off the board and into the hot oil. They hissed and browned.

"If the gas stops for a week, the village adjusts. If the gas stops for a month, people start cutting wood from places they know they should not cut. If the gas stops for six months, the forest starts to go. This has happened in other villages. Not in ours, because the gas has not stopped. But everyone knows which villages. And everyone knows the forest does not come back the way it went."

"Rafiq."

"Yes, Captain."

"You have been cooking on this ship for eleven weeks and I did not know this about your mother."

"You did not ask, Captain. That is not a criticism. You had a ship to run. I had a galley. We were both doing our jobs."

"I am asking now."

Rafiq turned down the heat on the dal and leaned against the counter. He looked at Sami for a long moment.

"My mother is sixty-three. Her oldest son, my brother, is an engineer in Chittagong. Her youngest, my brother Mahmud, is in his final year of engineering at a university in Dhaka. I am the middle one. The one who went to sea. She cooks three meals a day for whoever is in the house, which right now is my father and my brother Mahmud and a cousin who is staying with them while he looks for work. She cooks on a two-burner gas stove that my father bought for her in 2010 when the gas line finally came to the neighbourhood. Before 2010, she cooked on a kerosene stove my grandfather had given her as a wedding present. She still has the kerosene stove. It is in the back of the cupboard. She has not thrown it away."

Sami nodded slowly.

"My mother does not know what the Strait of Hormuz is. She does not read the international news. My father does. My father has already told her that something is

happening with the fuel. She has already checked the kerosene stove. She has already asked my father whether the tank in the shed is full." Rafiq turned back to the dal. "She did not tell me this. My father told me, on the phone yesterday. My mother did not tell me because she did not want me to worry while I am on the ship. My father told me because he thought I should know."

"Your mother is a wise woman."

"My mother is a woman who remembers how to cook with kerosene and with wood. Whether that is wisdom or whether that is survival depends on what you call wisdom, Captain."

Sami did not answer. Rafiq stirred the dal, added salt, tasted it, added a little more. Then he said, without turning around, "The ship is not a problem for me, Captain. I have a job. I have food. I have a place to sleep. My mother has food, and she has a stove, and she has a husband who is ahead of the news. We are not the ones I am worried about. I am worried about the families who did not keep their old stove. Those are the families that cut the wood that does not come back. That is a larger number than it was ten years ago."

"It is going to be a larger number again, by the end of this."

"Yes, Captain. That is why I am cooking the way I am cooking. I am using less oil than I usually use. I am using the vegetables we had yesterday instead of opening new cans. I am making bread instead of using the packaged loaves. I started doing this two days ago. Nobody has noticed."

"I notice now."

"You were not supposed to. That is the point. When a cook is cooking well, you do not notice what is not there. You only notice what is there." He gave the dal one more stir and put the lid on. "Go back to the bridge, Captain. Dinner will be at six."

Sami stood up from the stool. At the galley door he paused.

"Rafiq."

"Yes."

"When we get home, I want to meet your mother."

Rafiq did not turn around. But he nodded, once, slowly.

"She will cook for you," he said. "If the gas is still there, she will cook with gas. If it is not, she will cook with the kerosene stove in the back of the cupboard. And if there is no kerosene either, she will cook with wood, and it will be the best meal you have had in a long time, Captain, because that is how my mother is."

* * *

A few nights later, at the Evening Question, Jun brought up something that had been building since his first late-night conversation with Sami. The crew had been talking about fertilizer and cooking gas and jet fuel all week, the obvious dependencies, the ones the strait closure made loud. Jun wanted to speak to a quieter dependency that almost nobody in the room had considered.

"The rare earth metals in EV motors. China controls over 90% of the processing. Wind turbine blades are petrochemical composites. Solar panels use petroleum-derived materials." He looked at Sami. "The new system needs the old system to build itself. And nobody is talking about that."

"Because it complicates the story," Sami said.

"The story is already complicated. Pretending otherwise does not help anyone."

* * *

By the end of the first week, the crew of the *Noor Salaam* knew more about the global energy system than most people would learn in a lifetime. They knew it not from textbooks but from their own research, their own families, their own phones buzzing with news from home that got a little worse each day.

The mood had shifted. The first evening's cautious optimism, *these things happen, they end*, had given way to something more sober. Not panic. Not despair. Understanding. The crew was beginning to grasp the scale of what they were inside, and the understanding was changing them. Questions at dinner were different. Conversations on the bridge were different. Even the card games during the Evening Question had a different quality. People played more slowly, talked more between hands, asked each other things they would not have asked a week earlier.

Sami noticed all of this. He also noticed that Tendai's mother had not yet felt the full impact. The fertilizer price in Chipata had ticked up slightly, but her current crop had already been top-dressed weeks earlier, before any of this had started. The real test would come later in the year, when farmers across the region would be buying the fertilizer they needed for the next planting, and when the prices they would see on the bags would be prices nobody had budgeted for. The theoretical vulnerability Tendai had discovered in the first week would become personal then, many months from now, when his mother stood in front of a bag of urea and saw a number she could not afford.

That was still coming. For now, the crew was learning. The Evening Question continued. The connections multiplied. And every day the ship stayed anchored, the gap between what the crew knew and what the world on shore understood grew a little

wider.

* * *

7 March. End of the first week at anchor. QatarEnergy declared force majeure on March 4th. 20% of the world's LNG supply, gone. India received an emergency waiver to buy Russian oil on March 6th. The crew has been researching their home countries' energy dependencies, and the numbers are sobering. The Philippines at 98%. Bangladesh at 95%. Indonesia with 22 days of reserves.

These are the numbers before the real impacts have arrived. Viktor reminded us of that. He said in a week they will notice more. In a month, much more. He knows because he has lived through a supply chain collapse. Nobody asked him to elaborate, and that restraint might have been the wisest thing anyone in the room did all week.

The Evening Question is becoming something I did not plan. People arrive early. They bring things they found during the day. They ask each other questions I have not thought of. It is not mine anymore. It belongs to the crew.

Rafiq said three sentences about his mother's cooking fuel that I will carry for a long time. "If the gas stops, she cooks with wood. If there is no wood, she does not cook." I have read a hundred pages of research this week. Nothing I found was more precise than that.

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Part Two — The Cost of Not Knowing

"The doctrine was fragile long before the war."

Sami Kassar, in his cabin. Passage 8.

* * *

PASSAGE 4

The Road No One Paved

By the second week, Sami had stopped listening for the engines.

The body adjusts. What had been a missing heartbeat in the first days became, by the end of the first week, just another layer of the ship's quiet. The crew had found a rhythm that accommodated the stillness. Jun and Tendai worked through a maintenance backlog that had been deferred for months. Rafiq cooked more elaborately than he had in any of the years Sami had sailed with him, as if the galley was where he had decided to make his stand against the situation. Petros kept the bridge watch with the same steady attention he brought to everything. The ship had found a way to be a ship without moving, and the crew had found a way to be a crew without the reason that usually held them together.

What had not steadied was the news from home. Each crew member had a phone that buzzed more than it used to. Jun's wife was sending photographs, one a day, of the petrol station on her road in Cavite. The queue in the first photograph had been forty cars long. By day nine it went around two corners and out of frame. Arjun's mother had managed to buy an extra cooking gas cylinder before her shopkeeper began restricting purchases, and she was calling her son each morning to tell him what she had heard on the radio. Tendai's mother had heard nothing new. She was watching the bag of urea in Chipata the way a farmer watches weather.

Sami had told them, at the Evening Question on the seventh night, that he did not know how long this would last and that he had resolved to stop guessing. What he could do was keep reading, and keep sharing what he found, and hope that something in the reading helped them understand why their families were living what they were living.

* * *

That was what he was trying to do the afternoon Tendai came to his cabin.

For most of the week, Sami had been trying to answer a question he could not crack on his own. Why this strait, exactly? Why had the world let a single waterway become the single pinch point for a fifth of its oil? He had read what he could, between bridge watches and before the Evening Question and at the edge of sleep. He wanted the history. Not the geopolitics, which the news was already providing in wall-to-wall coverage, but the quieter question underneath the shouting. How did this happen? Who built it? When did the dependency actually form?

That afternoon he was still at his laptop, circling, when the knock came.

A knock, lighter than Jun's, at the open door. Tendai. He was in his work clothes and held two plates.

"Captain. Rafiq says you have not eaten lunch. He sent me with food."

"Did Rafiq send you, or did Jun?"

"Rafiq. Through Jun."

Sami smiled despite himself. "Come in. Sit. I want company for what I am trying to figure out."

Tendai put one plate on the desk and kept the other. He sat in the chair Jun had used on the first night, and the way he sat told Sami he had hoped to be invited to stay. He looked at the laptop screen. Sami had a browser open with a dozen tabs and a notepad beside him with a rough timeline sketched down one side.

"You are reading about oil again."

"I am trying to figure out how the world ended up depending on this strait. I have been at it for days and I am not getting anywhere alone. Help me."

"Captain, I am a deckhand. I do not have a background in this."

"You have curiosity. That is what I need. Tell me when something does not make sense. Ask me the questions that are not in the articles."

Tendai settled into the chair with the slightly stunned look of someone who had been promoted without warning. Sami turned the laptop so they could both see it.

* * *

"I have been trying to work out when this became the world we are sitting in. I started with today and went backward. Right now, the world uses roughly one hundred million barrels of oil every day. One hundred million, every twenty-four hours. A VLCC like ours carries two million. Fifty of our ships every day, on a loop that never stops."

Tendai was quiet.

"So I asked when it was half of that. When was the world using fifty million a day? The 1970s. When was it ten million? The 1940s. When was it one million? The 1910s. When was there no oil industry at all? Before 1859. I worked backward until I ran out of industry. Then I stopped, because the answer to my question was not back there. It was somewhere in between. So let us walk forward from 1859 and see how the pile got built."

"Why 1859?"

"Because that is when the first commercial oil well in history was drilled. Titusville, Pennsylvania. A man named Edwin Drake drilled a hole about twenty metres deep and hit oil that had been sitting under that field for a hundred million years, waiting for somebody to go looking."

"How did he know to look?"

"That is the question I would not have thought to ask. Thank you."

Sami searched for a minute, reading quickly, then looked up.

"A Yale chemistry professor named Benjamin Silliman had taken a sample of the black stuff that seeped out of the ground in western Pennsylvania and analysed it in his laboratory. He wrote a paper in 1855 saying that when you heated the stuff, you got a liquid that burned cleaner and brighter than whale oil. The paper made its way to investors. The investors hired Drake. Four years after the paper was published, Drake drilled a well where Silliman said the oil should be, and the oil came up."

"Whale oil."

"Whale oil. Until the middle of the nineteenth century, the main source of oil for lamps was whales. Sperm whales especially. Ships sailed from Massachusetts, chased whales across the Pacific for two or three years at a time, and came home with barrels of oil rendered from blubber. It was the lighting industry of its era. The fuel of the reading classes."

"What happened to the whales?"

"We hunted them faster than they could breed. By the 1850s, whale oil was getting expensive, the voyages were getting longer, and people who wanted to read at night were running out of light."

Sami paused.

"The oil industry began because we were running out of whales."

Tendai set his fork down.

"That is the whole beginning?"

"That is the whole beginning. For the first forty years, oil was for lamps. That is all. Nobody was thinking about cars, because there were no cars. Nobody was

thinking about plastic, because the word did not exist yet. Nobody was thinking about fertiliser or medicine or computer chips, because none of those things existed either. If you had asked in 1890 what oil was for, the answer was that it let people read at night."

"And now?"

"Now it is almost everything. That is what we have been learning. The question I am trying to answer is how we got from one to the other."

* * *

Tendai picked his fork back up, but did not eat.

"What changes?"

"In 1908, a machinist in Detroit named Henry Ford put a car on the market that ordinary people could afford to buy. The Model T. He did not invent the car. He invented the assembly line that made the car cheap."

"And the cars needed fuel."

"The cars needed fuel. Which was the strange part, because the fuel was already there, and nobody had wanted it."

Tendai looked up.

"When the refineries heated crude oil to make kerosene for lamps, several other things came off at different temperatures. One of them was a light, volatile, dangerous liquid called gasoline. The refineries had no use for it. It was explosive. It evaporated quickly. It fouled equipment. They could not easily store it and they could not sell it, so they poured it into rivers and let it evaporate into the air. In the 1890s, gasoline was one of the pollutants in the water supplies of Pittsburgh and Cleveland."

"The fuel everyone depends on now."

"The fuel everyone depends on now was the thing nobody wanted in 1895. Then Ford built a cheap car. The car needed a tank full of liquid that would burn in a small engine. That liquid was already coming out of the refineries in enormous quantities, and nobody had a use for it, which meant it was cheap. Within a generation, gasoline had gone from being an embarrassment to being the main event. The kerosene that started the industry became a side product."

Sami stopped and looked at Tendai.

"Nobody planned this. A chemistry professor tried to solve a lamp oil problem. A mechanical engineer built a cheap car. Those two decisions, made half a century apart by people who never met, produced a world in which every highway, every supply chain, every airport runway, depends on the thing nobody in 1900 knew what to do

with."

Tendai was looking at him with the look Sami had learned to watch for. The look of someone whose picture of the world was rearranging itself while he sat still.

"That is only the first fifty years," Sami said. "Keep asking."

* * *

"Who thought to make fabric out of oil?"

"A chemical company in Delaware called DuPont. In the 1920s, they decided to do something that no one in business had really done before. They hired scientists, gave them laboratories, and paid them to invent materials that did not yet exist. Most companies then sold what already existed. DuPont said, we will pay chemists to make things that have never been made, and then we will figure out what to do with them."

"Why?"

"Because they had more money than they knew what to do with, and because a few of their executives believed the next century would belong to whoever controlled the new materials. They hired a young chemist named Wallace Carothers. He was brilliant and he was depressive, and he killed himself in 1937 before he saw what he had invented become anything. But before he died, he figured out how to link small molecules from petroleum into long chains that could be drawn into fibres."

"Fibres for what?"

"For anything. Carothers was not trying to replace a specific material. He was trying to see what the chemistry could do. The long chains turned out to be a kind of molecule that does not exist in nature. His company called the result nylon. In 1940, DuPont put nylon stockings on sale across the United States. Four million pairs sold in four days."

"Four million."

"Four million. Women lined up around the block to buy a material that had not existed a decade earlier. Then the Second World War started and silk stopped coming from Japan, and the military took nylon for parachutes, ropes, mosquito nets, surgical sutures. By the end of the war, nylon and the materials that came after it were everywhere in the American economy, and in the decade after that, everywhere in the world economy."

Tendai was staring at his work shirt.

"The polyester in your shirt is descended from what Carothers did. So is the nylon in our mooring lines, and the polyethylene in Rafiq's food bags, and the polystyrene in the coffee cups in the mess hall. And Carothers was not doing it because the world

needed synthetic fabric. He was doing it because he was curious, and DuPont was paying him to be curious. The applications found themselves."

Sami set his hand on the notepad beside the laptop.

"One chemist solving one problem he had set himself. One solution that became the floor of everything else."

* * *

Tendai waited for him to go on.

"You told me about the fertiliser already, in the first week. Haber and Bosch. That is the same story again. Haber figured out how to pull nitrogen from the air in 1909 because he was curious about a specific chemistry problem. Bosch industrialised it because that was his job. By the 1950s the yields from a farmer's land could double or triple. That is where your mother's bag of urea comes from. Another curious person, another specific problem, another quiet decade in which the solution became the floor."

"And then?"

"And then it gets faster. I will not slow down for every decade because the pattern is the same. In the 1970s, the pharmaceutical industry rebuilt itself on petrochemical chemistry. The coatings on your pills, the sutures in our sickbay, the plastic housing on your asthma inhaler if you had one, all petroleum. In the 1980s, plastic packaging replaced glass and metal and wood, and food waste on supermarket shelves dropped by half. In the 1990s, computer chips went into everything, and the helium that cooled them and the resins that sealed them were all petroleum products. In the 2000s, shipping companies consolidated. A smaller number of refineries. A smaller number of pipelines. A smaller number of straits that everything had to pass through."

Sami let that sit for a moment before he went on.

"Each one was somebody's good idea. Each one solved a real problem. None of them was coordinated with any of the others. And the sum of them is a civilisation that cannot function for more than a few weeks if the flow of petroleum stops, and a strait that nobody ever chose as the chokepoint but which has ended up being the chokepoint anyway."

* * *

Sami turned to Tendai.

"Now. Look at your phone."

Tendai pulled his phone out of his pocket. It was a three-year-old Samsung with a cracked case.

"Plastic case, 1930s chemistry. Polymer screen, 1970s. Chips made with petrochemical solvents and cooled with helium during fabrication, 1990s. Battery with synthetic graphite from petroleum coke, 2000s. Your phone is not an invention. It is a stack of inventions, each from a different decade, each using petroleum in a different way, each solving a problem nobody had asked to have solved until somebody solved it."

Tendai looked at his phone as if he had not seen it before.

"Now look at your shoes."

Tendai looked. Standard-issue work shoes. Black rubber, steel toe.

"Synthetic rubber, 1940s. Polyurethane cushioning, 1950s. The adhesive holding the sole together, 1960s. The dye on the leather, petrochemical. The laces, polyester. Every decade you are walking on."

"Everything is oil."

"Almost everything. But the point I have been trying to work out with you is that no one ever decided that everything should be oil. Each of these things was invented because someone had a specific problem. A better lamp. A cheaper car. A stronger fabric. A bigger harvest. A pill that does not dissolve in the bottle. A chip that does not overheat. Each solution was small and reasonable. And over a century, the small reasonable decisions added up to a world where almost every object around you has petroleum in it, and most of the petroleum the world uses has to move through a handful of straits to get where it is going, of which the one we are anchored in is the tightest."

Tendai was quiet. Then he looked at the deck beneath his feet. Then at the paint on the bulkhead beside him. Then at his own hand.

"Even the paint on the ship."

"Even the paint on the ship."

"It was easier," Tendai said slowly, "when I thought someone had done this on purpose."

Sami understood what he meant. A thing designed could be redesigned. A thing accumulated over a century had a different kind of weight. You could not argue with it because there was no single decision to argue with. You could only begin to notice it and ask what to do next.

"Come back tonight if you are off watch," Sami said. "I think the Evening Question is going to be different."

Tendai took his empty plate and left without saying anything else, and Sami watched him go and knew the rest of the afternoon would be quieter than it should have been.

* * *

Sami did not get to read much more.

At 1510, the VHF on his bridge speaker, which he kept patched through to his cabin when he was working, began to carry traffic that was louder and closer together than the normal chatter of the anchorage. He listened for a moment without moving. Multiple vessels reporting activity. Multiple vessels checking on each other.

Then Petros on the intercom. "Captain. Bridge. Now."

Sami was on the bridge in ninety seconds. Petros handed him the VHF handset without a word.

A Thai-flagged tanker, anchored in the southern cluster of the fleet, had been hit. The news was still arriving in pieces, in the broken English that ship captains used with each other over the radio when there were no words for what was happening. There had been a drone. There had been a fire. The Royal Navy of Oman, which monitored the southern edge of the anchorage, had launched two boats. Twenty crew members had been pulled from the water. Three were missing.

The vessel's name arrived last, as these things do. A captain two kilometres away read it off her AIS transponder and said it twice so others could hear. The *Mayuree Naree*. She was still burning.

Sami went out onto the bridge wing. He could not see the *Mayuree Naree* from where the *Noor Salaam* was anchored. He could see a column of black smoke, low on the horizon to the south, and the grey silhouettes of two Omani patrol boats moving toward it.

"Twenty rescued, three missing," Petros said beside him.

"Twenty so far. They may still be searching."

"Yes. Twenty so far."

Petros had said it the way professional sailors say things they have lived through before. Sami went back into the bridge and picked up the ship's internal phone.

* * *

Sami found the bosun on the main deck, repacking a mooring line that did not need repacking.

On a ship like the *Noor Salaam*, the bosun was the senior deckhand, the man who ran every maintenance task and every young deckhand's working day. The deckhands took their orders from him before anything reached the captain. He was the foreman of every rope, paint stroke, and piece of deck gear, and he answered to the chief officer the way a factory foreman answers to a plant manager.

The man who held that job on the *Noor Salaam* was a big man in his fifties, broad across the shoulders in a way that made his work shirts strain at the seams. He worked with his hands in a particular way that Sami had always liked, not fast, not slow, but at the pace of something that could not be rushed because it was the only pace at which the work was done correctly. He had a quiet voice on the rare occasions he used it. The crew called him Boss, which was a running joke in three languages, and he had tolerated it for eleven years.

Somchai Pansawan had been the bosun on Sami's first VLCC command. When Sami moved to a new ship, he had asked the company if Somchai could move with him. He had asked again three ships later. They had been at sea together longer than some marriages.

Sami came up beside him and did not say anything for a moment. Somchai kept working on the line. He had heard the VHF through the mess hall speaker before Sami reached the deck. There was no need to tell him what had happened.

"Any names yet?" Somchai asked, still not looking up.

"Not on the radio. The company will post a list."

"My nephew," Somchai said, "was a bosun on a Thai-flagged tanker last year. I do not know if he is still on her. I have not heard from him in two months."

"Is there a way to find out?"

"I can try. My brother will know."

"Use the satellite phone. Use it for as long as you need. I will tell the office if they ask, which they will not."

Somchai finished the coil of line and set it down. He looked out at the smoke on the horizon. Sami did not ask him what he was thinking, because he had been a captain long enough to know that some questions were answered by not being asked.

"They will find the three," Somchai said after a while. "Or they will not. But the ones who were rescued will not sleep for a long time."

Sami stood with him until Somchai was ready to go back to work.

* * *

Sami went below to find Jun in the engine room.

Jun was on a maintenance task that did not require two hands, but he was using both. This had become his pattern in the second week. Finding work that could absorb his full attention, and doing it slowly. Sami had brought him coffee yesterday and the day before. Jun had taken both without looking up from his work.

Today Sami brought a second mug and sat on the step at the base of the ladder.

"*Mayuree Naree*," Sami said. "Thai vessel. She caught fire after a drone attack. Three of the crew are missing."

"I heard."

"I wanted you to hear from me before you heard it again from somebody else. Somchai's nephew may have been on a ship like that. We do not know yet."

Jun nodded. He did not look up from the valve he was disassembling. After a while he said, "My wife sent the photograph today. The queue goes around a third corner now."

"I know. I saw it when you showed me at breakfast."

"She asked if I was safe. I said yes. I did not tell her about the drone attack. I did not want her to worry more."

"You have until tonight to decide what to tell her."

Jun finished the valve and moved on to the next one. Sami stayed on the step for a few more minutes. They did not speak. The engine room, which had once been the loudest place on the ship, was a quiet space now, with only the auxiliary generator running at low load two decks down. Sami had read early in his career that engineers were people who felt most themselves in rooms full of sound. He thought about that sometimes, watching Jun work in the silence.

He left Jun with the coffee and went back to the bridge.

* * *

At the Evening Question that night, the mood had changed.

Twenty-one people came, which was all but two on watch. Rafiq had made sweet milky tea instead of coffee, which was his way of acknowledging something without saying anything about it. Somchai sat in the back with his arms crossed, the way he sat when he did not intend to speak but wanted to listen. His brother had confirmed that the nephew was on a different ship. He was not aboard the *Mayuree Naree*. Somchai had heard the news an hour before dinner and had not told anyone, though Sami thought the rest of the crew could see it in his face.

Sami took his chair in the circle. He did not begin with a question.

"You all heard about the *Mayuree Naree*. Thai tanker, drone attack. Twenty rescued, three still missing at last report. Somchai's nephew was not aboard. Somchai found out an hour ago."

Several of the crew turned to look at Somchai, who nodded once without smiling, which was his way of saying he was all right and did not want anyone to make a fuss.

"The question for tonight is short. Who decided that a fifth of the world's oil would pass through this strait? Who made that decision?"

Arjun answered first. "Geography. It was not decided. It is where the oil is."

"The oil is in the ground. The ground is not the decision. The decision is what we built around it."

Dani said, "The oil companies."

"Some of them. But the first commercial oil find in Saudi Arabia was 1938. The first in Kuwait was 1938. Iranian oil was earlier, 1908, but on a small scale. The Gulf became the centre of world oil over the next eighty years, one discovery at a time, and the shipping lanes were drawn afterward to serve what was already there. Nobody in 1910 could have told you this strait would matter. Nobody in 1950 was designing for what would happen if it closed."

Tendai, who had spent the afternoon with Sami, did not add anything. He did not need to. His face had the look of someone who had already walked the argument forward and was watching it land on the rest of the room.

"The *Mayuree Naree* caught fire today," Sami said, "because a ship was anchored in a strait that closed because a war was fought over a resource whose centrality no one alive had actually decided on."

Rafiq poured more tea. The room drank.

Then Viktor said, in the tone he used when he was about to say something he did not want to say, "In Ukraine, the gas pipes were laid before I was born. Forty years of pipes. By the time anybody under fifty understood whose pipes they were, they were already the only ones we had."

No one spoke for a while after that. It was quieter than the room had been all evening.

* * *

Later, Sami stood on the bridge wing in the dark.

The anchorage was a field of lights now. Over a hundred ships within visual range, each one a small community of people doing exactly what his crew was doing, which was waiting and trying to understand. Somewhere past the southern horizon, two

Omani patrol boats were still searching for three Thai sailors who had been in the water for nine hours. Military jets passed overhead every half hour or so, their navigation lights moving fast against the stars. The jets were not there for the ships. They were there for each other. The ships were incidental.

Sami had a small paperback of the *Tao Te Ching* in his cabin, bought years ago in the same Singapore bookshop where he had bought the Sun Tzu. He did not have it with him now, but he remembered one of the lines. It had landed the first time he read it, and it was landing harder tonight. Lao Tzu had written twenty-five centuries ago that water was the softest thing in the world, and that given enough time, water would carve the hardest rock. He meant that the gentle thing could overcome the unyielding thing if it did not stop. He meant that the patient accumulation of small actions would change what seemed unchangeable.

The century of construction Sami had walked Tendai through that afternoon was water. The world was rock. Each decade had added a few more drops of dependency. Nobody watching closely would have called any individual drop a problem. Nobody on shore looking at a bag of fertiliser or a plastic spoon or a phone or a bottle of aspirin was watching for what was being built underneath them. The rock was changing while everyone believed it was the same rock it had always been.

Lao Tzu had been writing about how a patient ruler could overcome a strong one. Sami thought he had been writing about something larger. The softest thing is the thing that no single person noticed being built. By the time anyone looks up, the canyon is already there. The water does not seem to have done much. Each drop was reasonable.

A fifth of the world's oil now flowed through this strait not because anyone had designed it to, but because a century of water had carved the riverbed where the oil happened to be. Now the river was dammed, and the world was discovering that it had forgotten what the land had looked like before the river.

He thought about Somchai on the deck that afternoon, repacking a line that did not need repacking, because a man whose work had just become dangerous needed something for his hands to do while he waited to hear about his nephew. He thought about Jun in the engine room, working slowly. He thought about Tendai looking at his own hand as if he had never seen it before.

The softest thing had carved the rock. It had also carved the people who stood on the rock. They were beginning to notice.

* * *

14 March. End of week two at anchor. Three days since the Mayuree Naree caught fire. Three missing, twenty rescued. Somchai's nephew was on a different ship.

The fire is still the thing the crew is carrying. I have watched it settle into them over the three days since, in the way sailors absorb news of other sailors. On the twelfth, Somchai was quieter than usual at meals, which for Somchai is already a small thing to notice. Rafiq cooked Thai food for dinner without announcing it and without explaining why. Somchai took his plate, nodded to Rafiq, and ate. That was the exchange. On the thirteenth, Tendai asked me at the end of the bridge watch whether ships like ours were now targets. I told him the honest answer, which is that we do not know yet. He absorbed it the way he absorbs things, which is to say he put it somewhere inside himself and went to find something useful to do with his hands. On the fourteenth, today, Jun ran the third test on the cooling pump and told me the bearing would hold another week. He told me this in a tone that suggested the bearing was not what he was actually thinking about.

The crew is steady. That is what I want to record tonight. They are taking the news in and carrying on, which is what professional crews do. But something has shifted, and I am marking it here so that I will remember when it matters. Before the eleventh, the conversation at dinner assumed the ship would move soon. Since the eleventh, nobody has said that out loud. The assumption has quietly changed and nobody has marked the change. That is usually how these things go.

I spent the week reading about how the dependency on this strait was built. I wanted to find a decision. A moment when someone chose to put a fifth of the world's oil through one waterway. I did not find one. There was no decision. There were a hundred thousand small reasonable choices over a century and a half, each of which was good for somebody, and the sum of them is the world we are sitting in the middle of tonight.

Tendai was in the chair beside me for three hours on the afternoon of the fire, while I walked him through what I had found. A chemistry professor. A car maker. A curious chemist at DuPont. A German who wanted to pull nitrogen out of the air. None of them were thinking about this strait. None of them were thinking about each other. Each of them solved a specific problem. Each solution became the floor of something bigger, and the floors stacked on top of each other until we were all standing on them without knowing.

Tendai said it was easier when he thought someone had done this on purpose. I understood him. A thing designed can be undone by decision. A thing accumulated by

a century of small steps can only be undone by another century of small steps, and nobody has started the second century yet.

Viktor said something at the gathering the night of the fire, about Ukrainian gas pipelines, that I have been carrying with me all week. Forty years of pipes laid before he was born, and by the time anyone his age understood whose pipes they were, they were already the only pipes they had. We have all been living in something like that, without noticing. Viktor noticed because his country was torn open and the pipes were exposed. The rest of us have not been torn open yet.

The softest thing makes the deepest cuts. Lao Tzu got there first, and I am sitting on a ship in a strait that proves him right. The water carved the rock. The rock is us.

* * *

* * *

PASSAGE 5

The Accelerant

Sami noticed the change in tone a few days into the third week.

It was nothing large. A short answer where there would usually have been a joke. Somebody eating alone who had been eating at the table. Arjun and Dani arguing about the satellite wifi, which was a real grievance because the connection dropped more often than it should, but the edge in their voices was new. Rafiq dropped a plate in the galley. The plate broke. He picked up the pieces without saying anything and put them in the bin, and he did not look up when Sami came in for a cup of tea.

Three weeks anchored was longer than any crisis any of them had been through before. Ships had been approached by pirates, held up by weather, delayed by engine trouble. This was different. There was no mechanical problem to solve, no destination to run for. The days came around in the same order. Maintenance, meals, news from home. Maintenance, meals, news from home. And the news from home was worse every time the cycle came around.

Sami had seen this pattern in shorter form over the years, on long transits and in bad weather, but never for three weeks and never with a war on the horizon. A crisis with a clear shape was something a crew could work through. Waiting without a clock was harder. It wore on people in ways that did not show until they showed all at once.

He started staying in the mess hall longer after meals. Not to supervise. Just to be there, so the crew could see that the captain was also here, also waiting, also not getting anywhere faster than they were.

* * *

Tendai came to his cabin on the morning of the sixteenth day.

"Captain. I have been turning something over since we talked on the eleventh."

"Sit down."

"You told me about the chemistry professor and the Model T and Carothers. Every answer you gave me was about somebody making something. But when I look at the

news now, everything is about fighting. I want to know whether oil was always worth fighting over, or whether that is something that came later."

Sami thought about the question for a moment. He had expected something like it eventually, and Tendai had got there a week before he thought he would.

"It came later. And it came fast."

"How fast?"

"Pearl Harbor. December 1941."

"The attack that brought America into the Second World War."

"That is the one most people know. What most people do not know is what was happening in the months before the attack, and why it happened."

Sami pulled up a summary he had been reading the night before, and Tendai leaned in.

"Japan in 1940 was trying to run a modern military and a modern economy on almost no domestic oil. They imported nearly all of it, and most of what they imported came from the United States. In July 1941, the United States froze Japanese assets and stopped oil exports to Japan. The reason was that Japan had been expanding violently in China and Southeast Asia, and the embargo was meant to punish that expansion. What the embargo actually did was give Japan's generals a clock. Without American oil, the Imperial Navy had about two years of fuel in reserve. When the fuel ran out, so did the military. And when the military ran out, so did the empire."

Tendai was following closely.

"So the generals had a choice. They could pull back from their expansion, make peace with the United States, and accept a smaller role in the world. Or they could attack the United States directly and try to seize the oil fields of the Dutch East Indies before their reserves ran out. They chose the attack. Pearl Harbor was not a war over territory. It was a war over fuel, launched because the Japanese military had been told to expand without being given the fuel to do it, and the men telling them to expand would not back down when the fuel started running out."

Sami paused.

"The attack that brought America into the Second World War was the response to an oil embargo."

Tendai set his mug down on Sami's desk.

"An embargo started it."

"An embargo started it. And embargoes never really stopped after that. In 1953, the American and British governments helped overthrow the elected prime minister of Iran because he was trying to nationalise his country's oil industry. The coup installed

a king. The king was overthrown in 1979, and the country became the Islamic Republic that is now, seventy-odd years later, the reason we are sitting at anchor in this strait. Every war and every intervention in the Gulf since 1941 can be traced back, in some part, to oil."

"You are saying we started this by fighting over it eighty-five years ago."

"We have been fighting over it longer than that. What is different is that since 1941, the fighting has been organised. Armies, central banks, treasury departments. The instruments got more sophisticated with each decade. Pearl Harbor was done with ships and planes. Most of what has been done since is done with sanctions and coups and financial pressure as well as with bombs. The shooting wars are only the eruptions. There are quieter wars underneath them, and the quieter ones are what I have not thought enough about."

"Quieter how?"

"That is what I want to spend this week on. Come back tomorrow. I will have read more by then."

Tendai nodded, took his mug, and left.

* * *

Sami did not leave his desk for most of the next two days.

Tendai had asked about wars. What Sami kept finding was that the wars were the simple part. The hard part was what came between the wars, in the decades nobody counted.

He made a list on a sheet of paper from his notebook. At the top he wrote: *the quieter wars*. Underneath he began adding what he had found.

Venezuela. In the late 1990s, Venezuela had been producing three and a half million barrels of oil a day. It held the largest proven oil reserves on earth, larger than Saudi Arabia. By 2023, its production had fallen to about one million barrels a day, less than one percent of world supply. The collapse came from many causes, most of them the country's own doing. But the United States had imposed a full embargo on Venezuela's state oil company in 2019 and kept it in place for seven years. Exports to India and the European Union were halted. The government lost access to dollars. Hyperinflation followed. Eight million Venezuelans, about a quarter of the country's population, left the country. Two United Nations rapporteurs called the sanctions economic warfare. On January third of 2026, two months before Sami had sailed from Ras Tanura, the American military arrested the Venezuelan president in his palace and flew him to New York for arraignment.

Iran. The United States had sanctioned Iran's oil exports in various forms since 1979. Forty-seven years of sanctions aimed at forcing a change of government that never came. What the sanctions produced instead was a generation of Iranian leaders who had learned to operate inside sanctions, and a population that had learned to hate the country doing the sanctioning. The war now being fought across the strait was, in part, the bill coming due for nearly five decades of pressure that had destabilised a region without replacing what it destabilised.

Russia. In 2022, after the invasion of Ukraine, the Western nations imposed oil price caps and trade restrictions. Europe cut Russian gas imports and found itself in an energy crisis of its own. Russia rerouted its oil to India and China at discounted prices. By 2026, Russia was financing forty percent of its federal budget from oil and gas exports that the sanctions had been designed to stop. The sanctions had hurt Russia and they had hurt the countries that imposed them almost equally. Yet the sanctions had stayed in place. Until, two weeks ago, when the United States Treasury had quietly given India a thirty-day waiver to buy stranded Russian cargoes, because Indian drivers could not afford to queue for petrol that was not coming from the Gulf.

Sami stopped reading when he got to that last line and looked out the porthole for a long time.

Three of the largest oil producers on earth, each placed under American or European sanctions. Venezuela, production collapsed. Iran, production restricted, now at war. Russia, production diverted but not reduced. Combined, these three countries held roughly a third of the world's proven oil reserves. For most of the last decade, the world had treated them as unavailable, or partly available, or available only on specific terms. The oil that the world actually ran on day to day, the oil that moved freely and could be counted on, came from the countries that had not been sanctioned. A fifth of the world's total oil, Sami now understood, moved through the strait the *Noor Salaam* was anchored in. That fifth had not grown in three decades. What had grown, round by round of sanctions, was how much the rest of the world's stability depended on this one waterway still functioning.

Sami underlined one line on his paper. Then he wrote, at the bottom:

Three of the world's largest oil producers restricted by sanctions. A fifth of global oil still moves through one strait. Every round of sanctions made that strait matter more, not because more oil went through it, but because less slack was left everywhere else. And now it is closed.

He looked at the sentence for a while. It was a simple sum, and almost nobody had been doing it out loud.

* * *

On the morning of the nineteenth, Sami made a decision about the evening.

He had been watching the crew for a week. Rafiq's plate. Dani leaving the card games. Petros drinking his coffee on the bridge now instead of bringing it down to the mess hall. Small things. But small things in a crew were how large things started.

He went to find Somchai on the main deck. Somchai was coiling a line from a maintenance check and stacking the coils along the bulwark. He looked up when Sami stopped beside him.

"Somchai. I want to move the Evening Question tonight. Out here on the foredeck. Bring out some chairs from the mess. Enough for the ones who want to sit. The rest can lean on the rail."

"Outside?"

"Outside. People have been looking at the same four walls every evening for three weeks. I want to give them different scenery for one night. See if it helps."

Somchai thought about it, then nodded. "The forward hatch, port side. Out of the wind. I will set it up before dinner."

"Thank you."

"Captain."

Somchai turned back to his coil. Whatever he set up that evening was going to be neater than the mess hall had ever been, because that was how Somchai did things.

* * *

The crew came out after dinner in ones and twos, the way they always did, but they arrived on deck differently from the way they arrived in the mess hall. In the mess hall, people sat down fast and arranged themselves in a circle that was almost the same every night. On the foredeck, they stopped when they came through the forward door, looked around, and chose where to put themselves. A few took the chairs Somchai had arranged. Others leaned on the rail. Jun stood by himself near the port-side lifeboat, looking out at the water. Viktor chose a coiled mooring line for a seat and settled onto it without hurry.

The sky was clearer than it had been in days. The lights of the anchored fleet spread to the horizon in every direction. The air smelled of salt and diesel and something sweet Rafiq had been baking earlier. Two military jets passed east to west

overhead, their navigation lights fast and high.

Sami did not take one of the chairs. He leaned on the rail next to Viktor's mooring line.

"I thought we could get some air tonight."

A few of the crew nodded. Tendai smiled.

"The question I have been working on is one Tendai asked me four days ago. Was oil always worth fighting over, or is that something that came later. I have been reading about it. What I want to do tonight is tell you what I found, because I think it explains why we are in the strait we are in, and not some other strait, or no strait at all."

He gave them the history first. Pearl Harbor. Iran 1953. The Gulf Wars. He kept it short, because he had walked Tendai through most of it already and he did not want to repeat to the whole room what was already circulating on the ship.

"Those are the wars that get counted. What I spent most of this week on are the wars that do not get counted. The sanctions. The embargoes. The coups and economic pressure that never make it onto a list of conflicts but that have done as much damage as any shooting war, and sometimes more."

He told them about Venezuela. The largest oil reserves on earth. Three and a half million barrels a day collapsed to one. Eight million people gone from a country of thirty. The American operation in January that arrested a head of state in his own palace. He told them about Iran. Forty-seven years of sanctions aimed at changing a government that had outlasted three American presidencies by the time he was a cadet. He told them about Russia. Sanctions in place for four years, circumvented by pipelines to China and India, financing forty percent of a federal budget that the sanctions had been designed to drain.

Then he gave them what he had found.

"Three of the largest oil producers on earth. Venezuela, Iran, Russia. Together, about a third of the world's proven reserves. Each one, for most of the last decade, under some form of restriction from the United States or Europe. The theory was that we could isolate those three and keep the system running on the oil that still moved freely. A fifth of the world's oil moves through this strait on any ordinary day. That share has not grown. What has grown is how much the rest of the system needs this strait to keep working, because the buffers everywhere else have been used up. And now this strait is closed."

The deck was quiet.

Arjun was the first to speak, and when he did he was not speaking to Sami. He was speaking to the part of himself that had said something similar two weeks earlier, in the mess hall, the morning he had brought Sami the news about the waiver.

"I said something like this to the crew the morning America gave India the waiver. Do you remember? The sixth. I remember because that was the date my mother had been trying to buy the extra cylinder. I told you America was telling my country it was acceptable to buy oil from Russia, a country America had been sanctioning for years, because there was no other option."

"I remember."

"I thought it was an irony. A moment. The kind of thing that happens once and you talk about it. I did not think of it as the moment the strategy failed."

"It was the moment the strategy failed."

"So for two weeks I have been watching my country buy oil from the country we were not supposed to be buying from, and I thought we were just improvising. But we were not improvising. We had been set up to fail."

"We had built the system in such a way that it failed as soon as this strait closed. The waiver was not an improvisation. It was the admission."

Dani, who had been quiet for weeks, spoke next.

"Then why have sanctions at all? If you can only sanction a country while you do not need their oil, you are not really sanctioning them. You are performing sanctions."

"That is the question the research does not want to answer clearly. Sanctions were meant to change behaviour. Mostly they did not. Venezuela did not change. Iran did not change. Russia did not change. What the sanctions did change was the buffer. The world's oil comes from many places. The Americas. The North Sea. West Africa. The Gulf. A fifth of it moves through this strait, and that share has barely shifted in thirty years. What did shift, round by round of sanctions, was how much slack the rest of the system had left to absorb a shock somewhere. We did not push more oil through the strait. We made the strait the place the world could least afford to lose. The oil kept finding the easiest path. We kept narrowing what counted as easy."

Arjun said, "Did anyone warn about this?"

"People warned. Analysts, a few journalists, some think tanks. The warnings were treated the way warnings are usually treated when the thing being warned about has not happened yet. They were noted and filed."

Then Rafiq, who rarely spoke at the Evening Question, said, "So we overplayed our hand."

The rest of the crew turned toward him. This was the longest thing Rafiq had said in a gathering since the first week.

"Yes," Sami said. "We overplayed our hand."

"Because we assumed what was left would always be there."

"Because we assumed it would always be there."

Rafiq looked back at his mug and did not add anything else. It was the most he was going to say.

Viktor, on his mooring line, had been quiet throughout. Now he shifted in the way that made the rest of the crew aware he was about to speak. On the deck, the signals were easier to read. In the mess hall, with everyone inside the same circle, some of them got lost.

"In my country," Viktor said, "the first winter after the Russian gas stopped, the electricity went off four hours every day on a schedule. Eight to ten in the morning. Seven to nine at night. We learned the schedule the way people learn a commute. Nobody rioted. People played cards by candlelight. It was the quietest winter anyone remembered. That was almost four years ago."

Nobody moved.

"We are now in our fourth winter of it. The schedule is worse. The grid is worse. My sister lives outside Kyiv. She sends me photographs of her children doing homework by head torch, because that is what people do now. The Russian drones still come. The power goes out when they hit the substations. She counts the hours until the light comes back, and when it does she puts the washing machine on, because she does not know when it will go out again."

He was not looking at Sami now. He was looking at the water.

"I will tell you what the sanctions look like from my side. The country being sanctioned is still there. Its leaders are still there. Its generals are still there. The people who feel the sanctions are the old woman who cannot afford her heart medication, and the farmer whose fertiliser costs three times what it did last year, and the mother who has to decide which of her children eats tonight. Sanctions do not make governments fall. They make ordinary people smaller. I have lived next door to a country that has been sanctioned for longer than my niece has been alive. The ordinary people of that country have not toppled anyone. They have got harder, and they have got angrier, and my sister's children are paying for the anger in the dark."

Viktor turned to Sami.

"The sanctions do not work the way the people who write them think they work. They do not change the government. They change the people. And the people

remember. My country will remember too, when this is over. If it is ever over."

The silence after Viktor's contribution was longer than any silence that had come before it on any evening. The two jets that had passed earlier had long since gone. Nobody looked up to check for new ones.

Sami waited as long as the silence wanted to continue, and then he said, quietly, "Thank you, Viktor."

Viktor shook his head once, which meant the thanks was unnecessary and also that he did not want more of it.

Rafiq had brought out a pot of sweet milky tea without anyone asking. He poured slowly into mugs. Tendai took one in both hands and held it without drinking.

The session ended without a question being asked. Sami did not ask one. He had not come to make anyone produce an answer. He had come to take the crew out of the mess hall, and something about being out of it had let them speak differently than they had in weeks.

Later, after most of the crew had gone back inside, Viktor was still on his mooring line. Sami leaned on the rail beside him.

"You said more tonight than I have heard you say in the twenty years we have been at sea together."

"The deck is a different room."

"I noticed."

"The mess hall is a room where we are inside a ship. The deck is a room where we are outside of everything. It changes what a man will say."

Viktor looked at the water, then back at Sami.

"I will not say it twice. The crew heard it. That is enough."

"That is enough."

Viktor stood, nodded, and went below.

* * *

Sami did not sleep much that night.

Viktor's last sentences had lodged somewhere he could not reach. *Sanctions do not change the government. They change the people.* He had been thinking about them at the wheel during the night watch. He was still thinking about them the next morning when he opened his laptop after breakfast.

He started looking for the country that had been sanctioned the longest.

The answer was not Iran, which had been sanctioned in its modern form since 1979. The answer was Cuba. The American embargo on Cuba had started in 1962, the

year before Sami's father was born. It was older than most of the people alive on the planet. It had been tightened, loosened, tightened again across twelve American presidents, and it had never been lifted. Cuba had spent sixty-four years as the test case for a question nobody in Washington had been willing to answer directly. What happens to a country when you cut it off from the world's economy and keep it cut off, not for a year or a decade, but for three generations?

Sami read through the morning and into the afternoon. He had known the broad outline. Most people who read the news had known the broad outline. What he had not done, until this week, was put the outline next to the outlines of Venezuela and Iran and Russia and look at all of them together.

Cuba had lost its first oil patron when the Soviet Union collapsed in 1991. The years that followed were called, in Cuba, the Special Period. Calories per person fell by a third. Cubans lost an average of nine kilos in the first two years. The buses stopped running. Pharmacies emptied. The country survived by tearing up ornamental lawns in Havana and planting vegetables on them, and by teaching itself, in a hurry, how to farm without the synthetic fertilizers the Soviet trade had been providing. It was one of the few national-scale examples in modern history of a country successfully growing its own food without petrochemicals, and it was forced, not chosen. They proved it could be done. Then, when a new oil patron appeared in the form of Hugo Chávez's Venezuela, they went back to depending on subsidised imports, because the old way was easier.

Venezuelan oil had kept Cuba running for the two decades since. In exchange, Cuba had sent doctors and security personnel to Caracas. It was not a trade relationship as much as a lifeline.

Then, on January 3, the American military arrested the Venezuelan president. The oil shipments to Cuba stopped within weeks.

Sami stopped reading for a moment when he got to that. Ten weeks before the *Noor Salaam's* engines had gone silent, Cuba's second oil patron in thirty years had been taken off the board by a country whose embargo had already been starving Cuba for six decades. The Special Period was beginning again. Blackouts were rolling through Havana. Sugar rations were down. The medical system that had been one of the country's few sources of pride was running out of the petrochemical inputs most of its medicines depended on. And now, with the strait closed, every alternative supply on the world market cost more than Cuba could afford.

Sami wrote a note to himself beside his Venezuela notes.

Cuba has been sanctioned for sixty-four years. Nobody there voted for the embargo. Nobody there chose the government the embargo was aimed at. The people who chose the embargo have all retired. The people who are living with it now were not yet born when it started.

He put down his pen.

He thought about Viktor's line on the deck. *They do not change the government. They change the people.* Cuba was the longest-running proof of the point Viktor had made the night before. Sixty-four years of an embargo meant to make a government fall, and the government had outlived American administrations that had put children into the White House who were now grandparents.

He thought about the Philippines, where Jun's wife was photographing a queue that went around three corners now. The Philippines had been hit by something none of them had chosen, something that had happened to them because of a war in a strait they had no soldiers in. The Filipinos in the queue were victims of circumstance.

The Cubans in the blackout were different. They were victims of choice. A choice that had been made before most of them were born, by people who did not have to live with the consequences, and renewed every four or eight years by politicians who had never been to Havana.

Sami closed his laptop and went for a walk on the deck to try to get some of it out of his head. It did not work. He came back to his cabin an hour later and read the same articles again, because he was not sure what to do with what he had read except keep reading it.

* * *

In the days that followed, the crew spoke about things they had not spoken about before. The deck session had opened something, and nobody wanted to close it.

Arjun reported on the twentieth that cooking gas in his mother's area was now officially rationed. One cylinder per household. His mother had bought the extra cylinder before the restrictions, and she was making it last by cooking fewer meals on the stove and more on the old kerosene burner her mother had kept in the back of the closet for forty years. "She says her mother used that burner during the Emergency in 1975. She thought she would never use it again."

Sami kept a coffee routine going with Jun through the week. He brought a mug down to the engine room in the late mornings, set it on the step at the bottom of the ladder, and left. Jun took it when he was ready for it. They did not speak. Some men did not need questions. They needed the presence of someone who was not going to

ask any.

Sami's own reading was getting darker as he went. The Niger delta, where oil extraction had funded a civil war and was still funding armed groups decades after the war ended. Iraq, twice. Sudan. Libya. Syria, where oil smuggling had paid for everything from government salaries to militia uniforms. Each of these was a place where the substance in the *Noor Salaam's* hold had been a cause of death, directly and indirectly, for years. And in each, the death did not end when the fighting stopped, because the economy and the state had both been built around the thing that was being fought over.

On the morning of the twenty-second, Sami found a figure that stopped him. The American military, the single largest consumer of petroleum in the United States, used twelve point six million gallons of fuel every day. If the Department of Defense were a country, it would rank thirty-fourth in the world in daily oil consumption. A single B-52 bomber burned three thousand three hundred gallons an hour.

He wrote the number down in his notebook and looked at it for a long while.

The warships enforcing the blockade in the strait were burning oil to keep a lane open for ships that were carrying oil. The bombers overhead, which he could see sometimes on clear nights, were burning oil at the rate of small countries in order to defend the supply of the substance they burned. It was not a contradiction. It was a loop. Each part fed the part after it, and the part after that fed the first part again.

He showed the B-52 number to Tendai that evening over the Evening Question, which was back in the mess hall but quieter than it had been in the first weeks. He did not tell the crew where he had got the figure. He just read it out.

Tendai looked up.

"How much fuel is that in my mother's village?"

"How much does your mother's village use?"

"Nobody there has a car. There are two diesel generators. They run about two hours a night when the grid is off, which is most nights. Maybe twenty litres a night for the whole village. Call it two hundred gallons a year."

Sami worked the numbers in his head, then again on his notepad to be sure.

"A single six-hour bombing mission by a B-52 burns about twenty thousand gallons. That is a hundred years of your mother's village, in six hours."

Tendai said nothing for a moment. Then he said, "One night. The planes do that every night, somewhere."

"Every night."

On the night of the twenty-third, Sami went to the bridge for the late watch.

Petros was there. They drank coffee without talking. The lights of the fleet spread out through the bridge windows. Above, military jets passed every half hour or so, high and fast. Below, on the mess deck, Rafiq would be putting out the flour for tomorrow's bread, and Jun would be in his cabin doing whatever Jun did in the hours when he was not in the engine room, and Tendai would probably be awake, reading on his phone.

On the shelf in Sami's cabin, the Sun Tzu was where he had left it. He did not need to open it tonight. He already knew the line he would have read if he had.

Sun Tzu had written that no nation had ever benefited from prolonged warfare. He did not mean that wars were bad in a moral sense, though Sami suspected he thought they were. He meant something more practical. A war that did not end quickly was a war that consumed the winners as thoroughly as it consumed the losers. Treasuries emptied. Armies wore down. Supply lines failed. Countries that had been strong at the start of a war became weak in the middle, and by the end, even the victors had been diminished to the point where their victory did not deliver what they had fought to gain.

Sami thought about what Sun Tzu would have said about sanctions. They were not war in the form Sun Tzu had known. But they were the same animal. A pressure applied continuously, intended to force a change of behaviour, which did not produce the change but did produce the exhaustion. Iran, forty-seven years. Russia, four. Venezuela, seven. Cuba, sixty-four. Each of them a slow war that had not delivered its stated goal, and each of them a slow war that the people doing the pressing had not had to pay for. Until now, maybe. Had they overplayed their hand?

A prolonged war, Sun Tzu had warned, did not stay where the fighting was. It reached outward, through the supply lines, and bled the distant countries that had not chosen the war but depended on what the war controlled. That was the strait right now. That was Jun's wife in Cavite. That was Tendai's mother in Chipata. That was Arjun's mother in Mumbai.

And somewhere out past all of them, past the strait and the anchorage and the closed routes that everyone was now arguing about, was an old woman in Havana. She had never been warned about any of this. She had no say in any of it. She was in her fourth decade of not having enough of almost anything, because of a decision made before she was born by people who would never meet her and would never

know her name.

Sami watched the jets pass.

The jets overhead were burning the substance this war was being fought over. As long as they flew, the war would not end, and as long as the war did not end, the jets would burn more of the substance they were flying to protect. Every link in the chain pulled the next link tighter.

He thought about Viktor on the mooring line four nights ago. The deck as a different room. Viktor's sister in a house outside Kyiv, where the children did their homework by head torch because that was what people did now.

Sun Tzu had been writing to generals. Viktor had been writing to whoever was listening. They were saying the same thing in different centuries. The protracted conflict exhausts everyone. The one who fights longest loses even if he wins. And the people who pay for the protraction are almost never the people who started it.

* * *

23 March. End of week three. Tomorrow is the twenty-fourth. I do not know why I am writing that. The twenty-fourth is just another day, and yet I have the feeling tonight that something is about to shift. I have had the feeling for three days.

We moved the Evening Question to the foredeck on the nineteenth. Somchai set up the chairs. Viktor talked longer than I have heard him talk in the twenty years we have worked together. He said his sister is in a house outside Kyiv where the children do their homework by head torch now, because that is what people do in the fourth winter of a war. He said sanctions change the people, not the government. I have been turning both sentences over all week.

Rafiq said we overplayed our hand. He said it the way he says things at dinner. He was right. Three of the largest producers sanctioned. A fifth of the world's oil still moving through this one thirty-kilometre gap, the same fifth it has been for thirty years. Every round of pressure on the three made that gap matter more, not because more oil flowed through it, but because less slack existed anywhere else. We did not build a system that could only fail in one place. We built a system in which the failure here would hurt most, because we had already spent the buffers that would have absorbed it. The failure was not a surprise. It was arranged.

Tendai asked a question about the village and we worked it out together. One B-52 mission. A hundred years of his mother's village. Six hours. He said it without expression, and then he said, one night, every night somewhere.

I spent the day after Viktor's speech reading about Cuba. Sixty-four years of embargo. An old woman in Havana who was not born when the policy that has shaped her whole life was written. She is not in the news. She is not in any of the research I have read this week. She is the person the embargo was never aimed at and the only person who has actually felt it for three generations. I do not know what to do with her, but I will not forget her.

No nation has ever benefited from prolonged war. Sun Tzu said it first. The men on this ship are proving it, and they have not fired a shot. The people in Havana have been proving it longer than any of us have been alive.

* * *

* * *

PASSAGE 6

Eight Days

On the morning of the twenty-fourth, Jun knocked on Sami's door before breakfast.

He was holding his phone. He looked like a man who had not slept, or had slept badly.

"Captain."

"Come in."

Jun sat on the edge of the chair. He did not settle into it.

"My wife sent this an hour ago. I wanted you to see it before I decided what to do."

He turned the phone so Sami could see. The photograph was of the queue at the petrol station on his wife's road. The queue Jun had been showing Sami a section of, every day, for three weeks. This morning the queue was not partial. The photograph had been taken from an upstairs window, and it showed the queue in full. It went down her road, around the corner, down the next road, and out of frame.

"Government announced at midnight. The Philippines is the first country in the world to declare a national energy emergency. Four-day government work week. Schools closed for two weeks. Power rationing by neighbourhood. Fuel reserves are down to fifty-three days for gasoline, forty-six for diesel, thirty-nine for jet fuel."

Sami handed the phone back.

"Your daughters?"

"Fine. Scared. My wife asked me this morning what to say when they ask her questions she cannot answer. I told her I did not know either. I told her we would say something when we knew something."

"That was the right answer."

Jun looked at the phone, then put it face down on the desk.

"Captain. For three weeks I have been telling her I was safe, and she has been telling me she was safe, and neither of us has been lying, exactly. But I think we were both waiting for the other one to say something. Now the thing is said. The emergency is declared. There is no more waiting to see if it will get bad. It is bad. And I am still

sitting on a ship full of the substance my country does not have enough of."

"I know."

"I do not know what to do with that. Other than what I am already doing, which is maintaining engines that have not run in twenty-four days."

"That is not small."

"It feels small."

Sami poured him coffee from the thermos on the desk and set the mug in front of him without comment.

"The ship you are keeping ready will move when she is allowed to move. When she moves, she will deliver her cargo to a refinery that will send products to dozens of countries, including yours. Nothing you are doing is small."

Jun took the mug without looking up.

They did not speak for a few minutes. Through the porthole, the anchorage was bright in the morning sun. The same hundred ships. The same stillness. The same absence of movement in either direction.

Jun stood.

"I will be in the engine room."

"You know where I am."

Jun left. Sami stayed at his desk with his hand on the surface where the phone had been.

* * *

Jun had been gone from Sami's cabin for less than ten minutes when the bridge intercom clicked on.

"Captain. You have a minute?"

It was Petros. Sami went up.

Petros had the VHF on low and a phone in his hand. He passed the phone to Sami without a word. It was open to a Reuters piece dated that morning, the formal version of the news Jun had brought to Sami's cabin a few minutes earlier. Sami scanned past the headline and the first paragraph he already knew. What Petros had wanted him to see was lower down. The emergency applied to hospital fuel deliveries. Emergency rooms in Manila would operate on reduced backup generator capacity until further notice.

"Jun know?"

"Jun knows the announcement. I do not think he has seen the hospital part."

"Do not put it in his way this morning. He will find it on his own, or he will not. Either way, not this morning."

"Understood."

Petros handed back the phone and went back to his watch.

Sami stood on the bridge for a few minutes looking out at the anchorage. A hundred and forty-three ships now, Petros had counted on his log that morning. Two more had arrived overnight. They came in at low speed with their anchor chains already prepared, and they dropped in the outer ring because the inner ring was full. Each new arrival meant another twenty or thirty people who had been told to wait, and nobody telling them for how long.

He went back to his cabin and opened the notebook. Under Venezuela, Iran, Russia, and Cuba, he wrote a fifth entry.

Philippines. Emergency declared 24 March. Hospital backup capacity reduced.

He looked at the four countries above it. He did not yet know, as he wrote it, that he would be adding a sixth before the day was out.

* * *

Lunch was rice and dal and a tomato curry Rafiq had put together from the last of the tomatoes. The mess hall was half full. Most of the crew had eaten early because the day had started early for everyone on the ship, not just Sami. Arjun came in when Sami was halfway through his plate. He took a portion from the serving counter, stood for a moment with the plate in his hand looking at the tables, and then came over and sat across from Sami without asking.

Sami had noticed Arjun for a few days. He had been quiet at meals. He had been slower at the card games. He had been going back to his cabin straight after the Evening Question instead of staying to talk. The edge that had been in his voice during the wifi argument with Dani last week had gone. In its place was a flatness that Sami recognised and did not like.

"Captain. The Indian government made it official this morning. Cooking gas is being rationed nationally. One cylinder per household per month. No exceptions, no waivers unless you can prove medical hardship."

"Your mother?"

"Still on the cylinder she bought early. She is making it last by using the kerosene burner."

"How long does she have?"

"She thinks four weeks. Maybe five if she is careful. After that, I do not know."

Arjun stopped eating. He looked at his plate for a long moment before he said the next thing.

"There is something else. Something I have not told you."

"Tell me now."

"My cousin Ravi is on dialysis. Three times a week at the district hospital in Thane. They cut him back to twice a week when the fuel rationing started. His numbers were bad at the last treatment. My aunt called my mother last night."

Sami put his fork down. The Philippines piece that morning had stayed with him. Hospital generators running short of diesel in a country that imported most of it. He had been worrying for a week about whose family on this ship it would reach. Arjun had been worrying longer, and about someone specific. He knew Ravi's dialysis schedule. He would have been watching the rationing news with a calendar in his head the rest of the crew did not have, working out how long before his cousin lost a session, and another. Sami understood now what had been bothering him for a week. Arjun had been carrying this alone. Saying it out loud would have meant saying he was at sea while his cousin ran out of time.

"What does the hospital tell her?"

"They told her they were prioritising the sickest patients. My aunt said, is my son not a sick patient. They said, yes, but there are patients on the machine who will die tomorrow if they come off it. Ravi has three or four weeks, if he stays on twice a week. They cannot promise even that, if the generator fuel runs low."

The Indian waiver on Russian oil was about crude imports, not about LPG or about the diesel that ran a hospital generator in Thane. Those were different supply chains. The Russian oil that kept Indian drivers on the road would not fill his mother's cooking cylinder, and it would not keep Ravi's dialysis machine running.

Arjun had not picked his fork back up. "She said, Arjun, if you can find a way to come home, come home. I told her the ship cannot move."

"She knows the ship cannot move. She is asking because she is frightened."

"Yes."

"Arjun."

"Yes, Captain."

"I am going to tell you something and you do not have to do anything with it. My younger brother died in Homs in 2012. I was at sea. I was on a tanker off Singapore when my mother called me. By the time I could get ashore and find a phone that

would reach her properly, three days had passed. I never saw him again after I left for sea in 1991. He was eleven the last time I held him. He was thirty-one when the shelling killed him. The last photograph I have of him is one my mother took on his eighth birthday."

Arjun was looking at him.

"I am not telling you this to compare. Your cousin is alive, and he has weeks, and that is not nothing. I am telling you because when my mother called me on that ship, the thing I needed was for somebody to tell me the truth about what was happening without telling me what I should feel about it."

Arjun looked at his plate.

"Ravi may not survive this. Your mother and your aunt know that. They are not asking you to solve it. They are asking you to be honest with them, and to call them when you can, and to let them hear your voice even when you do not have anything new to say. That is what your family needs from you right now."

"I have not been calling every night. I have been calling twice a week."

"Call more."

"I do not know what to say."

"You do not have to say anything new. Tell them what you had for lunch. Tell them what Rafiq cooked. Tell them about Tendai's question. They do not need information. They need your voice."

Arjun nodded.

"The other thing I learned, and this is the part nobody told me at the time, is that you are allowed to be afraid for Ravi and to keep working at the same time. The work is not the betrayal of the fear. The work is the only place you can go while the worse thing happens somewhere you cannot reach. You keep your watch. You check the engines with Jun when he asks. You call your mother every night. Those three things are what you do. They do not fix anything. They are what being a person in this situation looks like. Others have done it. You can do it."

"Yes, Captain."

"One more thing. When you call her, do not tell her what the ship can or cannot do. She does not need that. Tell her you love her. Tell her you are thinking about Ravi. Tell her you will call again tomorrow. Then call her tomorrow. The calling is more important than what is said."

"Yes, Captain."

"Go when you are ready. Find me later if you want to talk more. Or do not. Either is fine."

Arjun sat for a moment longer. Then he stood, picked up his plate from the table, took it to the counter, and left the mess hall without saying anything else. Sami stayed at the table for a while after he had gone. He had not spoken Majid's name aloud to anyone in three years, and he was surprised at how steady his voice had stayed. He suspected it had stayed steady because Arjun had needed it to, and the need of another person was one of the few things that could still hold Sami's voice steady when he was speaking about his brother.

He went back to his cabin. He opened the notebook. Beneath the Philippines line he had written that morning, he added a sixth entry.

India. National LPG rationing declared 24 March. One cylinder per household per month.

He looked at the list. Two countries in one day. It was the first time that had happened.

Then, in the back of the notebook where he kept the things he did not want to lose, he wrote the first name on a page he had not yet started.

Ravi, Thane. Dialysis cut to twice a week. Three or four weeks if the generators hold.

He thought about what he had told Arjun, and whether he had believed it when he told him, and he decided he had. Then he thought about whether he took his own advice, and he decided that was a harder question, and he closed the notebook without trying to answer it.

* * *

On the twenty-fifth, Sami read something that he did not finish in one sitting.

He had been working his way through the aviation coverage, which the news had been running heavily since the strike week began. What he kept finding, more than the operational updates, were analysts who had been writing the same warnings for years. A 2019 report from a European think tank. A 2022 paper from the US Energy Information Administration. A 2024 briefing from a London consultancy that had been mailed to every major airline's finance office. Each one said, in the careful language of analysts who did not want to be accused of exaggeration, that the world's aviation industry held around two weeks of jet fuel and depended on one maritime chokepoint for forty percent of it. Each one asked what would happen if the chokepoint closed.

They had asked in 2019. They had asked in 2022. They had asked in 2024. They had been answered each time in the way careful warnings are usually answered, which

was a polite acknowledgement and no change in policy.

Sami read for a long time. The pattern he found underneath the aviation reports was bigger than aviation. Strategic reserves existed for crude oil. The United States held sixty-nine days' worth. Australia held twenty-eight. Japan, after its 1979 experience, had built to two hundred and fifty-four days. But the reserves were all of raw crude, upstream of the refineries. Almost nobody had stockpiled the refined products that their populations actually used. Not the jet fuel. Not the diesel. Not the petrol. Not the heating oil. The refineries would keep running, the theory went, and the ships would keep arriving, and the products would flow through the system the way water flows through a pipe, continuously and reliably and without anybody needing to think about it.

He stopped reading.

The system assumed the strait would stay open.

He sat for a while without writing anything. Then he wrote three lines in his notebook.

Every reserve in the world is upstream of a refinery, which is upstream of a ship, which is upstream of a strait. The strait is closed. Nothing downstream of it has more than a few weeks.

He had known the Hormuz number for a week. Forty percent of the world's jet fuel. Twenty percent of all oil. What he had not understood until that afternoon was that the system had been built that way on purpose, and that the way of building had a name. He had found it in the third paper he read. The doctrine was called **just-in-time**. It was Japanese in origin, borrowed from Toyota's factory floors in the 1960s. American management had adopted it in the 1980s. By the 2000s it was the default assumption of every supply chain in the world. The idea at the centre was simple. Holding stock was waste. Every warehouse was a cost. Every reserve was capital sitting idle. The efficient system held as little as possible for as short a time as possible, and moved the rest through the pipes as fast as it could.

The doctrine had been taught in business schools. It had been written into World Bank loan conditions for countries that needed the loans. It had been called lean, which was a word that sounded like health. For forty years, officials had been trained to treat reserves as a sign of a backward economy, and a modern economy as one that trusted its supply chains.

None of the papers that taught it had pretended the system was shock-proof. They said plainly that a system without stockpiles was more vulnerable than one with stockpiles. The vulnerability was the price you paid for the efficiency. The efficiency

could be measured in every quarterly report. The vulnerability only mattered if something actually went wrong.

Sami closed the laptop and went for a walk on the deck to try to get some of it out of his head. It did not work. He came back to his cabin an hour later and read the same papers again, because he was not sure what to do with what he had read except keep reading it.

He thought about Arjun's mother and the kerosene burner from 1975. Her grandmother had known what it was to save against a bad year. Her country had known, at some institutional level, that the days might come when something would not arrive. Somewhere between her grandmother and her son, the habit of saving against the bad year had been replaced by the habit of trusting the supply chain. The people who designed the supply chain had been trained to tell her that trusting it was rational and saving against it was primitive. The supply chain was reliable until it was not, and when it was not, nobody had the old burner anymore.

Nobody except Arjun's mother, who had kept it in the back of her closet for forty years because she was that kind of woman.

* * *

The twenty-sixth was the day hope arrived.

It came over the VHF at 0740 local. An Iranian patrol vessel, broadcasting in accented English, announced that transit through the Strait of Hormuz was now permitted for vessels of the following flag states: China, India, Russia, Iraq, Pakistan. Additional flag states might be authorised at the discretion of the Islamic Revolutionary Guard Corps. Vessels wishing to transit were to stand by on a specific channel and provide flag documentation, cargo manifest, and destination.

Sami was on the bridge when the announcement came. Petros wrote it down verbatim in the log. Then they both looked at each other.

"We are Liberian-flagged, Greek-owned, Indian-destined," Petros said. "Two out of three are not on the list."

"The cargo is destined for India. The refinery that owns this cargo is Indian. The ship is owned by Greeks and flagged in Liberia for tax reasons. That is all. The oil is going to India."

"So we apply."

"We apply."

Sami drafted the message to the company office within the hour. The company told him they would pass the request up the chain and that they would be talking to the

Indian government's shipping ministry and to Reliance, the refinery's owner. They also told him, and Sami understood they were telling him so that he would not tell the crew, that the Iranian authorities had not yet authorised any non-Indian-flagged vessel to transit, even with Indian-destined cargo. The announcement was a door. Nobody had gone through it yet.

Sami did not tell the crew the second part. He told them the first part, because they deserved the news, and they had earned the right to hope for a few hours even if he could not promise them anything.

The bridge felt different that afternoon. Petros hummed while he worked, which was something Sami had not heard him do in four weeks. Tendai came up twice for no reason, checked the radar, and went below again. Rafiq cooked a proper dinner, a full galley's worth, chicken biryani with saffron and raisins, because for the first time since the engines stopped there was a reason to make something that took effort. The crew ate slowly. They did not talk about the announcement directly. They talked around it. Jun asked Petros about the main engine starting procedure, which both of them knew by heart, and which was a way of saying something neither of them had a way to say.

That night, when Sami made his last round of the deck, he heard singing from the engine room. Jun, in Tagalog, quietly, to himself. Sami stood at the top of the ladder for a few seconds and then went back to his cabin without saying anything. The engine room was Jun's space. Jun could sing in it or not sing in it as he wished.

* * *

The twenty-seventh brought two pieces of news, one after the other.

The first came at breakfast. Tendai, holding his phone, found Sami in the mess hall before either of them had finished their tea.

"Captain. The UN just announced they are allowing humanitarian shipments through the strait. Fertilizer is on the list. They are working with Iran and the shipping companies on a corridor."

Sami read the article Tendai held out. It was short. The UN had brokered a corridor for humanitarian and agricultural shipments. Fertilizer qualified because of the planting season. Grain qualified because of food security. Medical supplies qualified. The corridor would be monitored by a mixed group of Omani, Indian, and Chinese inspectors. Details were still being worked out.

"Your mother?"

"I called her. She has not bought yet. She was going to buy this morning because the price had gone up ten percent overnight and she was afraid it would go up again. I

told her to wait three days. If the corridor works, the price should come down. If it does not, she buys."

"How is she taking it?"

"She is tired, Captain. She is trying to decide how much to plant, and if she buys at the new price, she can only afford enough for half the plot. If she plants half, the harvest is half. If the harvest is half, she cannot pay Nambela's school fees for the next year."

"Nambela is your sister?"

"The younger one. She is fourteen. She is the first girl in our family to go to secondary school. The school is in Chipata town. My mother has been paying the fees from the maize crop for three years. If this year's crop is half, the fees are not there. My mother has not told Nambela yet. She is still hoping the price will come down."

"What does Nambela want to do?"

"She wants to be a nurse. She has wanted to be a nurse since she was eight. My mother had been saving the extra from the last two harvests to send her to the training school in Lusaka after she finishes. That saving is already spent on this season's inputs, so even without the price increase, the next two years were going to be thin. The price increase means the plan does not work at all."

"And your mother has not said this to her."

"No. My mother does not tell Nambela things that might stop her from studying. She waits until she has to."

Sami took a breath. He had met Tendai's mother only through Tendai's voice, across three weeks and a dozen conversations, and he had a picture of her in his mind that he knew was probably wrong in its details but correct in its essentials. A woman who had watched the price of urea climb every year since her own mother had farmed, and who kept planting anyway, because what else was there to do.

"She sounded different when she said we would survive it, you said. Different how?"

Tendai thought about the question. He was always careful about his answers when they were about his mother.

"She sounded like she was saying it to convince herself. Not me."

Sami nodded.

"Tendai. May I give you something, not as your captain, as a man twice your age who has had a few of these phone calls with his own mother."

"Please."

"When she called you and said what she said, and she sounded like she was convincing herself, she was not asking you to agree. She was asking you not to fall apart, so that she could keep going. Mothers who have had hard years do not need their sons to tell them everything will be fine. They need to hear that their sons are still steady. That is what she was calling for. She got what she called for, I think."

"I told her I was fine. I told her the crew was fine. I told her you were looking after us."

"You told her what she needed. That was correct. The next time she calls, tell her Nambela is going to be a nurse. Say it as a fact. She has heard people tell her it is impossible for three years. She does not need your agreement. She needs somebody to believe it with her, and you are her son, and you are the one who can. Say what she needs to hear. Then talk about something else, like the weather, or what Rafiq cooked, or a memory you have of being small. The belief goes first. Then give her something else to hold."

Tendai was quiet.

"I had a younger brother, Tendai. He wanted to be a teacher. He did not get to be one. The thing I would give away everything I have, to be able to go back and do, is tell him, in a call or a letter, that he was going to be a teacher. I did not say it, because I thought saying it was premature, or presumptuous, or that I would jinx it. I was wrong. Saying it was the only gift I could have given him in those last years, and I did not give it, and I am still sorry."

"Captain."

"Call your mother tonight. Tell her Nambela is going to be a nurse. Then let her tell you about the rain."

"I will."

Tendai did not leave immediately. He finished his tea slowly, and Sami let him.

Finally Tendai said, "She said something else, Captain. She said, tell the captain thank you. She said, I do not know who is deciding what passes through that strait, but somebody did the right thing about the fertilizer. Tell him I said so."

Sami looked at his tea for a moment.

"Tell her I heard her. Tell her the people who made that decision are probably in New York or Geneva and they will never meet her. But I heard her, and you heard her, and I am going to write down what she said."

"I will tell her."

Tendai went back to his station. Sami finished his tea and went to his cabin and wrote down what Tendai's mother had said, word for word, in the back of the

notebook where he kept things he did not want to lose. Then, on the next line, he wrote: *Nambela, fourteen, wants to be a nurse. Next time I see Tendai, tell him to say it to her again.*

* * *

The second piece of news came at 1420.

It came through Somchai, which was how Sami knew it was bad before Somchai said anything. Somchai had come to the bridge without being asked, and he had come from the radio room, which was not a room he normally visited.

"Captain. A word."

Sami stepped out onto the bridge wing. Somchai closed the door behind him.

"The *Mayuree Naree*. She has gone aground on Qeshm Island. The Iranians are holding the surviving crew on board, pending an investigation into whether the vessel was carrying material destined for Israel. Three of the missing are still missing."

Sami took a breath.

"Your nephew?"

"Still on the other ship. Still safe. But I know six men on the *Mayuree Naree*. One of them I trained on a previous vessel. He was one of the twenty rescued. He is now being held for questioning on the ship that nearly killed him."

"I am sorry."

"I am not telling you this for sympathy, Captain. I am telling you because the crew will hear about it on the news tonight, and I wanted you to hear it from me first. So that you can decide how to speak about it."

Sami nodded.

"How should I speak about it, Somchai?"

Somchai thought about this for a moment. He had earned, over eleven years, the right to be asked.

"Tell them the truth. She went aground. The Iranians are holding the crew. We do not know how long. Tell them the truth so they do not have to hear it somewhere else."

"I will."

"And Captain."

"Yes."

"Tell them there was a ship on fire two weeks ago that we have still not finished being told about. Tell them this is not over. We have been given some good news this week, and we should take the good news. But we should not forget that we are still

here."

Sami looked at him.

"I will tell them that."

"Thank you, Captain."

Somchai went back below. Sami stood on the bridge wing for a while longer. The water was the same colour it had been the morning the *Mayuree Naree* burned. The patrol boats had long since stopped running the search lanes to the south. Whatever had happened to the three missing men, nobody was going to find them now.

* * *

On the evening of the twenty-eighth, the crew put the map up.

Sami had not planned it for that night. He had been thinking about it for a week, since the deck session on the nineteenth. Somchai had produced a navigation chart that morning from the paper store on the bridge, a large Admiralty chart of the Indian Ocean and the Gulf, the kind the ship had twenty of in the bridge cupboard from before electronic charts replaced them. Somchai had pinned it to the mess hall wall after breakfast with four brass tacks, and by dinner everyone on the ship had walked past it at least once.

Sami did not have to say anything about it at the Evening Question. The map was already up. Rafiq had put a jar of markers on the table beside the coffee pot. Somchai had set out the chairs in the half-circle the crew preferred in the mess hall. The question the captain was about to ask was going to be a question the room had already begun to answer before he spoke.

He waited until twenty-one people had settled. Then he stood.

"There is a chart on the wall. Some of you have seen it. I want us to use it tonight. I want each of you to come up and mark where your family is. Put their town on the map. Then draw a line from the town to this strait. On the line, write what is happening to them right now. Not the percentage. Not the statistic. What is actually happening, in their kitchen, on their farm, in their hospital, at their petrol station, this week."

Nobody moved for a moment.

Then Arjun stood up. He walked to the chart and picked up a black marker. He found Mumbai on the west coast of India, circled it, and drew a line from the circle to the strait. On the line, he wrote in small careful letters:

Mother. One cooking gas cylinder per month. Four weeks of fuel left. Kerosene burner from 1975.

He paused. Then he wrote underneath, more slowly:

Cousin Ravi, Thane. Dialysis cut from three times a week to two. Three or four weeks if the hospital generators hold.

Then, below that:

Father's business: plastic parts from Mumbai factory delayed two weeks. Three orders cancelled.

He sat down.

Tendai went next. He took a green marker. He found Chipata on the eastern edge of Zambia, marked it, and drew the line. On it, he wrote:

Mother. Fertilizer up 40%. Planting half the plot. Waiting to see if UN corridor brings price down.

He paused. Then he added, in smaller letters:

Season starts in three weeks either way. Half a harvest means Nambela's school fees for next year are not there.

Then one more line, underneath:

Nambela is going to be a nurse.

Jun took the marker from Tendai. He did not sit down to do his line. He stood at the chart, one hand in his pocket, the other holding the marker. He found Cavite, south of Manila. He circled it, drew the line. He wrote:

Queue goes around three corners. Hospital generators on reduced capacity. Schools closed.

He stepped back. Then he stepped forward again and wrote two more lines, the second one smaller than the first.

Liza, nine, asked her mother last night why she is sad.

Mira, six, did not ask. She just stopped talking for the evening.

He put the marker on the table. Then he reached into his back pocket and took out his phone.

"My wife sent this yesterday morning," he said. He held the phone out toward the nearest seat, which was Petros. "She took it from our bedroom window. She is on the second floor. The petrol station is at the bottom right. The queue comes out of the station, down our road, turns the corner, and keeps going."

Petros took the phone. He looked at the photograph for longer than Jun had probably expected. Then he passed it to Somchai. Somchai passed it to Arjun. Arjun passed it to Dani. The phone went around the room. Nobody hurried.

Tendai, when he got it, held it for longer than the others. He was looking at something in the photograph that Jun had not mentioned. When he passed it back to

Jun, he said, "The woman in the blue shirt. With the child."

"Yes."

"She is not your wife."

"No. That is our neighbour. Her daughter is Liza's age. They play in our yard in the afternoons when it is not raining."

"She is in the queue."

"She has been in the queue since four in the morning, my wife says. When this was taken, she had been there two hours. She still had about three hours to go."

"And the child."

"She could not leave the child at home. There was nobody to watch her. So she brought her to the queue."

Nobody spoke for a long moment.

Jun took the phone back, put it in his pocket, and sat down without looking at anyone.

Dani went up. He drew a line from Jakarta, and he wrote:

Twenty-two days of fuel left in the national reserve. Government promised subsidies. Nobody knows for how long.

Arjun spoke without standing up. "If they have twenty-two days, why are they still promising subsidies?"

"Because they are Indonesian," Dani said, and a few people laughed, which was the first laugh in the mess hall in several days, and which did not travel very far because the weight of the rest of the room pulled it back down.

Eko, the older Indonesian cadet, who rarely spoke at the Evening Question, went up and added one line to Dani's entry. *Father is a fisherman. Diesel prices tripled. He is not taking the boat out.* He sat down next to Dani and did not say anything else.

Viktor did not go up. He sat with his arms crossed, watching.

Then Rafiq stood.

He walked to the map slowly, and he picked up a red marker instead of a black one. He found Dhaka on the east side of the Bay of Bengal. He made a small circle. He drew a line. Then he stood holding the marker for a long moment before he wrote anything. The mess hall was completely quiet by the time he started.

Eid al-Fitr was last Friday. The lights were off in every city.

He paused. Then he wrote:

My brother Mahmud was meant to take his final engineering exams in April. The university is closed indefinitely. Six months of studying. No date for when the exams will be rescheduled.

He paused again, longer. Nobody made a sound.

My mother cooks with gas. The gas has gone up three times this month. She is cooking with wood when she can find it.

He stepped back and looked at what he had written.

Then, without turning around, he said, "My brother is twenty-two years old. He was a month from finishing his engineering degree. The degree was meant to get him a government engineering post. Government posts come with housing. The plan was that when he had the post, he would move our parents out of the rented flat in Old Dhaka and into the posting, because my father's knees are bad and the flat is on the fourth floor and there is no lift. My mother has been carrying groceries up those stairs for nine years."

He paused.

"My brother forwarded me the email from the university yesterday. I want to read it to you. It was sent on the twentieth, which was Eid al-Fitr. It was sent at half past eleven at night."

He took his phone from his shirt pocket and scrolled until he found the message. He held the phone close to read it because the mess hall lights were not bright.

Dear Student. Due to the ongoing energy emergency affecting the operational capacity of our campus facilities, including insufficient fuel for backup generators and disruption to the public electricity supply, the scheduled final examinations for Semester 8 have been postponed indefinitely. A revised schedule will be communicated when conditions permit. We regret the inconvenience and thank you for your patience and understanding. Please continue your preparation. The University remains committed to your academic success. Signed, Office of the Registrar.

Rafiq put the phone back in his pocket.

"That is what he got. On Eid. At half past eleven at night. It does not mention Hormuz. It does not mention the war. It says the University is committed to his academic success. It calls an indefinite cancellation of six months of work an inconvenience. My brother has been reading it over and over for eight days."

He looked at the room.

"He understands the connection. He knows why the generators stopped, and why the grid is down, and why the university closed. Knowing does not help him. He keeps asking my mother why anyway. She does not know how to answer him. She tells him the university will reopen soon. He has not left his room since the email arrived. My mother tells me he is reading his engineering books. He is not reading them. He is

sitting with them open on his lap."

He put the marker back on the table and went back to his seat. He did not look at anyone on the way. Sami did not say anything. Nobody said anything. Rafiq had said what he had come to say, and the room owed him the silence afterward.

Then Viktor stood up.

He walked to the map. He picked up a blue marker. He found a town east of Kyiv that nobody in the room except Sami would have been able to find on a blank map. He drew a small line from the town to the strait. He wrote nothing. He drew a second line from the same town to a place on the map off the eastern edge, toward Russia. He wrote nothing. Then he capped the marker, put it back on the table, and returned to his chair.

The two lines were the contribution. No one asked him to explain. Viktor had told them the week before, on the deck, what the Russian sanctions looked like from his side. The two lines said the rest.

Somchai went up last. He drew a line from a town in northern Thailand to the strait. At the end of the line, he drew a small circle. Inside the circle, he wrote a single word.

Nephew.

Then, slightly to the right of the first line, he drew a second. At the end of it he wrote, in small letters:

Mayuree Naree. Aground, Qeshm Island.

He sat down.

Sami stood up and looked at the map for a long moment. Somewhere behind the circles and the lines was a paper-white chart of the Indian Ocean that he had sailed through two hundred times. The chart was almost invisible under what the crew had written on it.

"Thank you," Sami said. "All of you. I did not need to ask tonight's question. The map asked it."

Rafiq poured tea. The crew drank. Nobody went back to their cabin for a long time.

* * *

It was three in the morning when Sami sat down at his desk in his cabin.

He could not sleep. He did not try very hard. He took out his notebook and wrote for two hours.

He wrote about the papers he had been reading all week, and the analysts who had been writing them since before this war started, and the doctrine the analysts had been

trying politely to undermine. He wrote about the one cylinder per household in Mumbai. He wrote about the kerosene burner from 1975. He wrote about Tendai's mother, and how she was weighing how much to plant on a plot she had farmed for thirty years, with no way of knowing whether what she chose would mean enough food for the next year or not. He wrote Rafiq's three lines down word for word, because he did not trust himself to remember them accurately in the morning.

Then he wrote something he had been thinking about for four weeks and had not known how to say.

The crew of this ship has been figuring out how the world's oil dependency works for a month now, with phones and the wifi and a few articles I brought them. None of us have degrees in energy policy. None of us have been inside a ministry or a refinery boardroom or a strategic planning department. We are twenty-three men on a ship that cannot move. If we could see it in a month, the people whose job was to see it had decades. The analysts I have been reading this week were not writing their warnings for the first time. They were writing them for the fifth or tenth or twentieth time. Each paper sits on top of a stack of earlier papers, most of them unread outside the field.

The failure is not that nobody knew. The failure is that the knowing did not produce action.

He put the pen down and looked at the page.

He had read a Confucius collection years before, in the same bookshop in Singapore where he had bought the Sun Tzu and the Tao Te Ching. Confucius had been a civil servant before he was anything else. He had spent most of his life trying to teach officials how to govern well, and most of the officials had not listened. He had a saying Sami had carried with him ever since. *To see what is right and not to do it is the absence of courage.*

Confucius had been talking about individual moral failure. A man who sees a beggar and walks past is the example that is usually given. Sami had always read the line that way.

Tonight the line meant something larger. It meant an institution that had been warned for decades about a strait and had chosen not to build the buffers that would make its closure survivable. It meant the people who wrote the letters and the people who received the letters and did not act on them. It meant the governments that had stockpiled crude oil because crude oil was a commodity that could be traded, but had not stockpiled the products that their populations actually used, because stockpiling products would have cost money and would not have shown up on a quarterly economic report. The data had been available. The warnings had been written. The

papers had been published. The letter on Sami's laptop that night had been the hundredth letter of its kind, and the only thing that was new about it was that the disaster it warned about was now happening.

To see what was right and not to do it.

Sami wrote the line at the top of a blank page, and underneath it he wrote what he would have written if he had been an institution instead of a captain.

We saw the chokepoint. We saw that forty percent of the world's jet fuel passed through one strait. We saw that the refined products we actually used had no reserves. We saw that three of the world's major oil producers were under sanctions we had imposed. We saw that the strait we had grown most dependent on was a thirty-kilometre gap between two countries, one of which we had been in escalating hostility with for forty-seven years. We saw all of this, for decades. We did not build the reserves. We did not fund the alternatives. We did not map the refineries. We did not have the conversations. We saw, and we did not act, and the captain and the deckhand and the cook on a ship that cannot move are now paying the price of our not acting, along with their mothers and their brothers and their daughters.

He read what he had written. Then he closed the notebook.

The ship moved gently at anchor. Through the porthole, the running lights of the other anchored vessels were arranged around the horizon like a city, a city nobody in it had chosen to build, held in place by a decision nobody in it had made, waiting for news nobody in it could predict.

* * *

On the morning of the thirty-first, a drone hit the *Al Salmi*.

She was a Kuwaiti VLCC, roughly the same dimensions as the *Noor Salaam*, at full capacity with two million barrels of Arab Heavy crude in her tanks. She was anchored at the Port of Dubai. She was not in the strait. She was not in any of the lanes where the ships had been told to expect attacks. She was sitting at anchor in one of the wealthiest ports on earth, a few hundred kilometres from where the *Noor Salaam* had been anchored for thirty-two days.

The drone came in at low altitude and struck her port side amidships. The fire started in one of the wing tanks. Kuwait Petroleum Corporation issued a warning within the hour about a potential oil spill. The IRGC claimed the vessel had been identified as Israeli. The vessel was not Israeli. She was Kuwaiti, operated by a Kuwaiti state-owned company, flagged in Kuwait.

Sami was on the bridge when the news came in. Petros was already there. They did not speak for a while.

Then Petros said, "A ship like ours."

"A ship like ours."

"At anchor."

"At anchor."

"The strait does not matter anymore. They can hit us anywhere the drone can reach."

"The drone can reach here."

Sami made the announcement to the crew at lunch. He did not soften it and he did not extend it. He told them what had happened, where it had happened, and what it meant. He told them the *Noor Salaam* was still under the Iranian authorisation announcement from the twenty-sixth, and that the company was still in talks with the Indian ministry, and that as soon as he had any news about whether the ship would be allowed to move, he would tell them. He told them that until then, the anchorage was not automatically safe, and that they should all know that, because they deserved to know.

Jun asked the question the rest of the crew had not yet asked. "Are we a target, Captain?"

"We are not identified as Israeli. We are Liberian-flagged, Greek-owned, with Indian-destined cargo. Our manifest is public. There is no reason for us to be a target."

"The *Al Salmi* was Kuwaiti. The IRGC called her Israeli anyway."

"Yes."

"So the reason does not always have to be real."

"No. It does not always have to be real."

Jun looked at his plate.

"Jun."

"Captain."

"I grew up somewhere where the reason did not have to be real. I learned something there that I will tell you, and then I will not say it again. Being afraid and being paralysed are not the same. The first one is a thing a man can walk around with. The second one is the thing you have to refuse."

Jun was listening.

"You will be afraid for Liza and Mira until this ship is moving and for a long time after. That is correct. Do not try to stop being afraid. Try to keep working while you are afraid. The rest of us will do the same thing. Your daughters need you to do it,

because a father who is afraid and still doing his job is someone a child can watch and learn from. A father who has gone quiet is not."

"I have been quieter than I should be, Captain."

"You have been quieter than you usually are. That is fair enough. But write to your wife tonight. That is the one thing I am asking."

"About what."

"About Liza's question. Tell her what you would have answered if you had been there. She will keep that letter. Whatever happens."

Jun nodded. He did not say anything else. He finished his food and went back to the engine room.

* * *

31 March. End of week four. A week since the Philippines emergency. Twelve days since the Evening Question on the deck. Four days since Tendai's mother asked me to thank whoever decided about the fertilizer corridor. Three days since the map went up.

The Al Salmi burned this morning. A ship like ours, in a port we have used, attacked by a drone that travelled further than anyone had been told such a drone could travel. The IRGC said she was Israeli. She was not. The reason does not have to be real.

I have been reading the analysts again tonight. Their papers have been on my laptop for a week. 2019. 2022. 2024. Each one said, in the politest possible language, that the world's aviation sector held two weeks of jet fuel and depended on a single strait for forty percent of it. Each paper had a section called 'policy recommendations' that asked for strategic reserves of refined products. None of the governments that received those papers built the reserves. They did not refuse. They simply did not fund the line item. The papers were filed. The analysts kept working. A new paper came out every couple of years with slightly updated numbers. The habit of not acting became the policy.

Confucius wrote that to see what is right and not to do it is the absence of courage. He was writing about a man. Tonight I think he was also writing about a doctrine. The doctrine has a name. It is called just-in-time. It was taught in every business school in the world and written into World Bank loan conditions for countries that needed the loans. It said, plainly and in writing, that stockpiles were waste and flow was efficiency. It also said, plainly and in writing, that a system without stockpiles was more vulnerable to shocks than one with. The people who taught it knew the trade-off.

The people who applied it knew the trade-off. They decided the shock was hypothetical and the efficiency was real. They decided that for forty years.

Ravi is on dialysis twice a week because the diesel for his hospital generator was not stockpiled. Mahmud's exams are cancelled because the diesel for his university's generator was not stockpiled. Nambela's school fees are gone because the fertilizer for her mother's plot was not stockpiled. Liza and Mira's schools are closed because the petrol for their buses was not stockpiled. The lights of Dhaka were off for Eid because Bangladesh imports the gas it could have produced, because the efficient system asked it to.

None of these things were accidents. They were the predicted consequences of a policy that was taught and applied by people who could read. I have been looking at the same page in the same paper for an hour now, trying to understand how it is that a thing can be disclosed in writing, by the people who advocate for it, and still result in this much surprise when it happens. The only answer I have found is the one Confucius gave. The people who could see what was right did not do it, because they were not rewarded for doing it, and they were rewarded for the alternative. They taught the alternative to their students. The students taught it to theirs. By the time the thing arrived, no one with the power to prevent it was still in the room.

I do not know what to do with this except write it down. So I am writing it down.

I am keeping another list now, in the back of the book. Ravi in Thane. Nambela in Chipata. Mahmud in Dhaka. Liza and Mira in Cavite. Somchai's nephew, still unnamed to me because Somchai has not yet said his name out loud. Viktor's sister's children, doing homework by head torch. There will be more before this is over. I have not yet written the names of Rafiq's parents. I do not yet know the name of Jun's wife.

I told Arjun today that others have been where he is and they have come through, and that the work is the only place a man can put himself while the worse thing goes on at home. I told Tendai to say out loud to his mother that his sister is going to be a nurse, because his mother needed somebody to believe it with her. I told Jun that being afraid and being paralysed are not the same, and that he should write to his wife tonight. I meant all of it. I also realised, walking back from the mess hall this morning, that I have not written to anyone in three weeks. I tell the crew things I have forgotten to do myself. That is not new to being a captain. It is what captains should be ashamed of, and rarely are.

Rafiq wrote three lines on the map tonight. I have them in this book. I am not going to copy them twice. They have already cost him enough.

The map is still on the wall. I walked past it on the way to my cabin. Somebody had turned on the lamp above it before they left the mess hall, and had left it on. Probably Somchai. The map is the book the crew have written for themselves, and somebody on the ship thought it deserved light.

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Part Three — What the Ship Taught

"Kali kokha nkanyama. Aliawiri ndi anthu."

One alone is an animal. Two together are people.

Chewa proverb. Passage 10.

* * *

PASSAGE 7

The Rolling Disruption

Rafiq had started cooking things he had never cooked before.

On the first of April, he made Bengali shorshe ilish, which was hilsa fish in mustard sauce, a dish that required a fish he did not have and a mustard seed he did have and a great deal of patience he had developed over a life of making do. The hilsa was replaced with a tuna Somchai had brought aboard frozen from Ras Tanura thirty-five days earlier. The mustard had to be ground fresh because the pre-ground kind had lost its bite in the heat of the galley. Rafiq pounded the seeds in a small brass mortar for twenty minutes while the rice steamed. The mess hall smelled like a Dhaka wedding for an hour. When the crew sat down to eat, Rafiq put a single fresh lime on the table and stepped back into the galley without serving himself.

On the second, he made a Filipino adobo, which was a stew of chicken in vinegar and soy and bay leaves and black peppercorns. He had never cooked Filipino food before. He had asked Jun the day before for his grandmother's version of the recipe, and Jun had written it out on a piece of paper torn from his engineer's notebook, and Rafiq followed it exactly. When the stew was ready, he put it in front of Jun with a bowl of white rice and did not say anything. Jun ate slowly. Somebody, Petros maybe, said the chicken was the closest thing to what his own grandmother used to make, even though his own grandmother was Greek. The laugh that followed was not large but it was the first real laugh in a long time.

On the third, Rafiq made a Zambian nshima, which was a stiff maize porridge that Tendai ate by hand, rolling small portions the way his mother had taught him. Rafiq had never served nshima before. He had called his own mother in Dhaka three times over the previous week and had asked her what she thought a Zambian woman would have done with the last of the dried fish and the last of the green onions. His mother, who had never left Bangladesh, had told him to fry the fish hard with chilli and serve it with the nshima so that the richness carried the porridge. Rafiq did what she said. Tendai ate three servings and then sat in the mess hall for a while after the rest of the

crew had left, not doing anything, just being in the room where his mother's food had been cooked from memory by a Bangladeshi man on a Liberian-flagged ship anchored off the coast of Iran.

Sami watched all of this from the edge of the mess hall on each of the three nights. He did not say anything to Rafiq. He did not need to. Rafiq was doing what Rafiq did, which was hold the crew together with his cooking, because that was the thing he knew how to do.

It was a kind of preparation, Sami thought. Not the kind that got written about in newspapers. The kind that kept a ship's crew from coming apart in week five.

* * *

Sami had been quieter that week than he usually was.

The crew noticed. He was not joining the card games. He was eating at a smaller table. He was going to his cabin earlier. He was reading more than he wrote. When Petros brought the bridge log for his signature in the morning, Sami signed it without reading the entries, which he had never done in thirty years.

On the second of April, the crew learned about the Philippines clearance.

The news arrived over the VHF at 0910 local. The Iranian authorities had extended the list of authorised flag states. Philippine-flagged vessels were now permitted transit, and Filipino seafarers were permitted to crew any vessel carrying Indian-destined or Chinese-destined cargo under the previous day's authorisation.

Petros brought the announcement down to the engine room himself. Jun was on a maintenance task, lying on his back under the auxiliary generator. Petros read the announcement aloud. Jun set his tools down on the deck beside him and lay there for a moment without moving.

"Philippine-flagged."

"Yes."

"We are not Philippine-flagged."

"We are not."

Jun pushed himself out from under the generator and sat on the grating.

"My country's ships will move. Before ours."

"It looks that way."

Jun nodded slowly. He did not say anything else. He picked his tools back up and went back under the generator. Petros waited for a minute longer and then went back to the bridge.

Sami found Jun in the engine room an hour later. Jun was sitting on the step at the bottom of the ladder with his work shirt unbuttoned and his glasses off and a cloth in his hand that he had been using to clean the glasses and had now forgotten he was holding.

Sami sat down on the step beside him, at the lower end.

They did not speak for a while.

Eventually Jun said, "I should be happier about this."

"You should be whatever you are."

"My country's ships can move. That means the fuel will start reaching Manila again. The queues will shorten. The hospitals will come off the reduced generator capacity. Liza will go back to school. Mira will start talking again. My wife will not have to apologise to our neighbour for parking across her gate when she was queuing. All of that is coming."

"Yes."

"And I am not on a Philippine-flagged ship. I am on this one. And we are not moving yet."

"Not yet."

"I am glad for my country. I am angry for myself. I did not know a man could be both things at the same time with this much force."

"You can. You are."

Jun held the glasses up toward the overhead light, breathed on one lens, wiped it once more with the cloth, then did the same with the other. He put the glasses on.

"My wife sent a photograph this morning. The queue at the petrol station on her road is still around three corners. The announcement has not reached the petrol stations yet. It will take a week, she thinks. Maybe two. She is afraid to tell Liza yet because she does not want Liza to get her hopes up and have them crushed again."

"That is the right thing for her to do."

"Yes. I told her the same thing."

"Jun."

"Yes, Captain."

"You and I have been doing the work of people who are not allowed to be in the places they most want to be. It is harder than regular work, because regular work you can finish. This you cannot finish until the ship moves, and even then it will not be finished, because by the time we get home, what we were most afraid of will either have happened or not happened, and we will not have been there either way. Others have been here before us. They came through. You will come through too."

Jun did not respond for a long time. Then he said, "Thank you, Captain."

"Go and write to her. Tell her what you just told me. Reading it will help her when things are tough."

"Yes."

Jun stood. He buttoned his work shirt. He went to his cabin. Sami stayed on the lower step in the engine room for a few minutes longer, listening to the auxiliary generator running at low load two decks down, which was the one sound on the ship that had not changed since the first day.

* * *

Tendai came to Sami's cabin later that afternoon. He knocked twice, waited, and came in when Sami said to.

"Captain. The starboard winch is making a sound I do not recognise. Jun says it is probably a bearing. He wanted me to ask you whether we should pull it now or wait until we are moving."

"What does Jun think?"

"He said it was your call."

Sami thought about it. The winch did not need to work while the ship was at anchor. Pulling the bearing meant disassembly and reassembly by two engineers who were already doing more than they should be doing. Waiting meant risking that the winch would fail at a moment when the ship needed it, like during berthing at Jamnagar after a six-week shutdown.

"Pull it. Better to know what is wrong now than to find out later."

"Yes, Captain."

Tendai did not leave. He stood in the doorway for a moment longer than the errand required.

"Was there something else?"

Tendai looked at the floor. Then at Sami's desk. Then at the porthole. The light was falling through the porthole at the angle it falls through a porthole at four in the afternoon, a light every sailor knows, and it had been falling through the same porthole every afternoon for thirty-four days.

"She called again, Captain."

"Your mother."

"Yes."

Sami nodded and waited. He did not ask what she had said. If Tendai had come to tell him, he would tell him.

"She bought the fertilizer yesterday. The UN corridor is not enough. The price came down by three percent, which is not enough. She bought for half the plot. Nambela's fees for next year are not there. My mother told me not to say anything to Nambela."

"Yes."

"I have not slept properly in a week, Captain. I thought I was sleeping but I have been waking up at two in the morning every night and lying in the bunk until the watch change. I have not told anyone. I did not come in here to tell you any of this. I came in about the bearing. But I was going to have to say it to somebody."

"That is how it happens."

Tendai looked up. Sami gestured to the chair Jun had sat in on the first night, six weeks ago. Tendai sat.

"Captain, I do not know what to do."

"There is nothing to do."

"That is what I keep thinking."

"It is also what is true."

They sat for a moment. Sami did not reach for a lesson. He did not reach for a story about his own sister, because he did not have a sister and inventing one would have been a betrayal.

"Tendai."

"Yes."

"I am going to tell you the truth instead of something that will make you feel better tonight. This does not get easier. What changes is you. You will do things you were not sure you could do. You will not know you were doing them until after. Your mother has been doing this work since before you were born. She learned it from her mother. You will learn it the way she did, which is by doing it."

"I do not know if I can."

"You are already doing it. What you are calling not being able to do it is what doing it feels like from the inside."

Tendai did not say anything. They sat without speaking for a while.

After a while, Tendai said, "I am going to go and help Jun with the winch."

"Yes."

"Thank you, Captain."

"You did not come for thanks."

"No. But thank you anyway."

Tendai left. Sami stayed at the desk. He listened to Tendai's footsteps going down the corridor, and then down the ladder, and then fading into the lower decks. Then he opened the second drawer of the desk, which he had not opened in four weeks, and took out a sheet of writing paper that he had brought aboard at the start of the rotation and had not used.

* * *

He wrote the letter in under an hour.

It was addressed to his mother, who lived in a building on a quiet street in the eastern part of Beirut, and who was eighty-one, and who had not heard from him since the third of March. The letter was shorter than he meant it to be when he started, and longer than he meant it to be when he looked at the finished page.

Mama,

I am sorry. I have not written since the third of March. I told myself at first it was because I had no news, and later because I had no good news, and the truth is that I did not know what to write that was not the news. I am writing tonight because of something that happened this afternoon, which will come further down.

The ship is fine. I am fine. The crew is fine. We are still at anchor. There is a chance we will move in the coming week. I will not promise it, because I do not want you to wait on a day that may not come.

Tendai, the young Zambian I have written to you about before, came to my cabin this afternoon to ask a question about a bearing. He ended up staying an hour, because his sister is fourteen and wants to be a nurse, and the crisis has taken the money she was going to study on. His mother has not told her yet. While he was sitting there I remembered the night before I left for the academy. I was twenty. I was in the kitchen at three in the morning because I could not sleep. You came in and sat on the floor next to me and you did not ask me why I was there. You said that what we carry does not get lighter, but our shoulders remember. I had not thought of that sentence in forty years. This afternoon I needed it and it was there. I gave it to Tendai. I did not know, when you gave it to me, that you were giving me something I could pass on to another young man in another sea in another century. Thank you for leaving your sentences in your sons where we can find them later.

I have been thinking about Majid. I keep a notebook now with the names of people the crew are worried about, most of whom I will never meet. I wrote one of those

names down last night and then I realised I had not put Majid's name anywhere in the notebook, even though he is on a different list that has been with me every day for fourteen years. I think I was keeping him away from the names I might still save, because seeing his name there would have reminded me that names belong to the living until they do not. I am telling you this because you are the only person in the world who would not need me to explain it.

Please eat. Please walk. Please call Aunt Leila. She will tell you she does not need to talk and she will be wrong, the same way she has been wrong about needing to talk for sixty years. If I am home by June, I will take you both to Raouché. I owe you a day at the water.

I love you, Mama. I should have said it at Eid on the phone. I did not. I am saying it now.

Sami

He read the letter through once. He folded it, put it in an envelope, and wrote his mother's address on the front. The letter would go out with the next crew change, which might be in a week and might be in a month. He put the envelope in the drawer.

He sat at the desk with his hands flat on the wood for a while. When Petros knocked at six to tell him the bearing had been pulled and reinstalled, he had not moved.

"Are you all right, Captain?"

"I am fine, Petros. Thank you for asking."

"You have been quieter this week."

"I wrote to my mother. I had not written in a month. It took something out of me. I will be back to myself by morning."

"Do you need anything."

"I do not think so. But I am grateful that you asked."

Petros went back to the bridge. Sami washed his face in the small sink, combed his hair with his fingers, and went to the mess hall for dinner. The crew saw their captain as he had always been. Only Petros, and later Somchai, knew that on the second of April the captain had sat motionless at his desk for forty-five minutes.

* * *

On the fourth of April, Sami took the Evening Question into different territory.

He had been reading for three days about countries that had prepared for an oil shock and countries that had not. He had come in with a folder of notes. Rafiq had made a North African stew that smelled of cinnamon and lemon and was a variant of

something Sami's grandmother had made when he was small. Sami did not ask Rafiq how he had known to make it. He suspected Somchai had mentioned something.

Twenty-one of the twenty-three crew were present. Only Viktor and the night-watch helmsman were absent. Viktor was in his cabin with a slight fever. Rafiq had been taking him ginger tea every two hours and stopping by to check on him in between.

Sami took his seat in the circle.

"Tonight's question is the one I have been avoiding. We have spent a month learning what is broken. I want to spend an hour tonight on who had prepared. Not who prepared for this exact war. Who had built themselves a cushion against something like this."

He let that settle.

"You already know the Venezuela story from two weeks ago, when we were on sanctions. Three and a half million barrels a day down to one, and the largest reserves on earth. A country that had the oil and lost the use of it. Tonight I want to talk about the country that had oil, and kept it. Norway."

He paused.

"Norway struck oil in the late 1960s in the North Sea. They built a sovereign wealth fund. They put the oil money in the fund and wrote a rule that the government could only spend about three percent of the fund per year, regardless of how much oil was pumped that year. The fund is worth about two trillion American dollars today. They have come through every oil shock of the last fifty years without a crisis, and they still have their fund."

He sat back in his chair.

"I have been trying to understand why Norway succeeded, and I do not think I have a single answer. I have a few guesses. The first is that the oil did not arrive to a poor country. Norway was already rich before the oil. It was already a functioning country, with small towns where people knew each other, courts that worked, and a government that people mostly trusted. The oil came into a society that was able to hold it. In many places, the oil arrived first and had to build the society around it, which is much harder."

"The second is the winter. Norway is a cold country, far north. The winters are long. The summers are short. For every generation of Norwegians, the lesson that the summer's abundance had to feed the year was not taught in school. It was taught by the calendar. You do not eat the seed grain. You do not burn the last of the wood in November. By the time the oil came, they did not have to be told how to save. They

had been saving for five hundred years."

"The third is that they started early. They built the habit of saving when oil prices were still low and the fund had room to compound. A country starting now, in the middle of this crisis, cannot buy those decades back."

He looked around the room.

"Japan got to a version of the same place a different way. The 1973 and 1979 oil shocks humiliated them, and they built two hundred and fifty-four days of reserves after that because they never wanted to feel it again. China watched Japan and built one point four billion barrels of stockpile and shifted half of its electricity onto things other than oil. None of these countries did it the same way. They each had their own reasons. What they shared was that they started the boring work of preparing at a time when the preparing did not yet look necessary."

Dani asked the question first. "What about a country like mine, Captain? Indonesia. If we start now, is it too late?"

"I do not know, Dani. I think the honest answer is that the cost of starting has gone up. What Norway did in 1972 and what Japan did in 1980 cannot be done now on the same terms. But some version of it can still be started. The countries that will be least hurt in the next crisis are the ones that begin now, even late."

Arjun said, "India has institutions. We are one of the oldest democracies outside the West. But we did not prepare."

"You had institutions. You did not have the habit of setting aside the good year against the bad one. Norway had that habit before the oil. India did not. The institutions made the habit possible. They did not force it."

Jun spoke from where he was standing, near the port-side bulkhead. "The Philippines did not prepare either. We have fifty-three days of gasoline. That is not preparation. That is cargo in transit."

"Yes."

"An engineer looks at a system with one critical supply line and no backup and says the system is badly designed. We have been building a badly designed system for seventy years. The system is not badly designed in the way that means the engineer who built it made a mistake. It is badly designed in the way that means everyone who built it chose not to solve the problem they knew was there."

"Yes. That is it exactly."

Rafiq said, quietly, from the chair near the coffee pot, "Bangladesh had no oil to save. We could not have prepared the way Norway did. But we could have prepared the way Cuba did. We could have been growing our own food in a way that did not

depend on fertilizer from the Gulf. We did not. We followed what the World Bank told us to follow. We grew rice for export. We imported the fertilizer for the rice. We imported the diesel for the tractors. And when the diesel stopped, my mother was cooking with wood."

Nobody spoke for a while after that.

Then Tendai said, "Zambia too."

"Yes," Sami said.

"We never had the money to prepare the way Norway did. But we did not even keep the knowledge of how to farm without the fertilizer. My grandmother knew. My mother knows the kinds of things my grandmother knew. My sister will not know them. Not because she is not interested, but because nobody has taught her, and the schools teach the new way, and the new way needs the fertilizer."

Arjun added, "India stopped teaching traditional farming in the 1960s. The Green Revolution. Same story."

The room was quiet for a moment.

Sami did not speak at first. He sat forward in his chair and looked at the three men who had just spoken, one after the other, each from a different country.

"I want to tell you something I have been reading about for three nights and not saying out loud because I was not sure I was reading it right. I think I was. The three of you have each put down a piece of a larger thing. I want to put the rest of it on the table while we are all here."

He took a breath.

"For ten thousand years, every farming society on earth saved seed. You planted, you harvested, you kept the best of the harvest, and that was next year's crop. The system fed itself. In the 1960s, in most of your countries, a different kind of seed arrived. It was called a hybrid. It produced a bigger yield in the first season. The seeds you could save from that bigger yield did not breed true. If you planted them, you got a weaker plant, or an inconsistent plant, or almost nothing at all."

He paused.

"The hybrid seeds were given out as aid. That is the word that was used. Aid. In many places the old seeds were crowded out of the system within a generation. By the time anyone understood what had happened, the old seeds were not there to go back to, and the farmer was buying new seed every season from the company that had given the first bag away. The first bag was the door. The second bag was the business."

Nobody spoke.

"Since the 1990s the replacement has been genetic engineering. Seeds designed in laboratories to resist a specific weed killer, sold by the same company that sells the weed killer. Seeds coated with pesticides, held to the seed by a polymer shell. The pesticide is petroleum. The polymer is petroleum. The fertilizer those seeds need to deliver the promised yield is petroleum. The weed killer is petroleum. Tendai, your mother in Chipata plants a seed that was designed in a laboratory she could not find on a map. The seed will not deliver what it was sold to deliver without fertilizer, and the fertilizer is made from gas that moves through the strait that just closed. She borrowed money to buy the fertilizer against a yield she did not yet have. And the seed will not give her a seed she can plant next year, even if she tries."

He looked at each of them in turn.

"This did not happen by accident. Somebody in a boardroom did the math on dependency curves and decided that removing the farmer's option to save her own seed would be profitable. The math was correct. The companies I will not name because the names do not help us tonight have made more money from this arrangement than some of our governments have ever had. The farmers who refused the seeds were outcompeted on yield in the first few seasons and then pressured out by the banks who would only lend against the new varieties. By the 1990s, ninety-five percent of farmers in the developed world and forty percent of farmers in the developing world had crossed over. It is higher now."

He paused again.

"This is not the war doing this to us. The war is only closing the door. The room we are locked in was built slowly, over sixty years, by people who were paid to build it, and then sold to our parents and grandparents as a gift."

The room was very quiet.

Then Sami sat back.

"One more piece. The countries that saw this coming, the ones that funded their own agricultural science, that built seed banks, that kept the old varieties alive, that paid their universities to study the old ways alongside the new ones, those countries are insulated. Not completely. More than the rest. Norway has a seed vault. Cuba learned because it was forced to. India has built some local seed-saving programs in the last decade that might matter by the end of this. The countries that let their seed system become a product sold to them by a foreign company are the ones who will feel this crisis the hardest. Most of us on this ship are sitting in that second category."

Nobody spoke for a long moment.

Then Petros, who had not spoken all evening, said from the back, "May I add something, Captain."

"Please."

Petros took his time. He was not a man who used words when a nod would do.

"My wife is in Athens. She is not in a petrol queue yet. Europe's reserves have been holding so far, because Europe stockpiled more than most. But the papers the captain has been reading, which he has been telling me about on the night watch, say that what Europe has is around two weeks of jet fuel, and no strategic reserve at all for the refined products the cities actually use. Two weeks is better than Manila's situation, but it is not good. What we are watching in Manila now is what Athens will look like in three weeks, and what Frankfurt will look like in four."

He paused.

"My wife's pharmacist is German. The shop is near their flat. He does not know, any more than Jun's wife's neighbour in Cavite knows, that the shelves behind his counter are stocked from a supply chain that runs through this strait. The insulin, the blood pressure tablets, the antibiotics, all of them. When the reserves in Europe run out, his shop looks like the shops in Manila look now. The only difference between him and Jun's wife is that he has been at the front of the queue for longer. He does not know the queue is there."

Jun said, "That is what I have been trying to say to myself for three days. Thank you."

Petros nodded once.

Tendai, who had been on his phone for the last several minutes, spoke up from the chair by the coffee pot.

"Captain. I looked up the seed thing while you were talking. You were right about the petroleum. The coatings are neonicotinoids. Petroleum. The polymer that holds the coating to the seed is petroleum. Most of the fungicides are petroleum. Ninety percent of the corn grown in the United States and seventy-six percent of the soybeans are planted with coated seeds. The coating comes in a package. The whole package is petroleum, from the seed to the fertilizer to the weed killer they sell you after. You were not making any of that up."

Sami smiled. It was a small smile and it went as quickly as it came, but it was the first smile anyone on the ship had seen from him in a week.

"Thank you, Tendai."

"I did not look it up because you asked me to. I looked it up because I wanted to know if the article you read said what you said it said."

"That is the right reason to look something up."

Sami nodded, mostly to himself, and then he looked back at the room.

"Every country on this ship has a version of this story," he said. "The countries that came through earlier crises with something left were the ones that kept the old thing while they were building the new thing. Norway kept the habit of saving. Japan kept the memory of 1979. Denmark kept its villages involved in the wind turbines. Almost everywhere else, including every country represented on this ship, the old thing was called inefficient and let go of. The people who said the old thing might still be useful were not at the conferences where the decisions were made."

"So what do we do," Dani said.

"We do not do anything tonight. We are a crew on a ship at anchor. Tonight the job was to see the problem clearly. Tomorrow there will be something else."

"That feels small," Dani said.

"Seeing clearly is not small. It is where everything else starts."

The Evening Question ended later than it usually did. Nobody rushed to leave. Rafiq brought out the last of the lemon cake he had been making a little of every night for a week, and the crew ate it slowly, and the conversation continued in small clusters at the tables, and Sami sat in his chair and listened without joining any of them.

* * *

On the fifth, Iran extended the authorisation list again, adding Iraqi-flagged vessels.

On the sixth, the Pakistani foreign minister announced in Karachi that direct talks had been established between Tehran and Washington, mediated by his government, and that a ceasefire proposal was being circulated. The proposal was described as "provisional and without prejudice," which Viktor, who had recovered enough to join the Evening Question that night, said was the phrase you used when you wanted to announce something without committing to it.

On the seventh, the ceasefire was signed.

The announcement came through the company at 1415 local. The text of the agreement was brief. Hostilities would be suspended for an initial period of thirty days. The Iranian blockade of the Strait of Hormuz was "technically suspended," a phrase that Viktor again flagged, pending the implementation of mutual inspection and verification protocols. Merchant shipping could petition for transit through the normal channels. War-risk insurance premiums for Gulf transits remained extremely elevated. Both sides reserved the right to respond to violations.

Sami worked through the text carefully. As a captain, he was looking for operational consequences: what the ship was permitted to do, what it was not, what the insurance implications were, what the company's instructions would be. As a man who had been reading Seneca for a week, he was looking for something else.

Seneca had been a Roman Stoic who lived two thousand years ago, tutor to the emperor Nero, and, late in his life, the author of a long correspondence with a younger friend called Lucilius about how a man ought to live when everything he cared about was uncertain. His subject was preparation. Preparation for loss, for exile, for the night when good fortune turned in a single hour. A previous captain had left a translation of the letters in the ship's library. Sami had found them in the second week of the anchorage. They were two thousand years old. They read like they had been written that morning.

He went back to the opening paragraphs of the ceasefire text and read them again in the second light.

Seneca had written, in one of the letters to Lucilius, that the thing about shipwrecks was that they never happened to the people who had refused to put to sea. The point was not that you should refuse to sail. The point was that any decision made in the world was made under the possibility that the ship might not come back, and the wise man was the one who made his decisions knowing that. Seneca's wisdom was not about avoiding the sea. It was about living honestly in a world where the sea was always there.

Sami put the laptop down and went to find Jun.

* * *

Jun was in the engine room, going through the start-up checklist for the main engine. He had the checklist in his left hand and a pen in his right, and he had already worked through three of the eleven pages.

Sami sat on the step at the bottom of the ladder.

"How long until she is ready?"

"I want another thirty-six hours. The main bearings have been cold for six weeks. I want to warm the lube oil and run the pre-lubrication pumps for eight hours before we try to turn her over. I want to test the starting air compressors at full pressure. I want to check the cooling water temperature sensors, because two of them were drifting last week, and I want them to agree with each other before we put the main engine under load. I can do all of that in thirty-six hours."

"Good. Do it."

"Yes, Captain."

Jun went back to the checklist. Sami watched him for a minute longer.

"Jun. I want to ask you something that is not about the engines."

"Yes."

"You have been keeping this ship ready to move for forty days. You have been doing maintenance tasks that did not need doing, because they needed doing when the moment came. You have been doing the work that preparation looks like. I want you to know that I see it, and that the crew sees it, and that when we move, it will be because you made the ship moveable."

Jun looked at his checklist.

"Thank you, Captain. Now please let me finish this. She is going to be the most important thirty-six hours this ship has seen in five years, and I would like to be left alone to prepare for them."

"Of course."

Sami stood. He climbed the ladder out of the engine room and went up to the bridge. Petros was there. They stood together on the bridge wing for a few minutes without speaking. The anchorage spread out around them. A hundred and fifty ships now, Petros had told him that morning. More than at any point since the crisis began. The lights were coming on against the dusk. Somewhere among them, a vessel with a different flag was preparing to do exactly what the *Noor Salaam* was preparing to do.

"Viktor was right," Petros said.

"About which thing."

"The ceasefire is not peace. The ships are still here. The danger is still here. What has changed is that we are allowed to think about moving now, whereas yesterday we were not."

"Yes."

"That is still a change."

"It is."

They watched the lights for a while longer. Somewhere below, in the engine room, Jun was warming the oil in a machine that had been cold for six weeks. Somewhere in the mess hall, Rafiq was putting away the dinner plates and thinking about what he would cook tomorrow. Somewhere in a cabin, Tendai was calling his mother, and telling her that the ship might move, and that he would be home sooner than either of them had dared to hope, and that Nambela was going to be a nurse.

* * *

7 April. Pakistan-brokered ceasefire signed today. Thirty days initial. The strait is technically suspended. Nobody I have spoken to, including Viktor, believes the suspension will last. But the ships might move in the next week, and that is enough for tonight.

Jun has been preparing the engines. Rafiq has been cooking the crew's home food. Somchai has been setting out the chairs. Tendai has been doing his work with a weight on him that he did not know he could carry until it turned out he could. Arjun has been calling his mother every night, and his mother has been telling him about Ravi, and Ravi is still holding on. Dani and Eko have been doing the cadet work nobody else wants to do, and doing it better than their training required. Viktor has been telling us the truth in the doses we could take.

I wrote to my mother on the second. The letter is in the drawer. It will go with the next crew change. I told her I was fine, which is mostly true, and I told her about Tendai and about Majid, and I told her to eat and to walk and to call Leila. It was the first letter I have written her in a month. I did not know I had been waiting for a reason. It turns out the reason was my own advice to Arjun, returned to me slowly over the course of a week. I told him the calling was more important than what was said. I needed somebody to tell me that too. Arjun's unintended service to me was that by needing it himself, he let me hear it.

Seneca wrote that the wise prepare and the negligent hope. I have been reading him for a week and I keep finding myself thinking he was being harder on the negligent than he needed to be. The people who did not prepare did not usually choose not to prepare. The work of preparation looks, from the outside, like work that does not need to be done. Unglamorous, cheap, the boring repair of something that has not yet broken. Nobody gets promoted for it. Nobody puts it in a quarterly report. In the countries that did prepare, they did it because somebody with enough power to be heard said the boring work was worth doing. In most countries, that person was not there, or was not listened to, or was listened to for a while and forgotten when the next election came.

The crisis is not this war. The crisis is every month and every year for the last seventy years when somebody in a ministry or a board room knew the buffer was thin and did not fund the buffer, because funding the buffer looked like doing nothing, and doing nothing does not get you promoted.

Jun is warming the oil tonight. Thirty-six hours of warming oil in a machine that has been cold for six weeks, so that when the order comes, the machine is ready. Nobody will write about Jun in the newspapers. Nobody will put his name on a policy

paper. He will go home to Cavite when this is over and his daughters will run to the door and he will not tell them what he has been doing, because there are no words for it that would mean anything to a nine-year-old and a six-year-old. But the ship will move when she is allowed to move, because of him.

I am going to sleep now. If the ceasefire holds, we may move on the ninth or the tenth. If it does not, we will continue to wait. Either way, Jun's oil will be warm.

* * *

* * *

PASSAGE 8

The Transit

At four in the morning of the eighth of April, Jun checked the temperature of the oil for the third time in an hour. Forty-two degrees, climbing at the rate he had planned for. He wiped his hands on a rag, crossed the engine room to where Dani was tending the pre-lubrication pumps, gave him a small nod, and went up the ladder to tell Sami.

"Captain. Oil temperature is at forty-two degrees. I want to bring her to sixty before we open the fuel valves. Main bearings are pressurising normally. One of the starboard cooling water sensors is reading two degrees high compared to its partners, but both the others agree with each other and I think the high one is the wrong one. I will swap it out before we try to start."

"How long until you are ready?"

"Another thirty hours. Maybe thirty-two."

"Good. Take what you need."

"Yes, Captain."

Jun went back down the ladder. Petros, who had the bridge watch, waited until the footsteps had faded before he spoke.

"He has not slept in twenty-four hours."

"I know."

"He will not sleep until the engines start."

"I know."

"Should we make him?"

Sami thought about it.

"No. A man who has kept this machine ready for six weeks has earned the right to finish what he started. If we try to put him to bed now, he will not sleep. He will lie there thinking about the cooling water sensor."

Petros nodded and went back to his coffee. The running lights of the other anchored ships spread to the horizon. A hundred and fifty of them, by Petros's morning count. None of them moving yet.

The news arrived on the morning of the ninth in two pieces, a few hours apart.

The first was a newspaper story Arjun found on his phone while he was still in his bunk. The director general of ACI Europe, which represented the continent's airports, had sent a letter to the European Commission. His name was Olivier Jankovec and the letter ran to two pages. Europe's major airports, the letter said, had between eight and ten days of jet fuel remaining, measured in days rather than weeks. The Commission was being asked to coordinate emergency purchases from refineries outside the affected supply chains, to impose mandatory production obligations on those refineries that remained operating, and to urgently map the current state of European refinery capacity, because nobody had a current map. Arjun brought the phone to Sami at breakfast.

Sami read the article carefully. He had read the analysts who had warned about this, in different words, in papers going back six years. He had read them in his cabin through a long week in late March. He had written their conclusions in his notebook. Jankovec was not saying anything new. He was saying it publicly, for the first time, in a letter that the European Commission had to answer, because the thing the analysts had warned about had arrived.

"Captain."

"Yes, Arjun."

"There was another story, underneath this one. The head of the international airline trade body, Walsh, said yesterday at a press event in London that the disruption was comparable to nine-eleven. He said strategic reserves of jet fuel did not exist anywhere in the world. He said everybody was running on what was in the tanks."

"Yes."

"Is that true? That they did not exist anywhere?"

"It is true."

"Did they know? The people who did not build the reserves?"

Sami took a moment before he answered. He was still holding the phone.

"They knew, Arjun. They have been writing the papers that said so since before you were a cadet."

"Then why."

"Because the people who write the papers are not the people who control the budgets. And the people who control the budgets were told that efficiency was worth more than preparation. They were told this by people who had been trained at business

schools that taught efficiency as a doctrine. They taught it for forty years. Starting this week, they are going to have to explain themselves. They will blame the war, because the war is in front of them and the doctrine is behind them. Do not believe that explanation when you hear it. The doctrine was fragile long before the war."

Arjun nodded slowly.

"I am going to go and help Jun."

"Go."

Arjun left. Sami gave the phone back to the cadet it had belonged to before Arjun had borrowed it, and went to his cabin to read the article in full.

* * *

The second piece of news came at ten past eleven. Petros did not knock. He opened the door of Sami's cabin with a printed sheet in his hand, and when Sami looked up, Petros was wearing a small quiet expression that Sami had not seen on his helmsman's face in forty days.

"Captain."

"Tell me."

Petros put the paper down on the desk.

"We have clearance."

Sami read the note twice. He read it the second time because the first time he had been reading the words and had not yet let them be true. The company had confirmed it. The Iranian authorities had issued a transit clearance for the *Noor Salaam* under their Larak corridor protocol. She was to proceed when ready, transiting through Iranian territorial waters on the southern side of Larak Island, subject to IRGC visual confirmation at the checkpoint and subject to continuous VHF contact with the patrol vessels that would monitor her passage. The clearance was valid for forty-eight hours. If the ship did not transit in that window, she would have to apply again.

The company had also arranged an Indian Navy escort. A frigate called INS *Tarkash* would rendezvous with the *Noor Salaam* at the eastern end of the corridor and accompany her through the remainder of the Gulf of Oman and across the Arabian Sea to the Vadinar pilot station. The escort was not a guarantee of safety. It was a gesture by New Delhi that the cargo was important enough to the Indian state to be defended by the Indian state. Sami understood the gesture both for what it said and for what it implied. The Government of India considered the *Noor Salaam's* cargo important enough to send a warship for. The Government of India also considered the transit risky enough to require one.

Sami read the note a third time. Petros was still standing in the doorway. The two men did not say anything to each other for a while. What needed to be said had been said in the small quiet expression Petros had come in wearing, and in the reading twice and three times of the same piece of paper by a captain who had been ready to read it for six weeks. They were tanker men. Tanker men had reservations about saying things plainly, because the sea had taught them that plain things could be taken away. The relief stayed in the cabin with them anyway. It did not need to be named to be there.

"Thank you, Petros."

"Yes, Captain."

"I am going to the galley."

Petros nodded. Sami folded the printout, put it in his shirt pocket, and went to find Rafiq.

* * *

Rafiq was at the counter. He had rinsed the lentils twice and was on the third rinse. Beside the colander he had laid out three small storage onions, the last of the stock they had laid in at Ras Tanura, their papery skins speckled with the brown of keeping. He had set them out in a careful row, the way a man sets out the last of something when he is not sure he will get more.

"Rafiq."

"Captain."

"We have clearance. Forty-eight hours. Jun thinks we can be underway by tomorrow morning."

Rafiq stopped with his hand on the colander. He did not look up for a long moment. When he did, he nodded once.

"Good."

"I wanted you to know before the announcement."

"Thank you, Captain."

Rafiq poured the last of the rinse water through the colander. The water ran clear. He set the colander over the pot and picked up the paring knife. Sami did not leave. He was looking at the onions.

"Rafiq."

"Yes, Captain."

"What were you going to do with these?"

"Dal, Captain. With rice. A little turmeric if I can spare it. Why?"

Sami was quiet. He had come to the galley with a specific thing he wanted to say. It was not this. But the lentils were on the counter and the onions were next to them and he was fifty-seven years old, and thirty-five years had just collapsed into the kitchen of his mother's flat in Damascus.

"The night before I left for sea, my mother made me a dish called mjadara. Lentils and rice, slow-cooked together. Onions sliced thin and fried in olive oil until they were the colour of a chestnut. The onions went on top of the lentils and the rice. A drizzle of olive oil over the whole thing at the end, from the bottle she kept on the shelf above the stove, which was from her cousin's grove outside the city. I was twenty-three. I did not know at the time that she was making it as a goodbye. She knew. She did not tell me."

Rafiq had set the knife down.

"I have not thought about that meal in thirty-five years, Rafiq. I have been carrying it anyway."

Rafiq did not speak. He was looking at the onions.

"Food is the kind of thing that stays in a person without the person knowing it is there. Music. Dancing. The smell of a particular bread baked a particular way. My mother did not put her arms around me and tell me anything she wanted me to remember. She made me the mjadara. Thirty-five years later I am standing in a galley on the other side of the world looking at three old onions and a pot of lentils, and I am back in her kitchen."

"Yes, Captain."

"I came down here to tell you about the clearance. I wanted to tell you one other thing. The three dishes that week. The nshima. Your mother on the phone telling you what a Zambian woman would have done with the last of the dried fish. I have not said anything about any of it, because I did not know what to say, and I did not want to make it small by trying. You have been feeding us for forty-two days on a ship that had forgotten how to be glad. You have cooked three of the men back into their own mothers' kitchens. I am telling you now, before the announcement, because I do not want the announcement to be the thing that crowds this out. You have been a blessing to this crew, Rafiq. We have been able to find small joy because you have been making it possible. I am grateful."

Rafiq picked up the paring knife. He did not look at Sami. He peeled the first of the three onions. The peel came off in a single long curl and went into the steel bowl on the counter beside him.

"My grandmother fed a street during the war in 1971. My mother fed a block during the cyclone in 2007. There was no power for eleven days. Nobody had gas. She cooked over charcoal on the roof of our building and sent plates down to the families below. She did not know most of them. She did not ask if they were Muslim or Hindu or Christian. If they were at the door with a plate, they got a plate back. Nobody called it a blessing. Nobody called it anything. She would have been embarrassed if they had."

He sliced the first onion thin.

"When people ask what the dish is called, you tell them the name. That is all. It is just what you do."

"My mother would have said the same about the mjadara."

"I thought she would."

Sami did not leave. The onion on the board was releasing its sharpness into the galley air, and for a moment Sami could not be sure whether the stinging at the back of his throat was the onion or something else.

"Rafiq."

"Captain."

"If I can get olive oil."

"I have oil, Captain."

"Olive oil, Rafiq."

"Ah." Rafiq paused. He considered the three onions and the pot of lentils and the ship that would be underway in the morning. "I have a small jar, Captain. I was saving it for something. I had not yet identified what I was saving it for. I think I have now identified what."

"Thank you."

"Tonight, Captain. After the announcement. If you will tell me how your mother made it, I will try."

"Tonight."

Sami went back to the bridge. He did not stop to speak to anybody on the way. The word *oil* had meant one thing to him when he had come aboard the *Noor Salaam* at Ras Tanura on the fourteenth of February. It had meant two million barrels of Arab Heavy crude. It had meant the cargo. Six weeks later, standing in a galley doorway looking at three old onions and a small jar of olive oil that a Bangladeshi cook had been saving for a moment he had not yet identified, the word was beginning to mean something else as well. The old meaning was not gone. The old meaning was down in the tanks. But there was another meaning stacked on top of it now, older than the

crude, and the ship he was captain of had quietly been carrying both.

* * *

Jun started warming the cylinders at fifteen hundred.

By eighteen he had brought the oil to the temperature he wanted. He ran the pre-lubrication pumps at full flow for another two hours. He checked each of the main bearings by hand, which meant he placed his palm on the casing above each one and felt for the steady warmth that a properly pressurised bearing gave off, the kind of warmth that told an engineer everything he needed to know without a gauge. He did this for all eight main bearings. He checked the cooling water sensors a second time, and the new starboard sensor agreed with its partners. He checked the fuel injection pressures, the exhaust manifold thermocouples, the governor, and the auxiliary air compressors. By twenty hundred he had run out of things to check. He sat down on a camp stool in front of the main engine console and waited for the order from the bridge.

The order came at twenty-one hundred, local.

Sami said the words over the intercom, which he had not had occasion to use for starting the main engine in forty-two days. "Bridge to engine room. Start main engine, Jun, in your time."

Jun's voice came back. "Yes, Captain. Starting now."

There was a long silence. Then the compressed-air starters engaged with a hiss that became a roar. The engine turned over against its own inertia, which was considerable. The sound was the sound of a very large machine being persuaded to remember that it knew how to move. Then the first cylinder caught fire, and the second, and then all of them together, and the starters disengaged, and for the first time in forty-two days the main engine of the *Noor Salaam* was running under its own combustion.

The vibration travelled up through the hull into the decks into the soles of the feet of every man on the ship. Some of them were expecting it. Some of them had been asleep and woke up. A few of them who had been through other long anchorages in other crises recognised the feel of it as the feel of a ship that had remembered herself.

Petros was at the wheel. He did not say anything. He closed his eyes for a moment, opened them, and reached forward to confirm the RPM reading. The dial was steady.

Tendai came up to the bridge without being called. He stood at the back.

"Captain."

"Yes."

"I just wanted to be here for this part."

"Stay."

Tendai stayed. He watched Petros at the wheel and Sami at the window and the anchorage lights spread around them in every direction. The ship was not yet moving. The engine was running but the anchor was still down and the ship was holding her position on the water. Even so, something in the air had changed in the way the air changes when a person in a sickbed starts breathing differently. The *Noor Salaam* had come back.

* * *

They weighed anchor at zero one hundred on the morning of the tenth.

Sami gave the order quietly. Somchai acknowledged from the foredeck and the windlass started up. The chain came in link by link, clanking against the hawsepipe, dripping saltwater and the brown silt of six weeks of holding. A hundred and fifty other ships were still at anchor around them. The *Noor Salaam* was the third vessel to weigh that night. A Chinese tanker two kilometres to the north had gone first, twenty minutes before them, and a smaller bulk carrier of Pakistani registry had gone fifteen minutes before that.

The anchor came up at zero one forty. Petros eased the throttle forward. The ship took a moment to respond, the way a large vessel does when her mass is being persuaded rather than commanded. Then she started to move.

Nobody on the bridge said anything about the movement. Nobody needed to. The deck under their feet had been still for forty-two days, and now it was not still, and the unfamiliarity of the motion was the same shape, in reverse, as the unfamiliarity of the stillness had been on the twenty-eighth of February. A VLCC at sea is a building that happens to be floating. A VLCC at anchor is a building. They had been living inside the building for six weeks and had forgotten the feel of the building becoming a ship. It came back to them within a kilometre of movement. Underneath the apprehension about whether the clearance was real, and whether the patrol boats would meet them where the note said they would, and whether an IRGC officer somewhere might have reconsidered in the last four hours, there was the plainer and older sensation of a ship doing what a ship was built to do, with men on her bridge who knew how to make her do it. The sensation did not announce itself. Sailors did not announce sensations like that. It was there in the way Petros set his hands on the wheel, and the way Somchai came back from the foredeck without saying anything and went straight to his station, and the way Viktor, who was supposed to be off watch, was on the bridge anyway,

standing at the chart table, not doing anything except being there.

The water under the bow opened quietly. Nothing dramatic. A vessel of her size making three knots displaced a great deal of water but did not announce herself. The anchorage lights receded behind them at a speed that was barely a speed at all.

Somchai came up to the bridge at zero two hundred and thirty.

"Forward deck is secured, Captain. Anchor is housed. All crew in their stations."

"Thank you, Somchai."

He stood beside Sami. They watched the lights behind them shrink.

"A hundred and forty-seven ships still there."

"Yes."

"The ones we are leaving behind."

"The ones we are leaving behind."

"I wish we could have brought them with us."

"I wish that too."

Somchai did not say more. He went back to his station on the deck.

* * *

The transit of the strait itself took seven hours and forty minutes.

The corridor the Iranian authorities had approved ran along the southern side of Larak Island, close to the Iranian coastline, well inside Iranian territorial waters. The traditional route through the Omani shipping lanes was no longer available. The new corridor was narrower. It was also closer to IRGC patrol stations, which meant that the *Noor Salaam* was under continuous visual observation from the moment she entered the Iranian territorial sea until the moment she left it.

Petros was at the wheel for the first two hours. Viktor relieved him at zero four hundred and held the wheel through the tightest section, which was the passage through the gap between Qeshm Island and Larak itself. Sami was on the bridge throughout. Jun stayed in the engine room and slept there on a camp bed between checks, and took a total of two hours of sleep during the transit, spread across four twenty-to-thirty-minute periods.

The first hail came from an Iranian patrol vessel at zero four twenty.

"*Noor Salaam*, this is Revolutionary Guard Naval Command. Confirm flag, call sign, cargo, and destination."

Petros, who had been preparing for the hail, responded in accented English. "Revolutionary Guard Naval Command, this is *Noor Salaam*, Liberian flag, call sign Alpha Eight Yankee Golf Four, cargo two million barrels of Arab Heavy crude,

destination Vadinar Single Point Mooring Number Two serving Reliance Industries Jamnagar. Indian-destined cargo under clearance note dated ninth April. Over."

A pause.

"*Noor Salaam*, Revolutionary Guard. Confirmed. Maintain your current heading. You will encounter a second verification station at Larak Point in one hour. Proceed."

"Revolutionary Guard, *Noor Salaam*, roger. Proceeding."

The radio went quiet. Sami set the handset back in its cradle. Nobody on the bridge spoke for a moment. The ship continued at her slow speed through the corridor, the coast of Iran a dark line on the port side, the coast of Oman a darker line on the starboard. Viktor, who had been listening at the wheel, said to nobody in particular, "That was an officer. Not a conscript. The conscripts are the ones who read the script without stumbling. The officers make small mistakes in the grammar because they are trying to remember a rule they were taught in training."

Sami looked at him.

"I have been on the other end of Russian military radio for a long time," Viktor said. "The grammar of a tired officer is the same in any language."

"So they are letting us through."

"They are letting us through, Captain."

Sami nodded. He did not say anything more, because there was nothing a tanker man would say at this moment that was not better said by silence. But he felt, somewhere behind the sternum, the small steady gladness of a man who had been told something he had been waiting six weeks to be told. Petros at the wheel felt it too. Somchai on the foredeck, who had heard the exchange over the intercom, felt it. Joy had a different shape among men who had spent their working lives declining to trust an outcome until the outcome had completed itself. No embraces. No shouting. No raised glasses. A settling. A breath let out that the chest had been holding without knowing.

The second hail came at zero five twenty. It was shorter. The IRGC had them visually, from a patrol craft running parallel on the starboard side at a distance of two kilometres. The craft did not approach. It did not need to. Its mere presence was the message. The *Noor Salaam* was transiting Iranian waters with Iranian permission, and the permission could be revoked.

By zero seven hundred the ship had cleared the corridor. She turned south-east into the open Gulf of Oman and picked up her cruise speed, which was now thirteen and a half knots on a main engine that had been cold for six weeks and had every right to complain and was, so far, declining to.

Petros took the wheel back from Viktor.

"I have sailed this strait two hundred times, Captain. I have never sailed it this slowly. It felt like walking through a room where somebody is sleeping."

"Yes."

"Every time we heard the patrol boat on the radio I felt my shoulders come up."

"Mine too."

"I did not know we could be tired and afraid and relieved at the same time, with this much force in each of them."

"I have been learning that also."

Petros did not say more. They stood at the bridge windows and watched the coast of Iran slide away on the port side, and the coast of Oman come toward them on the starboard, and the sky in the east start to show the grey that came before the sunrise in early April. When the sunrise came, a few minutes after six, the *Noor Salaam* had already cleared the last of the territorial markers and was making south-east for the rendezvous with the Indian frigate.

* * *

The frigate hailed them at zero nine ten. She was called *Tarkash*, a word that meant "quiver" in Hindi, and she was a Russian-built Talwar-class guided-missile frigate of the Indian Navy. Her captain was a Commander Sanjeev Nayar, whose English was precise and who made his introduction in six sentences.

"Tarkash to Noor Salaam. Good morning, Captain. I have orders to escort your vessel to the Vadinar pilot station, ETA thirteen April zero six hundred local. I will remain approximately four nautical miles on your starboard quarter unless circumstances require otherwise. If you have any concerns, my bridge is available on channel sixteen twenty-four hours. I wish your vessel a safe passage."

"Commander Nayar. Captain Kassar speaking. Thank you. We are grateful for the escort."

"The gratitude is on our side, Captain. Your cargo has been missed."

The formality of the exchange was the formality of two men who had done this many times and who both understood that the formality was a courtesy. Nayar's frigate fell into her station on the starboard quarter. The two vessels ran south-east together across the Gulf of Oman.

Arjun came up to the bridge at ten.

"Captain. The Tarkash."

"Yes."

"I know that ship. My cousin served on her in 2019. He was a junior electrical officer. He was on her for two years before he transferred to shore duty."

"That is a nice coincidence."

"I wanted to call him and tell him. Is that all right?"

"Go and call him."

"Thank you, Captain."

Arjun went to the radio room. Sami heard him later, through the open door, speaking in Hindi to his cousin, who was presumably in Mumbai and was presumably also about to be calling his own mother to tell her that the escort the Indian Navy was sending for the strategic cargo returning from the Gulf was a ship he had served on, and that his little cousin was on the tanker. The *Noor Salaam* had entered the news cycle of one particular family in Thane and Mumbai. It was a small victory of a kind Sami had not expected.

* * *

The crossing of the Arabian Sea took three days.

The first day was the day the engines settled. Jun slept for six hours in the afternoon, which was the longest he had slept in a week. When he came back on duty the main engine had been running steadily at thirteen and a half knots for twenty hours, and none of the readings had drifted, and the oil consumption was at the rate Jun had expected. He went round the engine room once without touching anything, which was the engineer's version of saying a prayer of thanks.

The second day was the day Viktor broke open. He had held himself close all through the anchorage, speaking in short sentences, offering his observations at the Evening Questions in the bleak and informative style the crew had come to rely on. On the second day of the Arabian Sea crossing, a Thursday, Rafiq made him a bowl of borscht. He had called his mother in Dhaka two days earlier to ask how a Bangladeshi cook might approach a Ukrainian dish. His mother, who had never made borscht, had told him to go and find an Eastern European sailor on another ship at anchor and ask. Rafiq had not had access to another ship, so he had gone to Viktor directly and asked him how his mother would have made it. Viktor had written out a list of ingredients, half of which Rafiq did not have, and had told Rafiq to do what he could with what he had.

What Rafiq had was beets from the forward cool store, a single potato that had been sitting in his personal locker for three weeks, the last of the cabbage, and some yogurt that Rafiq had been saving for exactly a moment he had not yet identified. He

did what he could with what he had. The borscht was not Viktor's mother's borscht. It was nobody's borscht. It was a red cabbage stew with beet liquor and a spoon of yogurt on top, served in a deep bowl to a Ukrainian man at a table on a tanker in the Arabian Sea on the twelfth of April, and when Viktor saw it he put down his spoon before he picked it up, and sat still for a moment.

Then he picked up the spoon and ate.

He did not say anything. Rafiq did not hover. He went back to the galley and started on the main course, which was chicken that had been in the freezer for six weeks and needed to be used. Viktor ate the borscht slowly. When he finished, he stood up, carried the bowl himself to the galley door, handed it to Rafiq, and said, "Thank you. It is not my mother's. But it is close enough that I can pretend."

Rafiq took the bowl.

"I will not try again," he said. "It is not a dish you can make a second time."

Viktor nodded. He went back to his cabin.

* * *

The third day was the day the crew began, quietly, to talk about what they would do when they got home.

Nobody had talked about it during the anchorage. Everyone had known that talking about it would be a way of daring the thing not to happen. Now the ship was making thirteen and a half knots across the Arabian Sea toward a port they would reach in thirty-six hours, and the superstition that had kept them silent was eroding.

Arjun would go to Thane first, not Mumbai, because Ravi was still in the hospital and the hospital was in Thane, and his mother would want him there before she wanted him home. Jun would go to Cavite. His company had told him he would have two weeks of leave before the next rotation, and he intended to take all of them and then some. Tendai would go to Chipata. His mother had already told him that she would kill him herself if he tried to go to Lusaka or Kitwe first. Rafiq would go to Dhaka. His mother had not said anything, because his mother never said anything, but she had cooked a new thing for Mahmud that morning, a sweet vermicelli that she usually saved for celebrations, and Rafiq understood what the sweet vermicelli meant. Dani would go to Jakarta. Eko would go to a village outside Surabaya. Petros would go to Athens. Viktor would not say where he would go, and nobody asked.

Somchai would go to Bangkok. His nephew, whose name Sami had still not been told, was due to dock in Singapore on the fifteenth, and Somchai would fly there from Bangkok to meet him and to hear, in person, about the friends who had been held on

Qeshm Island and who had been released three days earlier without incident and without explanation, and who were now on a flight back to Bangkok via Dubai. The Iranians had fed them, as Somchai had said. They had not been harmed. The situation was not resolved because no situation involving the Iranian state was ever resolved. But his nephew's friends were going home. Somchai would see them before the month was out.

Sami would go to Beirut. He had not been to Beirut in eleven months. His mother's letter was still in the drawer of his cabin desk, and he would deliver it himself. He had decided that a letter that had been in a drawer for eight days deserved to be hand-delivered by the man who had written it.

* * *

On the evening of the twelfth, Sami read for two hours.

He had been working his way through a collection of essays by a Lebanese-American writer he had been introduced to years earlier by another captain, in a bar in Singapore, during a layover that had lasted three nights instead of two because the vessel's pilot had been delayed. The writer was called Nassim Nicholas Taleb. He had written a book in 2012 about systems that broke under stress and systems that did not. He had invented a word to describe the third category, the rarest one, the systems that got stronger from stress. The word he had coined was *antifragile*, because no language on earth had a word for that category, because most human cultures had not thought to separate it out as its own category, and Taleb had thought it was important enough to separate. A system that broke, Taleb said, was fragile. A system that held its shape under pressure was robust. A system that was improved by pressure, that came out of the pressure stronger than it had gone in, was antifragile.

The just-in-time supply chain, Sami thought, was fragile in the technical sense Taleb used. It had been designed for the absence of stress. It broke the first time stress arrived. The analysts who had warned about its fragility had been using Taleb's categories without always citing him, because by now the categories had passed into the common language of people who thought about systems for a living.

But Taleb's framework was not only about systems. It was about people. It was about cultures. It was about what a civilisation chose to build when it had the choice. A civilisation that had built itself to rely on stability, and that treated every shock as an exception, was fragile. A civilisation that had built itself to survive shocks without being changed by them was robust. A civilisation that treated shocks as information, and that used them to rebuild into something better adapted than what had come

before, was antifragile. Most of the world had spent the last two centuries becoming more fragile, not less. Efficiency had been the word they called it. Fragility was what it produced.

Sami put the laptop down.

He thought about the crew he was sitting on a ship with.

They had gone into this anchorage six weeks ago as a crew that had worked together but had not really known each other. They had come out of it knowing each other in a way that was different from the knowing of friendship and closer to the knowing of family. They had learned more about the world's energy system in six weeks than most people learned in a lifetime, and they had learned it because the world had stopped moving and they had been forced to ask why. Arjun had kept a cousin alive through calls he had thought did not matter. Tendai had said a sentence to his mother that had given her permission to believe his sister could be a nurse. Jun had kept an engine ready to move for forty-two days and had then moved it. Rafiq had cooked three dishes he had never cooked before and had made a borscht for a Ukrainian man on the second day of the Arabian Sea crossing.

They had not been broken by the anchorage. They had been made by it.

They were not robust, exactly. Robust would have meant they came out unchanged, and they had not come out unchanged. They had come out as something else. Something more carefully assembled, more honest with each other, more aware of what they were and what they could do. Taleb would have said they had become antifragile. Sami was not sure that word was right either, but it was closer than any word he had.

He wrote in his notebook for the first time in four days.

The ship is three hundred nautical miles off the coast of Gujarat. The Tarkash is on the starboard quarter. Jun is in the engine room. Rafiq is in the galley. We will arrive at Vadinar in the morning. The cargo will be discharged. I do not know what will happen after that.

Taleb wrote about fragility and robustness and antifragility. The supply chain was fragile. The men on this ship have become something more than robust. I do not know which word applies to what we are now. I know only that we are not what we were when we dropped anchor on the twenty-eighth of February. A shock that teaches is different from a shock that breaks. A shock that teaches is what Taleb meant by antifragile. Most shocks do not teach. Some do. The difference is whether the people inside the shock are willing to learn from it, and whether the people around them are

willing to let them learn.

We have been let learn. I am grateful for that. I have not always been grateful during the six weeks. I have been afraid and tired and angry, and at times ashamed that I had so little to offer the people our cargo will reach. Tonight I am only grateful. The gratitude will pass. The learning will not.

In the morning we will see Vadinar.

* * *

At zero six hundred on the thirteenth, the *Noor Salaam* reached the pilot station at Vadinar.

The weather was clear. The Gulf of Kutch spread out to the east, flat and grey in the early light, the coast of Gujarat a low dark line on the horizon. A pilot boat came alongside, and the pilot came aboard. His name was Mehta. He had piloted this terminal for nineteen years and he shook Sami's hand with a firmness that communicated more than any welcome speech could have. The *Tarkash* peeled off to port, her mission complete, and Commander Nayar sent a final message of goodwill over the VHF before heading north to her own berth.

The ship made her way in through the approach channels to Single Point Mooring Number Two, which was one of Reliance's two crude import buoys. She was made fast by eight in the morning. The hose connections were made by nine-thirty. Discharge began at eleven.

Two million barrels of Arab Heavy crude began flowing out of the *Noor Salaam's* tanks and into the shore tanks of the Reliance Jamnagar refinery. The refinery would process the crude into diesel and petrol and jet fuel and naphtha and a hundred other refined products. Those products would be loaded onto other tankers and trucks and pipelines and would flow outward across India and to export markets beyond India. Some of the diesel would go to the generator at the district hospital in Thane that was keeping Ravi's dialysis machine running. Some of the cooking gas would go to the cylinders that Arjun's mother was trying to stretch. Some of the diesel would go to the tractors at the planting farms of eastern Zambia, delivered through a supply chain that would need weeks to reach Chipata. Some of it would not reach any of these places. It would be held back, or diverted, or absorbed by refineries further upstream in the system. The flow would resume, but it would not resume evenly, and some of the people the crew had been worried about for six weeks would be reached by it and some would not.

Sami stood on the bridge wing during the first hour of discharge. He watched the manifolds through binoculars. He watched the sea for the sheen that would indicate a leak, and the sea was clean, and the hose connections held.

Petros came up beside him.

"Captain."

"Yes."

"The company office has just called. The Americans announced it overnight. A full naval blockade on all Iranian ports, effective this morning. No vessels in or out of Iranian territorial waters except with United States authorisation."

Sami lowered the binoculars. For a long moment he did not speak. Petros waited beside him at the rail.

"The strait."

"The strait will be closed again by evening, Captain. Differently this time, by us instead of by them, but closed just the same."

Sami looked out at the sea. The discharge manifold was running cleanly. Two million barrels of oil were flowing into a refinery in a country that could use them. A hundred and forty-seven ships were still at anchor in the strait they had just left, and the strait was closing around them again, and this time the ones holding the door shut were the Americans. The *Noor Salaam* would finish discharging and then the ship would be empty, and an empty ship with no clearance to enter the waters where her next cargo was waiting was an expensive piece of steel floating in the Gulf of Kutch.

"Get Jun."

"Captain?"

"Get Jun. I want him to hear it from me, up here, before I tell the rest of the crew."

Petros went.

Jun came up to the bridge wing ten minutes later. He had come straight from the engine room and had not changed out of his work coveralls. His hands were dark along the knuckles from a morning of checking valves. He stood beside Sami at the rail, and Sami gave him the news in the same two sentences Petros had used.

Jun was silent for a long moment. He looked out at the sea, then at the discharge manifold, then at the low brown line of the Gujarat coast.

When he spoke, his voice was steady in the way a voice is steady when the man speaking with it has already decided not to let it do anything else.

"Captain. On the second of April I told you I did not know a man could be glad for his country and angry for himself at the same time with that much force. I was wrong about which side of that sentence was the hard part."

Sami waited.

"I am out of the strait. My country is back in the queue. The fuel that was starting to reach Manila will stop again within a week or two, because the refineries that feed Manila feed on crude that comes through the chokepoint that just closed again. The buses in Cavite will be back where they were in March. My wife will tell Liza that the school is suspended again after all. Exactly the same conversation my wife had with her three weeks ago, in exactly the same kitchen, with exactly the same ending. I am here, Captain. I am standing on the deck of a ship that made it out. My wife and my daughters are on the side of the door that just shut."

"Yes."

"For six weeks I told myself I could carry the anger I had about not being on a Filipino tanker when Filipino tankers were being sent home. I told myself the anger was the price of being in the right place to do the work. I was not a selfish man. I was a useful one. Today my country is back inside the closed door, and I am outside it, and I am the one who is free to go. The arithmetic I was doing has inverted. The anger has not gone anywhere. It was waiting for me on this side."

Sami did not answer at once. The sea was bright around the ship. The sun was climbing.

"I am angry, Captain. I do not know where to put it."

"Nowhere. You carry it. I will carry some of it with you. The man at the hospital generator in Thane does not know the name of the man on the ship that brought him his diesel. The man on the ship does not know the name of the man at the hospital generator. Both of us are carrying the same arrangement. That does not make it lighter. It makes it shared."

Jun nodded. He did not say thank you, because what Sami had said was not a gift. It was a description of the arrangement they were in.

"I will go back to the engine room, Captain. I will think about this there."

"Tell Petros to gather the crew in the mess hall before dinner. I will tell them together. They should hear it from me."

"Yes, Captain."

Jun went. Sami stayed on the bridge wing for a while longer. The sun was climbing. The sea was bright. Somewhere on the shore, in a control room at the refinery, an operator he would never meet was watching the crude flow into the right tanks. Somewhere in Thane, Ravi was receiving his dialysis, or was not. Somewhere in Chipata, Tendai's mother was planting maize on half of her plot. Somewhere in Dhaka, Mahmud had walked around the block once and come home. Somewhere in

Cavite, a nine-year-old girl was being told by her mother about the school buses. Somewhere in Beirut, Sami's mother was reading, or was not, or was sleeping, and would not know for another few weeks that her son was going to bring her the letter he had written her, because of the flow that had briefly opened and was about to close again.

The crisis was not over, and the ship had not escaped anything. She had completed her one task, and the world her task had been meant to serve was still broken. Her next cargo, wherever it was supposed to come from, was on the far side of a door the Americans had just shut.

Sami went back to his cabin. The letter to his mother was in the drawer. He took it out, turned it over once in his hand, and put it back in. It would keep.

* * *

* * *

PASSAGE 9

The Cycle

On the afternoon of the thirteenth, Petros assembled the crew in the mess hall.

Sami had asked him to deliver the news. Sami was the captain, and the captain was the one who brought bad news to the crew; that was the custom and Sami had held to it every other time. But on the afternoon of the thirteenth, a few hours after the first million barrels had been pumped into the shore tanks, Sami had asked Petros to do it instead. Sami did not give a reason. Petros did not ask for one. He went to the galley, and the engine room, and the deck, and gathered the men and women who were not on active duty for the discharge, and brought them to the mess hall at fifteen hundred.

Somchai sat at the back. Rafiq stood in the galley doorway so he could hear without leaving his station. Jun came up from the engine room, wiped his hands on a rag, and sat in the third row. Viktor sat in the front. Arjun was on the bridge, because somebody had to be, and he had asked Petros to come and tell him privately afterwards. Tendai and Dani and Eko sat together. The rest of the crew filled the remaining seats.

Petros waited until the room was settled.

"The Americans announced a naval blockade of all Iranian ports at zero three hundred this morning, our time. The blockade is comprehensive. No vessels enter or leave Iranian territorial waters without United States authorisation. The stated reason is the interdiction of military shipments to Iran and to Iranian-affiliated organisations in the region. The operational reason is to apply pressure for a new round of negotiations. The timing is not a coincidence. Our ship was the last Indian-destined cargo to transit through Larak yesterday. The Americans wanted us out before they put the door down. I do not know whether to be grateful to them or not."

Somebody laughed, very quietly, at the back. It was Eko. The laugh did not travel.

"What this means for us is the following. The discharge will finish tonight or tomorrow morning. After that, we will be an empty ship in Indian waters with no next cargo. Our next cargo would normally be loaded at Ras Tanura, Kuwait, or Basra. All

three of those ports are now behind the blockade. We cannot reach them without American authorisation, and I do not know what the American authorisation process is or how long it will take. The company is working on it. Until they tell us otherwise, we will remain at Vadinar after discharge. The anchorage here is deep and we will move about half a mile offshore to wait. The company has confirmed that our wages will continue."

Jun raised his hand. Petros nodded at him.

"How long, Petros."

"I do not know, Jun. Honestly. It could be a week. It could be two months. The Americans will negotiate. Iran will negotiate. Somewhere in the middle, a deal will be made. We are not the priority in any of those negotiations. We are a tanker in a port, and there are tankers in a port all over the world right now, and the companies that own the tankers are talking to their governments, and the governments are talking to the Americans, and nothing is going to happen very quickly."

"My family," Jun said. "My family is going to see this on the news."

"Yes."

"I have told them the ship is moving. They are going to be afraid when they see the blockade on television."

"Call them tonight. Tell them what is happening. Tell them you are safe in an Indian port. Tell them the company has confirmed wages. Tell them you will know more in a few days. That is what you can tell them. Do not promise anything you cannot deliver. Do not hide anything they are going to find out from a television anchor."

"Yes."

Jun lowered his hand. Petros looked around the room.

"There is one more thing I am authorised to tell you. The company is arranging crew rotations for any of you who want to go home. Flights will be paid. Those of you who stay will be compensated at the port rate. The decision is yours. You have until tomorrow evening to decide. I will be in my cabin tonight if anyone wants to talk through it. The captain will also be available, though he has asked me to handle the initial announcements."

Rafiq spoke from the galley doorway, without turning around, while he continued chopping something Sami could not see from the front of the room.

"What about the ones on the other side."

"The other ships in the Gulf, you mean."

"Yes. The hundred and forty-seven."

"They are still at anchor. I do not have more specific information than that. My assumption is that the Iranians will continue to issue clearances for Indian-destined cargo under the Larak protocol for as long as the Americans permit it, and that the Americans will try to negotiate a process for allowing at least the humanitarian-classified cargoes through. But I do not know, Rafiq. The situation is changing by the hour."

Rafiq nodded without looking up.

"I wanted to know."

"Yes."

Petros let the room sit for a moment before he closed.

"The last thing. Over the next two days we are going to feel many things about what has happened. Relief that we are here. Guilt that we are here when the others are not. Anger that we are going to be stuck again, after we thought we were moving. Fatigue that we thought the waiting was over. If any of you need to talk about any of it, come and find me, or the captain, or the person you have been relying on these weeks. Do not hold it. We have been holding enough."

He looked at Sami, who was standing against the bulkhead near the door.

"Captain. Anything to add."

Sami shook his head.

"No. You said it."

The room broke up slowly. Jun went back to the engine room. Rafiq went back to his chopping. Somchai came up the aisle and passed Sami on the way out and paused for a moment, said nothing, and continued on. Viktor remained seated. He did not move for a long time. Then he stood up and went to his cabin.

Tendai came up to Sami as the room emptied.

"Captain."

"Yes, Tendai."

"I am going to stay with the ship."

"You do not have to decide now."

"I know. I am deciding now. My mother said on the phone yesterday that Nambela passed her mathematics test. She said to tell everyone. I am telling you. I am also telling you I would rather stay here and keep my wages going into that account than go back and have the wages stop. If I go back now, the money stops, and if the money stops, Nambela does not eat next week."

"Tendai."

"It is not a sacrifice. It is a choice. I am telling you so you know I made it with my eyes open."

"Nambela is going to be a nurse."

"She is going to be a nurse."

"Stay, then. I am glad."

Tendai nodded and went out.

* * *

The discharge finished at zero four hundred on the fourteenth.

The hose connections were broken at zero five. The pilot boat came out at zero six and took Pilot Mehta off. Sami signed the final paperwork in the bridge. The *Noor Salaam* moved off Single Point Mooring Number Two at zero seven and reanchored half a mile offshore in twenty-eight metres of water, bow pointing east into the rising sun, empty.

An empty VLCC sat high in the water. Her freeboard was now the height of a four-storey building above the waterline, where it had been the height of a two-storey building the day before. The main deck felt different underfoot. The ship rolled differently in even the small chop of the anchorage. Petros noticed it first and mentioned it to Sami over coffee.

"She is light."

"Yes."

"She rolls like an empty ship rolls. It has been a long time since I felt that with this vessel."

"Have you ever been anchored empty in a friendly port?"

"No. Usually we discharge and go back out for the next load. Or we stay a day while paperwork clears. This is different."

"It is different."

They drank their coffee. The shoreline was busy with the traffic of a working refinery: other vessels loading, tugs moving, helicopters lifting personnel in and out. The *Noor Salaam* was the only tanker in the anchorage not working. She sat half a mile offshore, empty and waiting, and the crew on her decks had the strange experience of being on a ship with nothing to do.

* * *

On the evening of the fourteenth, Somchai came to Sami's cabin.

He knocked. Sami said come in. Somchai did not sit. He stood just inside the door with his cap in his hand.

"Captain."

"Yes, Somchai."

"My nephew's name is Anan."

Sami nodded. He had been waiting for this for five weeks.

"Anan."

"He is twenty-four years old. He is a third officer now, and he will take his second officer's examinations next year. He arrived in Singapore at eleven this morning, local. He sent me a message. His friends from the *Mayuree Naree* arrived in Bangkok last night. They are at their homes. They are thin and they have nightmares, and one of them does not want to go back to sea, but they are at their homes."

"That is good news, Somchai."

"It is good news, Captain."

Somchai turned his cap in his hands.

"I did not tell you his name until today because I was superstitious. I am old enough to know better. I do not know when I started thinking that saying his name aloud to you would be the thing that caused something to happen to him. I think it was sometime around the *Mayuree Naree* grounding. I apologise."

"There is nothing to apologise for."

"There is, a little. The rule on this ship is that we name the people we are afraid for. I broke the rule. I was the bosun, and the rule should have applied to me first."

"Somchai."

"Yes, Captain."

"The rule is for the people who need it. If you did not need it, it did not apply."

Somchai looked at the floor. Then he looked up.

"I needed it. I did not want to need it. I have been learning, this past year, that those two things can sit closer together than I used to believe."

"Sit down."

Somchai sat on the edge of Sami's desk chair.

"I have been thinking, Captain. Anan's friends are home. My nephew is safe. The ship has discharged. We are anchored in a friendly port. And I feel worse today than I did on the day the *Al Salmi* burned. I do not know what to do with that."

"It has a name."

"I do not want the name."

"You can have the name or not. It is called the collapse after the survival. You were running on adrenaline for six weeks. Now you are not running on anything. The feelings you were holding at arm's length because you had to are now in the room with you."

"Is that what you have been going through."

"Yes, Somchai. For two days."

"I thought so. You asked Petros to do the announcement."

"I did not trust my voice for the announcement."

"Now you know."

"Now I know."

Somchai stood up. He put his cap back on.

"I am going to call Anan again tonight. I am going to tell him his old uncle is all right and he does not need to worry about me."

"Tell him he has been a good influence on his uncle this year."

"I will tell him that."

Somchai left. Sami sat for a while before he went back to his notebook.

* * *

On the fifteenth, the Americans enforced the blockade in full.

The news coverage was continuous. NPR reported that the US had deployed over a hundred aircraft and a dozen warships to enforce the cordon. Only two vessels were observed transiting the strait in either direction in the twenty-four hours after the blockade began. CNBC counted hundreds of tankers stranded in Gulf waters. European outlets focused on what the blockade meant for the reserves Jankovec had written about a week earlier. A Spanish airline cancelled half its routes. A German pharmaceutical distributor warned of shortages of four generic antibiotics by the end of April. The Indian press covered the domestic angle: fuel prices stabilising because of the cargo arriving at Jamnagar and a second Indian-flagged tanker arriving at Kochi, combined with a government announcement that the strategic petroleum reserve would be released to the commercial market at a rate that should hold prices through the end of May.

Sami read all of it on his laptop. None of it surprised him. He had been watching the same pattern since late February. The pattern was now running in reverse, with the Americans holding the door shut instead of the Iranians, but the mechanics of the closure were the same, and the effects were landing on the same people in the same order.

He opened his notebook.

The Americans announced the blockade overnight. The strait is closed again. The closure is now enforceable by the only navy in the Gulf with enough force to enforce it, which means the closure is comprehensive and the negotiations will take weeks. Possibly months. The pattern is the pattern. It was never about who was doing the closing. It was about a shipping lane that a hundred and fifty countries had arranged their supply chains around and that six powers were entitled to close. Six powers. Any of them. On any day. For any reason. The supply chains had been arranged around the assumption that the powers would not close the lane. The closure was treated as impossible. It had been impossible for long enough that it looked like a law of nature. It was not.

I am starting to wonder whether I should write this down. Not in the notebook. As a book.

* * *

On the morning of the sixteenth, the first crew rotations left.

Arjun flew to Mumbai. A company car took him to the airport at Jamnagar at zero five. Sami walked him to the gangway.

"Your mother knows you are coming?"

"She knows. She has been cooking since yesterday."

"What about Ravi?"

"He was at the hospital last night for dialysis. The generator at the hospital is running now. The fuel that came from our cargo is the fuel that is running it. I will see him this afternoon."

"Tell him a stranger on a ship off the Gujarat coast has been thinking about him."

"I will tell him. Captain."

"Yes."

Arjun stood at the top of the gangway. He did not speak for a moment.

"I have had a hundred thoughts I wanted to say to you on the walk from my cabin. I cannot think of any of them now."

"You do not have to say them. Go home. Call me when you are there. Tell me Ravi is alive or tell me Ravi is not alive. That is all I need."

"Yes, Captain. Thank you."

He went down the gangway. The company car took him up the access road. Sami watched it go until it turned the corner.

Jun flew to Manila that afternoon.

He did not come to find Sami before he left. Sami found him in the engine room, where he had been since the discharge ended, running a last inspection on the main engine and the auxiliaries and writing a three-page handover note for the relief engineer the company would send out the following week.

"Jun."

"Captain."

"You are going."

"I am going. The company car is in an hour. My flight is this evening."

"The engine is in good order."

"The engine is in good order. I have written everything I think the relief engineer will need to know. I have also written the things I do not think he will need to know, because I would rather he be annoyed at the length of the note than surprised by something I did not mention."

"When do you come back."

"Three weeks. Maybe four. Depending on the company. I would like it to be four."

"Four is fine."

Jun finished the note. He put the pen down. He took a breath and then he did something Sami had not seen him do in the eleven years they had sailed together. He put his hand on Sami's shoulder.

"Captain. When Liza asked her mother why I was not coming home, and her mother said the man on the radio had told her ships were not leaving the Gulf yet, and Liza said no I meant my father. I told you that story on the deck on the second of April. Do you remember."

"Yes."

"I want you to know that when I walk through my door tonight, Liza is going to run to me, and Mira is going to hide behind her mother because she does not remember me, and then she is going to come out from behind her mother when Liza does, because she trusts Liza. And I am going to hold both of them, and I am going to think about this moment, and I am going to think about you standing in the engine room of a tanker in the Gulf of Kutch, and I am going to remember that you were the one who told me to write to her that night, and that she kept the letter. She has shown me the letter. She read it twenty times while I was gone. She did not know I had written it the night the Al Salmi burned. She thought I had written it the night before the ship

cleared. She thinks I had a premonition. I did not have a premonition. I had a captain."

Sami looked at him.

"Jun."

"Captain."

"Go home. Hold your daughters. Come back in four weeks."

"Yes, Captain."

Jun lifted his hand from Sami's shoulder. He picked up his duffel bag. He went up the ladder.

Sami stood in the engine room for a long time after Jun had gone. The engine was not running. The silence in the engine room of a ship at anchor was the silence of a machine that had been shut down correctly. He ran his palm along the casing of the main engine. It was cool. The machine had done what it had been built to do and was now resting. He did not know when he would hear it run again.

He went back to his cabin.

* * *

Rafiq flew to Dhaka the next morning.

He came to the bridge before he left. He was carrying a small parcel wrapped in a clean tea towel.

"Captain."

"Rafiq."

"This is lunch. For today. I have left dinner in the galley, with a note. Eko knows how to warm it. Tomorrow the replacement cook will arrive. His name is Krishnan. He is from Kerala. He will cook differently. I have left him a note too, about what the crew has been eating and what they like."

"Rafiq."

"Yes."

"Thank you."

"You are welcome, Captain."

He held out the parcel. Sami took it.

"There is one more thing. My mother told me to tell you something."

"Your mother told you?"

"I told her about you. About what you said in the galley that day. About your mother and bread. She wanted me to tell you something."

"What did she say."

"She said to tell the Syrian captain that bread keeps, and so does a letter, and so does a mother. She said to tell him to come to Dhaka after Beirut if he has time. She will feed him."

Sami did not speak for a moment.

"Tell her I will come."

"I will tell her. Captain."

"Rafiq."

"Yes."

"You have fed a crew through a war. Most men do not get the chance. You do it as if it is ordinary. It is not ordinary."

"It is ordinary, Captain. It is just that most people who are fed have never been through a war."

Sami smiled for the first time in three days.

Rafiq went down the gangway. The car took him away.

* * *

By the evening of the sixteenth, the *Noor Salaam* had fifteen crew aboard. Sami, Petros, Tendai, Viktor, Somchai, Dani, Eko, the replacement cook who would arrive the next morning, and seven others who had chosen to stay. The ship had become quieter. The mess hall seated the remaining crew in half the chairs. The watches covered the necessary ground. The rest of the time, the men and women who had stayed read, or slept, or walked the deck.

Sami opened the Meadows essay that evening.

He had been told about Donella Meadows by a systems-thinking consultant whose workshop he had attended in Manila six years earlier. The workshop had been dull. He had not picked up the essay at the time. Now, with the ship empty and the blockade in place and the world he had spent seven weeks trying to understand arranging itself in front of him with unusual clarity, he read it properly for the first time.

Meadows's argument was simple. Any system could be changed, if you knew where to push. But not all pushing was equally effective. Meadows listed twelve leverage points, ranked from least effective to most effective. At the bottom of her list were the parameters of the system: the numbers, the subsidies, the tax rates, the quotas. These were the things governments usually adjusted, because they were easy to adjust, and because adjusting them produced visible changes in the short term. They were almost never the source of the problem. Above parameters were the buffers and stocks, the material structures of the system, the delays in the feedback loops. Above

those were the feedback loops themselves, the rules of the system, the information flows, the distribution of power, and the goals the system was optimising for. At the very top of her list, the highest-leverage intervention any human could make in any system, was the paradigm: the mental model, the set of assumptions, the shared understanding of what the system was for.

The paradigm, Meadows said, was where the whole architecture lived. Change the mental model, and the goals that flowed from it changed. Change the goals, and the rules changed. Change the rules, and the structure adjusted. Change the structure, and the numbers followed. A paradigm shift was the single most powerful change any reformer could achieve, and the single hardest.

Sami read her examples. Copernicus had shifted a mental model. The germ theory of disease had shifted one. The idea that slavery was not a natural condition but a choice enforced by law had shifted one. Each of these shifts had taken decades. Each had been resisted. Each had been delivered by people who were largely dead before the shift had completed.

The mental model of just-in-time, Sami thought, was the mental model of efficiency as a virtue. Holding reserves was waste. Flow was good. The supply chain would hold because the supply chain had always held. This way of seeing the world had produced a world in which every country, including the countries that had built themselves on resource extraction and on the movement of resources, held inventories measured in days rather than months. It had produced a world in which a strait thirty kilometres wide could starve forty percent of the world's aviation in three weeks. It had produced a Ravi on dialysis twice a week instead of three. It had produced a Mahmud reading his engineering textbooks without being able to sit the examinations. It had produced a Jun's neighbour in a petrol queue with her child.

The model would not change.

Sami understood this with the clarity he had been avoiding for a week. The model would not change, because the people who held it were the same people who had to authorise changing it. The ministers who had been trained in lean thinking. The treasury officials who had been taught to treat stockpiles as inefficiency. The consulting firms that had built their practices on optimising just-in-time supply chains. The business schools that had made the doctrine into a curriculum. The central banks that measured productivity through the lens of capital efficiency. Every institution capable of authorising the shift had been built by and was run by people whose careers had been made by the model as it stood.

Meadows had said the same thing, in different words. She had known paradigm shifts were the highest-leverage interventions. She had also known they were the ones nobody with power was willing to make. She had therefore proposed that the best a reformer could usually do was to work one or two levels below, on the goals, or on the feedback loops, or on the information flows, and to hope that these smaller interventions added up over time.

The smaller interventions, Sami thought, were the ones being announced now. The strategic petroleum reserve release. The emergency coordination meetings in Brussels. The promise to look again at aviation jet fuel reserves. A hearing in the American Senate on the vulnerability of the global supply chain. A white paper from the International Energy Agency. These were parameter adjustments and, at best, information flow improvements. Meadows had ranked them near the bottom of her list. She had also known they were what people did when they did not want to admit that the underlying model was the problem.

He closed the essay.

He wrote in his notebook for an hour.

Meadows ranked the paradigm as the highest leverage point in any system. She also ranked it as the hardest to move. The model that produced this crisis was the model of efficiency. Efficiency has been the organising principle of global supply chains for forty years. It will remain the organising principle for the next forty years, because the people with the power to change it are the people whose careers are built on it.

What will change is small things. A bit more Strategic Petroleum Reserve. A bit more coordination at the EU level. A new office somewhere with the word 'resilience' in its name. A report. Another report. A working group. Meadows knew. She told us. She told us in 1999, and we did not listen, because the model we had built was comfortable and the shift she was asking for was not.

The shift would be easier if the crisis lasted long enough to discredit the old model. This one will not last long enough. The Americans will negotiate, Iran will negotiate, a deal will be made, the strait will reopen, oil prices will settle, aviation will resume, and within a year most people will be telling each other that the system was broken but it is fixed now. It was not broken. It was built this way. The system delivered exactly what it was optimised to deliver, which was efficiency under conditions where no shock occurred. When a shock occurred, it delivered the predicted consequences of the trade-off that had been disclosed in every textbook that taught it.

The next big shock will land on the same system, slightly upgraded at the parameter level. Some reserves will have been built. Some agreements will have been signed. The underlying model will be unchanged. The vulnerability will be unchanged. The people who paid the bill this time, the Ravis and the Mahmuds and the Nambelas and the Jun's-neighbours, will pay the bill again. Some of them will not survive the second bill.

I do not know how to end this paragraph. So I will stop.

* * *

On the evening of the seventeenth, Sami told Petros.

They were on the bridge wing. The sun had set and the lights of the refinery had come on. Petros was smoking his one cigarette of the day, the one his wife in Athens had made him promise to reduce to after his last checkup. They stood against the rail and watched the lights.

"Petros."

"Yes."

"I am going to write a book about this."

Petros took a pull on his cigarette. He did not answer for a moment. He exhaled.

"Yes, Captain."

"That is all you have to say?"

"I have known you for fourteen years. You have been making notes since the twenty-ninth of February. You have been talking to the crew as if you were rehearsing how you would explain it to somebody else. You have been reading philosophers who do not usually end up in the cabins of ship captains. You have also started asking me questions about Athens that are not about Athens, they are about how Athens is organised, and I have noticed. I have been wondering when you were going to tell me."

"That obvious."

"That obvious."

"Will anybody read it."

"Somebody will read it. Whether enough somebodies will read it to make a difference is another question. You do not have an answer to that. I do not either."

"No."

"I have a question for you. Why are you writing it."

Sami considered the question.

"Because when I tell the story of these seven weeks to my grandchildren, if I have any, which I do not yet, I want to tell it with the facts I learned. Not just the story of the crew and the ship. The story of the supply chain that put us where we were. The story of the doctrine that built the supply chain. The story of the men who taught the doctrine and the men who followed the doctrine and the people who paid for the doctrine. I want my grandchildren to know why, when the next strait closes, the adults in the room were surprised, even though the analysts who should not have been adults in the room had been warning about it for twenty years. I am writing it so that my grandchildren do not have to learn the story from the adults who will have forgotten it."

Petros nodded.

"That is a good reason. The only problem with it is that your grandchildren will not read your book either. They will live in a world where people have forgotten. They will assume the adults knew what they were doing."

"I know."

"But write it anyway. There will be one grandchild. There is always one."

"Yes."

They stood at the rail for a while longer. Petros finished the cigarette and put it out in the small sand-filled can he kept for the purpose.

"Captain."

"Yes."

"I think you should ask Rafiq to read it first. He will tell you the truth about whether it is any good."

"I thought of that."

"Rafiq will not be kind."

"I thought of that also."

* * *

On the morning of the eighteenth, the news about the *Sanmar Herald* arrived before breakfast.

Arjun sent Sami a message from Thane at zero six twenty. The *Sanmar Herald*, an Indian-flagged VLCC, had been fired on by two Iranian gunboats at zero five local, six nautical miles off Larak Island, while attempting to transit the corridor that had taken the *Noor Salaam* through eight days earlier. The *Sanmar Herald* had been authorised. She had called in with the correct flag and call sign and cargo manifest. She had been hailed by Revolutionary Guard Naval Command and had confirmed her

clearance. Four minutes later, two gunboats had approached her from the east and had opened fire with light cannon on her bow and bridge. The ship had stopped. No crew had been killed. Two had been wounded. The gunboats had withdrawn without boarding. The *Sanmar Herald* had declared an emergency and was now dead in the water two kilometres inside Iranian territorial sea, waiting for a tug that Indian Navy was negotiating with Iranian authorities to allow.

Sami read the message. He did not move for a long time after he had finished. He read it again.

Petros came up to the bridge at zero seven. Sami passed him the phone.

"Read this."

Petros read it. He handed it back.

"We got lucky, Captain."

"We got lucky."

"The *Sanmar Herald* did exactly what we did. She followed the same procedure. She called in the same hails. She had the same clearance. She was fired on anyway."

"Yes."

"The clearance we received was not a clearance. It was a coin flip that came up our way."

"Yes."

Petros looked out at the anchorage. The lights of the refinery were still on in the early light.

"If they had fired on us the way they fired on the *Sanmar Herald*, we would have had two wounded instead of zero, and we would be sitting at anchor dead in the water with the IRGC waiting to decide what to do with us. And Jun would be on his way home instead of already home. And Liza would be growing up without her father."

"Yes."

"I have been grateful for the last week. I am more grateful now."

"So am I."

They stood at the bridge window. The company office called at zero nine. The message was that the *Sanmar Herald* had been successfully taken under tow by an Iranian vessel and would be escorted to Bandar Abbas for inspection and eventual release, timeline uncertain. The Government of India had lodged a formal protest. The wounded crew members were in stable condition. The Iranian Ministry of Foreign Affairs had issued a statement apologising for what it called a communication error between the Revolutionary Guard naval command and the patrol squadron. The Government of India had noted the apology and asked whether the communication

error was going to be a feature of the remaining clearances. The Iranian ministry had not yet responded.

Sami wrote in his notebook that afternoon for the last time before the passage closed.

The Sanmar Herald was fired on this morning. She had the same clearance we had. She followed the same procedure. She is in the same waters we transited. The only difference is the eight days between her passage and ours. Eight days, in which the internal politics of the Revolutionary Guard shifted, or the coordination between the IRGC and the Tehran government shifted, or some officer's tolerance for a directive he had received but chosen not to follow ran out. Whatever changed, changed between her ship and ours. The strait that let us through was not the strait that let her through. The strait changes by the hour. The ship that gets across is the ship that happens to be moving during the hour the strait permits her to move. The rest wait.

The Americans have closed the strait from their side. The Iranians have closed it again from theirs. The two closures overlap. Between them, the strait is shut. A hundred and forty-seven ships are still at anchor in the Gulf, and more have arrived since we left, and none of them are moving at the rate they were a week ago, and the cargo they are carrying is needed somewhere that does not yet know it will not arrive. The system has no feedback loop fast enough to convey, to the ministers and the analysts and the business school deans, the information that a new generation of Ravis is starting to form.

Meadows said the paradigm was the highest leverage point. I said earlier this week that the paradigm would not change. I want to correct myself. It will change eventually. It will change when enough shocks have landed on enough people that the cost of defending the old model exceeds the cost of abandoning it. That is how these shifts usually happen. Not because the evidence arrived, but because the evidence piled up so high that pretending not to see it became more expensive than acknowledging it.

I do not know how many shocks it will take. I hope it is fewer than I am afraid it will be. I am writing this down so that when the next one lands, somebody, somewhere, has a book on their shelf that said it was coming.

The ship will remain at anchor off Vadinar for the foreseeable future. I will remain with her. Petros will remain with her. Tendai will remain with her. The rest of the crew have gone home or are going home. When the strait reopens, we will go back in. Another anchorage will form. Another shock will be absorbed. Another book will need

to be written.

I hope somebody else will write it next time. I am tired.

* * *

* * *

PASSAGE 10

The Watch

On the morning of the twentieth of April, Sami opened a new notebook.

The old one was full. He had filled it over seven weeks, in the margins of every day, in his cabin at night, on the bridge wing at dawn, once in the galley doorway while Rafiq chopped onions and did not look up. The old notebook now sat on the corner of his desk, closed, with a rubber band around it to hold the loose pages in. He would keep it for the rest of his life. He had already decided that.

The new notebook was a hardcover one his mother had sent him for his fiftieth birthday, seven years earlier. He had never used it. He had been saving it for something he did not yet have a name for. That morning, on an empty ship at anchor off Vadinar, with the Indian sun coming up over the Gulf of Kutch and the lights of the refinery still burning against the grey sky, he understood what he had been saving it for.

He opened it to the first page. He wrote, in the small careful handwriting he had used all his life for the things he did not want to revise later, the sentence that he had been carrying without knowing he was carrying it.

On the twenty-eighth of February, the Noor Salaam dropped her anchor fourteen nautical miles south of Bandar Abbas, in the Strait of Hormuz, and did not move for forty-two days.

He looked at the sentence. He did not change it.

He wrote for an hour. Then he put the pen down, because he realised there was something he had to do before he could keep writing. He had to call the crew. One by one. Not to check on them. To ask them the questions he had not thought to ask while they were on the ship, and to hear the answers, because he could not write the book without the answers.

He went to the bridge. The satellite phone the company had authorised for extended use during the anchorage was on the chart table. He picked it up and dialled the first number.

The call to Jun went through at seven in the morning, Cavite time, which was nine-thirty in the Gulf of Kutch. A child answered.

"Hello."

"Good morning. Is this Liza?"

"Yes."

"This is Captain Kassar. I am calling from your father's ship."

A pause.

"You are the captain my father works for."

"I am."

"My father told me about you."

"He has told me about you too."

"What did he tell you."

"He told me you read the letter he sent you."

"Twenty times. Mama counted."

"That is a lot of times."

"It was a good letter. I have it in my room, under the pillow. I keep it there because I do not want to forget that it came."

"Do you want to put your father on, Liza? I would like to talk to him."

"Yes. He is outside. I will run and get him. Do not hang up."

"I will not hang up."

Sami heard the phone put down on a counter. He heard footsteps running. He heard a door open and a voice calling, and a man's voice answering, and footsteps coming back. Jun picked up the phone.

"Captain."

"Jun."

"You met my Liza."

"I did."

"She will remember you called. She will tell her classmates at school next week that the captain of her father's ship called her on the telephone. They will not believe her. She will not care whether they believe her."

"That is the right instinct."

"Yes."

There was a small silence on the line. Sami heard wind moving in the background.

"Jun. I am calling to ask a question."

"Ask."

"The night the Al Salmi burned. I told you to write to your wife. You wrote to her. She kept the letter. Liza read it twenty times. I have been thinking about that sequence. The advice I gave you came from somewhere. I have been trying to remember where. I cannot."

"You told me, Captain, in the engine room. You told me that a father who is afraid and still doing his job is someone a child can watch and learn from. A father who has gone quiet is not. Those were the words."

"I remember the words. I mean the source. Where the advice came from before it came from me. It did not start in my head."

Jun was quiet for a moment.

"My grandfather told me something like it, once. When my uncle was in the hospital and my grandfather was working on the boat. I asked him why he was not at the hospital. He said that if his son was going to live, it would not be because he was in the chair at the bedside. He said the boat was the way he stayed useful. He said being useful was the thing a son can see a father doing, and that the son would remember."

"Your grandfather told you this."

"Yes, Captain."

"And you carried it."

"I carried it. I did not remember carrying it. When you spoke to me in the engine room, I did not remember what my grandfather had said. I remembered it last night, when Liza was asleep, because I was thinking about the letter."

Sami did not speak for a moment.

"Jun. I am going to write a book about the seven weeks. About the strait and the crew and the supply chain. I am going to put you in it."

"I thought you might."

"Is that all right."

"It is all right, Captain. Write what you want. Put in what you want. I have one request."

"Name it."

"Put my grandfather in. Not by name. Just the part about the boat. The boat and the son. I would like him to be in a book."

"He will be."

"Thank you."

"Jun."

"Yes."

"Give Liza my regards."

"I will, Captain."

* * *

Sami wrote for two hours after the call with Jun. He wrote the story of Jun's grandfather and the boat, even though it did not yet have a place in the book. He knew it would find its place later, the way material did when you trusted it enough to keep it.

He called Rafiq at noon.

Rafiq's phone rang four times before he picked up.

"Captain."

"Rafiq."

"You caught me at the market."

"I can call back."

"No. I have my bag. I have everything I came for. What do you need."

"I am writing the book."

"I assumed."

"I have been thinking hard about the question of why I am writing it. I thought I had an answer. Now I am not sure. I want to ask you what you would write about these weeks if you were writing a book. Not what I should write. What you would write."

A pause. Sami heard traffic. He heard a vendor calling something in Bengali. He heard Rafiq walking.

"Captain. I am not going to write a book."

"I know. I am asking hypothetically."

"I am going to answer what I would answer if I were going to write a book. Is that what you want."

"Yes."

"I would write about my mother. And my grandmother. Not because they are interesting. Because they are the people I was cooking for on the ship."

"You were cooking for them."

"I was cooking for them. The crew was in front of me and the crew ate the food. But I was cooking for my mother and my grandmother. They had taught me. They taught me by making food for other people for fifty years each. The time I was cooking on your ship, I was finally doing the thing they had been trying to teach me. I

did not notice I was doing it until I had been doing it for three weeks."

"And now."

"Now I am home, and my mother is cooking for Mahmud, and I am watching her. I am the one being cooked for. I do not mind. She has been waiting a long time to cook for me again. I will go back to the ship in three weeks and I will cook again. But right now my job is to eat her food, and to tell her it is good, and to let her feed me the way she used to."

"Rafiq."

"Yes."

"You said something to me in the galley. You said that talking about the cooking would make it into something it was not. I have been trying to understand that sentence for two weeks. I do not think I understand it yet."

"It is simple, Captain. When you talk about a thing that is just what you do, you make it a performance. The food stops being food and starts being a story about food. Stories about food are for other people. The food is for the people who eat it. If I had started to talk about why I was cooking, I would have been cooking for the story. I was not. I was cooking for the crew. Same for my mother. She does not say why she cooks. If she started saying why, she would stop cooking for Mahmud and start cooking for whoever was listening to her."

"That is a distinction I need to write down."

"Write it down, Captain. But do not put it in the book."

"Why not."

"Because putting it in the book makes it a story about food. Do you see."

Sami laughed, once.

"I see."

"Put in my mother. Put in the cyclone in 2007. Put in the neighbours who came to the door. Do not put in what I just told you about why we were cooking. That is between you and me."

"All right, Rafiq."

"Captain."

"Yes."

"My mother still wants you to come. Bread keeps. So does a letter. So does a mother. She said it again this morning, without my prompting. She asked me when I was going to ring you."

"Tell her I will come. After Beirut. It will be in June."

"June is good. June is mangoes."

"I have not had a Dhaka mango in twenty years."

"You have been missing out."

* * *

The call with Arjun was on the evening of the twenty-first.

Arjun answered on the second ring. He was in a corridor of the hospital in Thane. He walked out to the car park to take the call.

"Captain."

"Arjun. How is he."

"He is in the chair. Three hours to go. He is asleep. Mother is with him. I came out to talk."

"How has he been."

"He is better than he was when I arrived. He had dropped seven kilos in the weeks before the fuel came back. He is slowly putting some of it back on. The doctor says the long-term kidney function may or may not recover. We will know in a month. He does not want to talk about that."

"Understandable."

"He is talking about everything else. His old stories. His work. His wife, who died eight years ago, who he had not talked about since her funeral. It is as if being close to the end reminded him of what had been there before the end. I have been taking notes. Not in front of him. Afterwards. I want to have the stories when he is gone. Even if he is not gone for a long time."

"That is wisdom."

"I got it from you, Captain."

"How."

"You told me once on the ship that the calling was more important than what was said. I thought it was about keeping someone alive. Now I see it was also about remembering them. If I had not been calling every night, I would not know the stories he is telling now. Mother had heard them. I had not. The calling was the way I got to hear them."

"Arjun."

"Yes."

"Your mother was right to pick up for you. I want to tell you that."

"She told me you said that. I did not fully understand it until I saw her with him in the chair. She told me, the day I arrived, that there had been nights during those weeks when she had thought she would not be able to keep going. She said hearing my voice

had been the thing that had kept her from going under. She said this in the same flat way she says everything. I did not know what to do with it. I still do not."

"You do not have to do anything with it. You already did it. The doing was the phone calls."

"Captain."

"Yes."

"I have a question I have been wanting to ask you for a week. Ravi wanted me to ask it too."

"Ask."

"Will you come to Thane."

"Yes."

"When."

"Let me finish the book. Then I will come. It will not be more than a year."

"Ravi may not be here in a year."

"I know. I will come sooner if he is not. I will come in July either way. Can your mother hold a guest for a few days in July."

"She has already decided she will. She told me yesterday she was going to invite you. I was going to ask at the right moment. Now I have asked."

"I will be there in July."

"Thank you, Captain."

"Arjun."

"Yes."

"The samosa story. The one where you told Ravi the cargo was running the generator. Can I put that in the book."

"Please put it in the book."

"Thank you."

* * *

The call with Somchai was on the morning of the twenty-second.

Somchai was in his kitchen in Bangkok. Sami heard a teapot whistling in the background.

"Captain."

"Somchai. You are home."

"I am home. Anan came yesterday. He is sleeping now. He slept twelve hours."

"He needed it."

"He did. The friends from the Mayuree Naree were supposed to come for dinner tomorrow. The one who does not want to go back to sea has cancelled. He says he cannot face a room full of people yet. The other three will come. Anan is glad they will come. He is not angry at the one who cancelled. He said he would have done the same, if he had been the one held on Qeshm."

"How is Anan."

"He is Anan. He laughs at things. He eats. He drinks his mother's tea. Yesterday he took me fishing, which is something we used to do when he was a boy. He was a boy the last time we went fishing. Yesterday he was a man taking his uncle fishing. I did not catch this reversal until we had been on the boat for an hour. When I caught it, I did not say anything. I let him take me fishing."

"Somchai."

"Yes."

"I have been thinking about the name thing. The rule. The one you said you had broken."

"I remember."

"I want to tell you something. The rule is not a rule. I invented it. I invented it ten years ago after a young crewman on my third ship disappeared off the deck in a storm and I had not known his family's names. I had not known anything about him except his job. I told myself after that, that on my ships I would know the names. It was for me, Somchai. It was not for the crew. When you told me you had not said Anan's name because you had been superstitious, I wanted to tell you the rule was a personal rule I had extended to the crew without explaining why. I did not tell you because it seemed too small a correction to make, and because you had needed the absolution. I am telling you now because you called it a rule and it was not a rule."

"It was a rule, Captain. You made it a rule. Any captain can make a rule for his ship."

"I suppose so."

"I would rather it be a rule. A rule is easier to follow than a preference. A preference bends with the weather. A rule does not. Keep it a rule."

"All right, Somchai."

"Captain."

"Yes."

"Anan woke up. He is standing in the kitchen. He wants to say hello."

"Put him on."

Somchai handed the phone to Anan. Anan's English was good. He introduced himself. He thanked Sami for what he called his uncle's long friendship, which was more formal than it needed to be, and then Anan laughed at himself for saying it that way, and said he had been practising the phrase because he had known Sami might call. Sami laughed with him. They talked for a few minutes. Anan said his friends from the Mayuree Naree had all asked how Somchai was, every day on the flight home, and that Somchai was a figure in their stories now. Sami told him Somchai had been a figure in the story of the *Noor Salaam* too. Anan said he was glad of that. He passed the phone back.

Somchai said, "Captain."

"Somchai."

"Write the book."

"I am writing it."

"Put Anan in it."

"I will."

"Put his friends in it too. The ones who came home. The one who did not come to dinner."

"I will."

"Good."

* * *

On the evening of the twenty-third, Sami called his mother in Beirut.

She picked up on the fourth ring, which was her usual ring count, because she did not like to seem too eager.

"Saeed."

"Mother. It is your other son."

"I know who it is. I recognise your breathing. What is wrong."

"Nothing is wrong. I am calling because I can."

"That is a reason. I will accept it."

He heard her moving something in the kitchen. She was probably putting the kettle down. She made tea every time any of her children called, and drank the tea while they talked, and used the tea as a prop for the silences she did not know how to fill.

"Mother."

"Yes."

"I am coming home."

"When."

"In three weeks. After the ship is handed over to the next captain. I have written the company. They are sending a captain to relieve me. I will fly out on the fifteenth of May."

"Three weeks."

"Yes."

"That is a long time."

"I know. I wanted to tell you now, so you have it to wait for."

"Saeed."

"Yes."

"You have been gone a long time."

"I know."

"I am not asking why. I am saying it. You have been gone a long time."

"Yes, Mother."

"I will be here when you come."

"I know."

"Bring me something from the ship. Something small. I have been wanting a small thing from your ship for a long time. I do not know why."

"I will bring you something."

They did not speak for a moment. He heard her drinking her tea.

"Saeed."

"Yes, Mother."

"Eat something before you hang up. You sound thin."

"I will eat something."

"Good."

She hung up. She always hung up first. It was her own form of keeping the upper hand in conversations with her sons, and Sami had long ago stopped resisting it.

He put the satellite phone down on the chart table.

He did not pick up the notebook for a while.

* * *

On the morning of the twenty-fourth, Sami walked the deck at five.

The watch that morning was Tendai's. Tendai had resumed his duties the day he returned from Chipata, and he was on the foredeck at five, checking the hawsepipe for wear, because a ship at anchor for two weeks put strain on different parts of the chain than a ship underway, and a bosun worth his pay checked those parts daily.

Tendai saw Sami approach.

"Captain."

"Tendai. All well on the forward deck?"

"All well. The chain is in good order. The port windlass motor is running slightly hot. I have logged it. The next chief engineer will look at it when the relief comes."

"Good."

They stood at the rail. The sky in the east was beginning to turn.

"Tendai."

"Yes."

"When I hand over the ship to the next captain on the fifteenth of May, you will be the one who briefs him on the foredeck. The anchor chain, the windlass, the winches, the state of the paint, the places that need attention. He will come aboard and you will walk him around and you will tell him what he needs to know to keep this ship as well as we have kept her."

"Yes, Captain."

"That is a watch handover."

"It is."

"Every ship is a line of handovers. From the captain who built her to the captain who scrapped her. The ship is continuous. The men on her are not. The handovers are what keep the ship herself. A ship without handovers is a ship that forgets. A ship that remembers is one where every watch is passed carefully to the next."

"Yes, Captain."

"The book is my handover."

Tendai looked at him.

"The next captain will not know what happened on this ship in February and March. He will not know why the crew is the way it is. He will not know why Krishnan is cooking for us instead of Rafiq, or why Jun wrote a three-page handover note that will annoy the relief engineer, or why Somchai keeps a record of names in the back of his bosun's ledger. He will not know. The book is how I will tell him, and the readers who are not him, what happened here. So that when they are on ships of their own, in crises of their own, they will have been handed over to."

Tendai nodded slowly.

"Captain. When you write my chapter."

"Yes."

"Write it for Nambela."

"I will."

"Write it so that when she reads it, she understands that what her brother did on this ship was what her mother had been doing in Chipata for ten years. That it is the same work, at a different scale. That she is not alone in her plot when she is the one planting it."

Sami looked at him.

"Tendai. That is the sentence I needed for the chapter."

"Write it down, Captain."

"I will. I have something to do this morning first, and then I will write it down tonight. It will go in as you said it."

"Tonight is fine."

Sami stood at the rail with him for another moment. Then he went back to his cabin, and he wrote the sentence down in the notebook on a fresh page, where it would wait until night, and he closed the book. The sun was coming up over the Gulf of Kutch. He had a call to make.

* * *

At nine o'clock in the Gulf of Kutch, which was seven in Chipata, Sami picked up the satellite phone again. Tendai had told him on an earlier call that his mother woke early, and that if Sami wanted to speak to her, the morning would be the time.

Sami wanted to speak to her. He had been building up to it for two days.

Tendai answered on the first ring.

"Captain."

"Tendai. How is the planting."

"The second half is in. All of it, now. The rains have not started yet but they will soon. My mother is pleased."

"How is Nambela."

"She passed the test she took last week. She wanted me to tell you. She also wanted me to tell you that she has decided she will do mathematics at a level higher than is required for the nursing school. She says it is because the nursing school will be easier if she has already done the hard mathematics. I think it is because someone told her once that a girl like her would not be able to do it, and she has been wanting to prove him wrong for two years. She will not say which one it is."

"Probably both."

"Probably both."

"Tendai. I am calling because I would like to speak with your mother. I know she does not speak English. You can translate. I will keep the questions short. I understand

if this is too much to ask."

"It is not too much to ask. She has been expecting you to call. She has told me twice in the last week that she would not be upset if the Captain wanted to speak to her."

"Twice."

"Twice. The second time was yesterday. She was specific about the hour. She said she would be in the kitchen between seven and eight, and if you called then, she would be ready."

"She was ready for this."

"She has been ready, Captain."

Sami heard Tendai walking somewhere in the house. A door opened. He heard a voice in the background, a woman's voice, speaking Nyanja. Tendai answered in the same language. There was a short exchange. Then Tendai's voice came back.

"She is here. I will hold the phone between us and translate both ways. She says good morning."

"Good morning, Mrs Banda. Please thank her for taking this call. And please tell her that her son has been one of the finest young men it has been my privilege to sail with."

Tendai translated. Sami heard the mother respond. Her voice was low and steady. It had the rhythm of someone who chose her words carefully and did not fill the gaps between them.

Tendai came back.

"She says she is glad you think so. She says she raised him as best she could, but that a son becomes a man at sea, not at home, and that she is grateful to the men who took him the rest of the way. She is including you in that."

"Tell her the thanks is mine. He became a man on our ship because he was already nearly one when he arrived."

Translation. The mother laughed once, quietly. Response.

"She says that is a kind thing to say, and that she will not argue with a captain about his own crew."

"Tendai."

"Yes."

"Ask her, please, how the weeks were for her. While you were at sea. I would like to understand."

A pause. Tendai translated. Sami heard her speaking for longer than before. When Tendai came back, he spoke slowly, getting it right.

"She says the weeks were long. She was afraid more than she slept. She did not tell Tendai she was afraid, because a mother does not add to her son's burden when her son is on a ship. She told Nambela she was not afraid, and Nambela believed her, and that was what mattered. She says she woke every morning and she went to the plot, and she worked the half she had planted, and she watched the half she had not, and she prayed for both halves equally. She says a mother can hold a lot, but she does not hold it alone, even when she looks like she is."

Sami was quiet for a moment.

"Ask her who was holding it with her."

Translation. The mother spoke more slowly this time. Tendai listened, nodded, said something short back, listened again, then translated.

"She says she will tell you a thing her grandmother used to say. She wants to say it to you in her own language first. She says you will not understand the words but you should hear them."

"Yes. I would like that."

A pause. Then her voice, unhurried, with the measured rhythm of something said many times before.

"Kali kokha nkanyama. Aliawiri ndi anthu."

Tendai translated.

"Alone, one is an animal. Two, they are people. She says her grandmother said it when Tendai's father died. She said it to her own children, who were small then. She has said it to Tendai and Nambela many times over the years. She is saying it to you now because she thinks you will understand it, and because she has been thinking about it during these weeks."

"Thinking about it."

Translation. Mother spoke.

"She says she was thinking about it because she realised, during the weeks Tendai was at sea, that she had not been alone. The people she thought were holding it with her were not all in her house. Some were in her house. Nambela was. Her neighbour was. The woman who sells her the maize seed was. But there were others she had not been able to see. She says you were one of them, Captain. She did not know your name until Tendai came home and told her. She only knew that there was a man on the ship her son was on, and that man was doing his best by her son. She says she was holding her half of the plot, and you were holding the other half, even though you did not know that was what you were doing."

Sami did not trust his voice for a moment.

"Tendai."

"Yes."

"Please tell your mother that I did not know I was doing that. That I am grateful she would tell me. And that she was holding her half of the plot for me also, without knowing she was doing that."

Translation. The mother was quiet when Tendai finished. Then she said one word, slowly, and Tendai translated it.

"She says: yes."

Sami let out a breath.

"Tendai. I have one more question. May I ask it to her directly."

"Ask it."

"Mrs Banda. When I write this book, I would like to include the thing your grandmother said. The words you said to me in your own language. I would like to put them on a page of the book in the way you said them. Is that something you would allow."

Translation. The mother did not pause. She responded immediately, and the response was longer than Tendai had expected, because he had to ask her to repeat part of it.

"She says yes. She also says that she wants to add one thing to the book, if you will permit it. She says that when you write the proverb down, you should write it as her grandmother wrote it, and that her grandmother was a woman called Esnatch Phiri, who lived in a village north of Chipata, and who died in 1978, and who had fourteen children of whom nine lived. She says the proverb is a proverb of the Chewa people, but the grandmother who gave it to her was a particular person, and the person should be remembered, because proverbs do not come from nowhere. They come from women who sat on overturned buckets at the edges of plots and watched their daughters work, and who said what they said so that their granddaughters would say it too. She says if the proverb is to be in the book, Esnatch Phiri should be in the book with it."

"Tell her Esnatch Phiri will be in the book."

Translation. The mother said something short. Tendai laughed quietly.

"What did she say."

"She said thank you. Then she said that if you are going to put her grandmother in the book, you should also put in the fact that the grandmother made the best *ndiwo* anyone in the village remembered making, and that her recipe is written in her granddaughter's handwriting in a book on the shelf in this kitchen. My mother will not

give me the recipe. She says I will get it when she decides it is time."

"Please tell her that if her grandmother made the best *ndiwo*, then her grandmother belongs on the same page as Rafiq's mother, who fed her block during the cyclone in 2007."

Translation. The mother's response was short.

"She says she would like to meet Rafiq's mother one day. Not in this life, probably. But one day."

"One day."

A small silence.

"Tendai. Can you ask her if there is anything she wants from me before we hang up."

Tendai translated. The mother answered quickly.

"She says yes. She wants a photograph. Of the ship. Of the anchorage where you were. Not of the sea, which she can imagine. Of the deck, she says. Where her son spent the days. She wants to see where he was standing during the hours she was on the plot."

"I will send it. Tendai has my address. I will post a photograph. I have one of the foredeck that your son himself took during the anchorage, and I will send that one."

Translation. The mother said something. Tendai translated.

"She says a photograph her son took is the right one. She says she knew the captain would choose the right one."

Sami smiled.

"Tell your mother thank you. For everything. For the proverb, for your grandmother, for raising your son, for holding her half of the plot, and for trusting me with the photograph."

Translation. The mother said one thing more. Tendai held the phone for a moment and then translated.

"She says her last thing is this. She says: *Chikuni chimodzi sichipsetsa mphika*. A single log does not cook the meal. She says she has been cooking all her life and she knows this. She says you should keep it with you, because she thinks you are a man who sometimes forgets that he is not the only log in the fire. She says she means this kindly."

"Please tell her I will keep it with me."

"She knows you will. She is handing me the phone now. She says she is going back to her cooking."

Sami heard the phone passed. He heard the mother moving in the background. He heard Tendai breathe out, the way a son breathes out when he has just watched his mother do something he did not know she was capable of.

"Captain."

"Tendai."

"That was more than I expected."

"It was more than I expected also."

"She has not spoken that long to anyone outside the family in a year. I do not think even my father's brother has heard her speak that much at once. She chose her words for you, Captain. I hope you heard them the way she meant them."

"I heard them."

"Good."

"Tendai. I am going to end the book with what your mother just said to me. If that is all right with her and with you."

"It is all right with her. I will tell her. It is all right with me."

"I am going to go and write now."

"Write it well, Captain."

"I will try."

* * *

Petros came up to the bridge wing that night and stood beside Sami without speaking for ten minutes, which was their standard procedure for being together without being in the way of each other's thinking. Then Petros asked, "How is the book."

"The first chapter is finished."

"What is the first sentence."

Sami told him.

Petros repeated it back to himself, quietly, the way he repeated courses back when they were given to him at the helm.

"That is a good sentence."

"Thank you."

"Captain."

"Yes."

"When I asked you last week why you were writing it, you said it was for your grandchildren. You told me that was a good reason. I agreed it was a good reason. I did not tell you what I thought the better reason was."

"Tell me now."

"The better reason is the crew. Not your grandchildren's crew. This crew. Every man who was on this ship for those forty-two days is going to go back to sea sooner or later. They are going to go back with what they learned. They cannot explain what they learned, because sailors do not explain things. They will carry it. When they are helmsmen on other ships, or chiefs on other engines, or bosuns on other decks, they will be slightly different men than they would have been. Your book is the one place where what they learned will be written down in words they did not have. You will be writing the language they were not able to write."

"That is a generous way to put it."

"It is accurate, not generous. Sailors do not write books. Captains do, sometimes. When a captain writes a book for his sailors, he is doing a kind of work that is not in any contract. You are doing that kind of work."

"Petros."

"Yes."

"You are in the book also."

"I assumed."

"You are not surprised."

"I have been helping you think about it for fourteen years. I am going to be in the book. I expect that. I am not going to read the parts where I appear. I will trust you."

"Why not read them."

"Because I prefer to be surprised by myself. When I read other people's descriptions of me, they never match my description of myself, and I do not want to have to reconcile the two. Leave me unread. I will enjoy being in the book without knowing what I said."

"All right."

They watched the lights of the refinery for a while longer. The night was clear. The moon had risen over the coast. The *Noor Salaam* rolled gently at her mooring. Somewhere below them, in the mess hall, the reduced crew was eating the dinner Krishnan had made. Krishnan's cooking was different from Rafiq's. The crew had noticed. Somchai had told Sami, before he had left, that Krishnan was a good cook but he would not cook for the crew the way Rafiq had. Rafiq had cooked for them. Krishnan would cook for the ship. There was a difference, Somchai had said, and Sami would feel it when Krishnan started.

Somchai had been right. Sami had felt it. Krishnan's food was good. It was not Rafiq's food, because Rafiq's food had been made by a man who had been cooking for twenty-two people who had become his mother's block during a cyclone. Krishnan

had not yet been on a ship during a crisis. One day he might be. Then his cooking would change, too.

* * *

Sami went back to his cabin and wrote for four hours that night.

He wrote the chapter on Ubuntu. He had been carrying it in pieces since the first night of the anchorage, when he had stood at the window of his cabin with the ship just stopped and the Iranian coast visible in the distance, and had reached for a concept he had first encountered years ago as a student, and had come across again over decades in port cities along the east coast of Africa. *I am because we are*. He had liked it then as an idea. On night one of the anchorage, it had stopped feeling like an idea and started to feel like a description.

He had left it there in his notebook for most of seven weeks. He had not known how to write the chapter. He had not known what could stand alongside the word and carry its weight in a book written for sailors.

Now he knew. Two things would stand alongside it. One had been given to him by Viktor weeks earlier. One had been given to him by Tendai's mother that morning.

He wrote Viktor's speech from the foredeck first. The sentence he had heard Viktor give about the first winter after the Russian gas stopped, and the schedule of the four-hour blackouts, and the card games by candlelight, and the quiet dinners. Viktor had said that the sanctions had not changed the government; the sanctions had changed the people; and the people would remember. That sentence had lodged in Sami's mind for six weeks without his knowing it had lodged.

It took Sami until after midnight to see why. What Viktor had been describing was not only sanctions. Viktor had been describing a climate. A climate in the meteorological sense, which was what a population lived inside without noticing it lived inside anything. The four-hour blackouts were not weather. Weather was what happened to one family on one night. The four-hour blackouts for a whole winter, the quiet dinners, the card games by candlelight, the schedule that became a commute, those were a climate. A country living inside a climate discovered, over time, what the climate had made them become.

Sami wrote it down, in the quiet cabin of the empty ship. A person, he wrote, was a weather system. A crew was a climate.

He rested his hand on the notebook and thought about what he had just written.

A weather system was a thing in its own right. It had its pressure and its temperature and its winds. It had its own history, its own causes, its own

consequences. But a weather system was not an independent thing. It was part of a climate. The climate was the sum of a thousand weather systems, interacting over a hundred years, producing the conditions that each individual weather system rose and passed within. A weather system could not understand itself without understanding the climate. A climate could not exist without the weather systems that composed it.

A person was a weather system. A crew was a climate. And what a crew taught a person was that the person's weather was not only the person's weather. It was also the climate's weather. The climate shaped the person, and the person shaped the climate, and neither could be fully described without the other.

That, Sami wrote, was what the word Ubuntu meant in the plain language of seamen.

Then he turned the page.

On the new page, he wrote the second of the two things that would stand alongside Ubuntu in the chapter. He wrote what Tendai's mother had said that morning, and he wrote the story of her grandmother Esnatch Phiri, who had lived in a village north of Chipata, and who had died in 1978, and who had made the best *ndiwo* in her village, and whose proverb had been carried forward by her granddaughter, who had repeated it at the kitchen table of a small house in Chipata to a Syrian captain on a satellite phone eighteen hundred miles away.

He wrote the proverb first in the language in which Tendai's mother had said it to him, because those were the sounds in which it had been made.

Kali kokha nkanyama. Aliawiri ndi anthu.

Alone, he wrote on the line below, one is an animal. Two, they are people.

He looked at the two sentences on the page. One in a language he did not speak. One in a language he had been thinking in for forty years. The two sentences said, between them, what his chapter had been trying to say.

A person was a weather system, and a crew was a climate. A person alone was an animal, and two were people. These were not two different claims. They were one claim, arriving in two languages, separated by eighteen hundred miles and by fifty years and by a ship and a plot of land and a strait that had closed and opened and closed again. The claim was older than Ubuntu and older than the proverb of Esnatch Phiri. It had been travelling in the mouths of people who had never met each other for as long as people had lived in groups. It had arrived at him because he had been on the right ship with the right crew, and because a woman he had never seen had chosen to say one sentence to him on a morning she had been preparing for.

He had not written the book to prove a thesis. He had written the book to record a climate. The Ravis and the Mahmuds and the Nambelas and the Liza-and-Miras were weather systems. So were the Jankovecs and the Walshes and the treasury officials and the business school deans. Every one of them was doing what the climate had taught them to do. If the climate did not change, the weather would not change. If the weather did not change, no individual Ravi would be saved the next time.

The climate, Sami wrote, was the thing the book was trying to teach the reader to see.

And the proverb of Esnatch Phiri, spoken to him in Chewa by a woman he had never met on the morning of the twenty-fourth of April, was the sentence with which the book would end. Because it had been earned by a mother on a plot of land, and not by him, and because it had been given to him by a crewman's mother in a language she had chosen carefully, and because it said in eight words what the book had taken him seven weeks and forty thousand to circle.

He closed the notebook at midnight. The chapter was not finished. He had not yet written the paragraph about Esnatch Phiri's *ndiwo*, which he would write in the morning, and he had not yet connected her proverb to Rafiq's mother's cyclone, which was a connection Tendai's mother had already made for him. But the turn the chapter needed to make had been made. The rest would come.

* * *

There were many things still to do. The book would take a year, maybe more. The crisis would resolve and recur. The strait would open and close and open and close. The Ravis and the Mahmuds and the Nambelas of the next crisis would appear, and some of them would not survive it, and some of them would, and the ones who survived would be the ones who had been carried by their climates even when their climates were thin and tired.

Sami would go home to Beirut. He would see his mother. He would give her the letter. He would eat what she made him and listen to her talk about people he had been gone too long to remember. He would stay three weeks and then he would go to Thane and see Ravi, if Ravi was still there. He would go to Dhaka after Thane. He would eat mangoes with Rafiq's mother. He would hear the stories she had been waiting to tell. He would write them down.

Then he would go back to sea. His next command would be a different ship, with a different crew, and he would learn their names the way he had learned this one's, and he would remember Ibn Khaldun, who had taught him that civilisations rose and fell

on the strength of their *asabiyyah*, their bond of togetherness, and that every vessel, whether a ship or a nation, was held together by a thing older than its machinery. He would remember Lao Tzu, who had taught him that the softest thing made the deepest cuts. He would remember Sun Tzu, who had taught him that prolonged warfare made nothing rich. He would remember Confucius on the absence of courage, Seneca on preparation and negligence, Taleb on fragility and what came after it, Meadows on the points of leverage, and Ubuntu, which had taught him everything the others had taught him in one word.

He would remember the crew of the *Noor Salaam*.

And he would keep writing, because a book was a watch handover, and every watch deserved to be handed to the next man carefully.

* * *

On the morning of the fifteenth of May, Sami walked down the gangway at Vadinar for the last time as captain of the *Noor Salaam*.

The relief captain was a Greek named Kostas, whom Petros had known for twenty years. They had served together on a ship in 2004. Kostas shook Sami's hand at the bottom of the gangway. Petros walked him aboard.

Tendai was on the foredeck. He would brief Kostas on the forward gear when the handover round began. Krishnan was in the galley. The reduced crew was at their stations. The *Noor Salaam* was a working ship again.

Sami stood on the quay with his bag. The company car was coming in twenty minutes. He looked up at the ship. She was a good ship. She had carried them through a crisis that nobody on board had been fully ready for. She had not asked anything of her crew that they had not been able to give.

Petros came back down the gangway. He stood with Sami.

"Captain."

"First mate."

"Write the book."

"I will write the book."

"Make sure the first sentence does not change."

"It will not change."

"Keep the watch."

"I will keep the watch."

They did not embrace. They did not need to. They had been on a ship together for fourteen years, and the ship was their embrace, and the handover was their goodbye.

The car came. Sami put his bag in the boot. He looked up at the *Noor Salaam* one more time. Then he got in.

The car drove up the access road. Sami looked out the rear window. The ship grew smaller. The refinery grew smaller. The Gulf of Kutch opened and then closed behind the coast.

He turned around and faced forward.

He took the notebook out of his bag. He opened it to the last page he had written. He read what he had written there. He underlined one sentence, small, with the pencil he kept in the spiral.

Kali kokha nkanyama. Aliawiri ndi anthu.

He closed the notebook.

The road went on.

* * *

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* * *

Epilogue

This is the third book I have written about the same problem.

I did not know it was the same problem when I started. I thought I was writing three different books about three different subjects, and I only understood that I was tracing one shape when the third book began to announce itself on a news cycle I was watching with the same attention I had given to the first two.

The first was *The Cost of the Machine*. I wrote it because I had been working with artificial intelligence every day for years, and I had watched the pace of the science accelerate in a way that most people outside the field had not yet registered. The systems were getting more capable faster than we were building the institutions to govern them. Alongside AI, quantum computing and biological computing were advancing on paths that would, within this decade, produce tools so powerful that the people who had them would be able to do things the people who did not have them could not. I wrote the book because I was worried that the gap between those two groups was about to become something larger than a gap. It was about to become a separation, in the way that gods and mortals are separated in older stories. One group shaping reality. The other group living inside it. This is untenable.

The second was *The Referendum*. I wrote it because I had been looking at the resource chains that run out of Africa, and had come to the same conclusion a great many Africans had come to long before me, which is that the continent is not poor. The continent has been priced poor. Africa's resources are real. Her people are real. The labour and the knowledge and the earth all exist, and the reason the continent stays behind is not a shortage of what is valuable. It is a cycle. Resources are extracted at a price the extraction country does not set. They are processed somewhere else, where the value is captured by people who had nothing to do with the extraction. They are packaged, branded, marked up, and sold back to the continent they came from at prices the continent pays in currency it had to borrow. The borrowing becomes debt. The debt justifies the next round of extraction. The cycle tightens. *The Referendum* was written because this cycle is not an accident and it cannot be fixed with a single intervention. The inequities have to be addressed at every stage of the chain: at extraction, at processing, at the middleman's markup, at the final sale, and in the debt arrangements that hold the whole thing in place.

The third book is the one you have just read. I did not plan to write it. I was watching the news, the same as everyone else, when the strait closed on the twenty-eighth of February, and I began to see the same shapes I had spent three years tracing in the other two. The weapons referred to in *The Cost of the Machine* were present here. The resource-chain inequities documented in *The Referendum* were present here. What was new was the immediacy. The first two books were written about trends that would arrive fully within a decade. This book was written about something that is happening now, on a timeline of weeks.

Three books. The same argument seen from three different angles. Each one asking the reader to come to roughly the same place. Who holds what. Who cannot act because of what they cannot see. Who has been priced out of the decisions that shape their lives. What the cost is when these things continue unexamined.

* * *

As I finished this book, I was preparing for a trip.

The itinerary reads: New York, India, Indonesia, Singapore, Hong Kong. Westbound. Across the Pacific. It is not the route I would normally fly. The usual route for someone going from the East Coast of the United States to South Asia and onward runs east through Europe and then down through the Gulf, with a connection at Dubai or Abu Dhabi. That route is not viable right now. The airlines that used to connect through the Gulf have been reducing flights, rerouting others, and quietly

declining to quote me a fare on the ones that are still operating. So I am flying the long way. Against the direction of the earth's rotation, into a sequence of cities that all sit inside some version of the crisis this book has just described.

Each of those cities carries something the reader has just watched unfold in the passages.

In India, petrol stations are limiting purchases at the pump. LPG cylinders for home cooking are being rationed at the shops. The cost of fertilizer is being openly debated in the press because the next planting season is already being discussed in terms that, until recently, were only used in years with actual drought. Some of what I will hear there I have already written.

In Indonesia, the government has made public assurances about the country's fuel reserves while quietly accelerating discussions about what rationing would look like if the strait closure continues. The fishing villages on the north coast of Java, the ones Dani in this book is from, are real. Their boats, and the diesel that powers them, and the families they feed, are real.

In Singapore, the refinery economy that sits at the centre of the region is operating with increased cost of crude and tightened margins. The city is running. The city will keep running. Singapore has always been good at sitting at the centre of storms. That is a kind of lesson, and I intend to listen to it.

In Hong Kong, a city that already knows what it means to live inside a disruption it did not choose, I will hear things I will not repeat, because they were told to me in confidence, and because some of the people I speak to still need to work tomorrow in the systems they are describing to me today.

I will learn. I will write about what I learn. Whether it becomes another book or an essay or just the private education of a man who wanted to understand what he had been writing about, I do not know yet.

* * *

I should say one more thing honestly.

In some of the places I am going, I have given thought to how visible I want to be.

I do not mean physical safety. The cities are safe enough. I mean something quieter. In a crisis, people often take their anger out on whatever face of the cause they can reach. The face they can reach is rarely the face of the actual cause. The causes of a crisis like this one are spread across governments, industries, and decisions made decades ago by people who are no longer alive. The face in front of a suffering person is usually a visitor, a stranger, someone who looks or sounds or carries documents that

mark them as associated with whatever the suffering person has decided is to blame.

I have thought about what to say in which city. I have thought about what to wear and what to carry and what to keep out of sight. I have thought about which parts of my work it is useful to talk about and which parts it is better to leave unmentioned until the conversation has found its own balance. These are not large compromises. They are small, ordinary, the kind of adjustments thoughtful travellers have always made in places under strain. But I notice that I am making them, and I notice why I am making them, and I offer the observation because it is part of what this book is actually about.

I wrote a book that says we are all on the same ship. I am about to fly through what that means when the ship is stopped.

* * *

Kali kokha nkanyama. Aliawiri ndi anthu.

One alone is an animal. Two together are people.

My grandmother said this, in Chewa, on a small farm north of Chipata, many years before this book was written. She has been gone since 1978. The proverb is older than her. It will be older than this book. The wisdom we need is almost always already available. What has been missing, and what is still missing, is the willingness to listen to it, across the borders we have drawn and the languages we have stopped learning.

I hope the next ship you board carries you, and the people next to you, where you all need to go.

Pumulo Sikaneta

April 2026

* * *

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The five appendices that follow are reference material. They can be read in order, consulted as needed, or skipped by the reader who wants the narrative to stand on its own. Each one exists because a reader who finishes the book deserves access to the facts the book rests on.

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Appendix A — The Philosopher's Guide

This book draws on eight thinkers. They are not here to decorate. Each one is named because a specific insight of theirs helped Sami make sense of something happening on the *Noor Salaam*, and that insight is offered to the reader in the form it arrived, through the crisis rather than the classroom.

What follows is a short guide to each thinker. Who they were. What they saw. Where in the book they appear. And one book or essay you can read if you want to carry the insight further.

* * *

Ibn Khaldun (1332–1406)

A Tunisian-born historian and statesman who served several North African courts before retreating to a remote fortress to write the *Muqaddimah*, an introduction to history that is considered by many scholars to be the first work of social science. He was trying to understand why civilisations rose and fell. The word he settled on for the thing that held them together, or the thing whose absence pulled them apart, was *asabiyyah*. The usual English translation is "group feeling" or "social cohesion," but neither is quite right. *Asabiyyah* is the felt obligation between people who know they belong to one another. It is the thing that lets strangers act like kin in a crisis.

Ibn Khaldun argued that every polity begins with high *asabiyyah*, because it had to in order to form at all, and that over time, as the polity becomes comfortable and its elites grow distant from the labour that built it, *asabiyyah* thins. When the next crisis arrives, the thinned polity cannot respond, because it no longer has the bonds of obligation it once had. Another polity, with higher *asabiyyah*, will replace it.

In this book, Ibn Khaldun arrives at the end. Sami, leaving the ship, remembers him. He remembers him because the crew of the *Noor Salaam* had *asabiyyah* for forty-two days, and because he understands that most of the systems the ship serves, the governments and the companies and the markets, do not.

Reading recommendation: *Ibn Khaldun, *The Muqaddimah: An Introduction to History*, translated by Franz Rosenthal (abridged edition by N.J. Dawood, Princeton University Press). The abridged version is accessible for a modern reader. The sections on *asabiyyah** are in Chapter 2.

* * *

Lao Tzu (6th century BCE, approximate)

A Chinese philosopher whose existence as a single historical person is uncertain. The text attributed to him, the *Tao Te Ching*, is eighty-one short chapters of compressed observation about how the world actually works, as opposed to how it appears to work. His central teaching is that the softest thing overcomes the hardest, that water carves stone by yielding to it, and that the ruler who wants to hold power must hold it lightly.

In this book, Lao Tzu anchors Passage 4. Sami is watching a crisis that was not built on a battlefield but built over forty years by the steady action of markets and the slow erosion of national resilience. There was no general decision to arrive here. There was only a long series of small efficiencies, each one rational on its own terms, each one carving away a little more of the redundancy that a previous generation had maintained without being able to explain why. The ground the ship now sits on was carved by water, not by war.

*Reading recommendation: *Lao Tzu, Tao Te Ching, translated by Ursula K. Le Guin (Shambhala). Le Guin is a novelist, not a scholar of Chinese, but her version is the one a reader who has not read the Tao* before should start with. It reads as a text that was written for people who worked with their hands.*

* * *

Sun Tzu (544–496 BCE, approximate)

A Chinese general and military strategist whose *The Art of War* is a thirteen-chapter treatise on how to fight, and how to avoid fighting, and how to know which of the two is the right answer. The book has been read for two thousand five hundred years by soldiers, businesspeople, politicians, and almost anyone else who needs to think about competition under uncertainty.

Sun Tzu anchors two passages in this book. He arrives first in Passage 2, in the form of a line about terrain: the general who does not know the ground he is fighting on will lose. Sami reads it and understands that he has been sailing for thirty years without really knowing what was in his hold. He returns in Passage 5, with one of Sun Tzu's least-quoted lines: there has never been a case of a country benefiting from prolonged warfare. The strait has been closed for two weeks by then. Nobody is

winning. Nobody will win. That is Sun Tzu's point, and he made it in the fifth century BCE.

Reading recommendation: *Sun Tzu, *The Art of War*, translated by Thomas Cleary (*Shambhala*).* Cleary's translation is readable, his introduction is concise, and his commentary tradition draws on classical Chinese scholars rather than modern business gurus. Avoid the versions that market themselves as manuals for corporate executives.

* * *

Confucius (551–479 BCE)

A Chinese teacher who lived through the collapse of the Zhou dynasty's moral order and spent his life arguing that a society held together by correct conduct between its people could withstand what a society held together only by power could not. He did not leave a book. The *Analects* are fragments of his conversations, written down by his students after his death, and they are the closest thing to his voice that remains.

In this book, Confucius anchors Passage 6. The line Sami carries from the *Analects* is: *to see what is right and not to do it is want of courage*. He carries it because the governments of the world could see what the closure of the strait was doing to their populations, and they could see what policies would have prevented it, and they did not do those things, and the want of courage was visible across every institution. Sami reads the line as an indictment. Confucius meant it as one.

Reading recommendation: *Confucius, *The Analects*, translated by Annping Chin (*Penguin Classics*).* Chin's translation is the most readable modern version, with short commentaries that do not overwhelm the text. You can read the whole book in an afternoon.

* * *

Seneca (4 BCE–65 CE)

A Roman Stoic philosopher, statesman, and tutor to the emperor Nero. He wrote in letters and essays rather than systematic treatises. His subject was how to live. His answer was that living well required preparation. The person who had thought carefully about loss before loss arrived was the person who could bear it when it came, and the person who had not was the person who would be broken by the smallest thing.

Seneca anchors Passage 7. The distinction Sami takes from him is the distinction between preparation and negligence. A country with six months of grain reserves is a country that has prepared. A country with eight days of jet fuel reserves is a country that has been negligent. The difference is not about wealth. It is about whether the country has taken loss seriously as a possibility that would arrive, rather than as a theoretical scenario that would not. Seneca would have said the country with eight days of jet fuel was a country ruled by people who thought themselves immortal. He would have been right.

Reading recommendation: *Seneca, *Letters from a Stoic*, translated by Robin Campbell (Penguin Classics).* The letters are short, practical, and written as if to a friend. Letter XCI, on the fire at Lyon, is the one most directly connected to the themes of this book.

* * *

Nassim Nicholas Taleb (b. 1960)

A Lebanese-American essayist, former derivatives trader, and probability theorist. His books, including *Fooled by Randomness*, *The Black Swan*, *Antifragile*, and *Skin in the Game*, have introduced a set of concepts about risk, uncertainty, and systems that have become part of how many people now think about the world, though often without credit to him.

In this book, Taleb anchors Passage 8. The frame Sami uses is his three-part classification of systems. A *fragile* system is one that breaks under stress. A *robust* system is one that survives stress. An *antifragile* system is one that gets stronger from stress. The *Noor Salaam* is robust. She survived the anchorage. The global oil supply chain is fragile. It broke. An antifragile system would have used the closure of the strait to become more resilient than it was before. Almost nothing in the global energy system is antifragile. Denmark's community wind sector is one of the few exceptions, and only partially.

Reading recommendation: *Nassim Nicholas Taleb, *Antifragile: Things That Gain from Disorder* (Random House).* The book is long and digressive. A reader who wants the core argument without the full six hundred pages can read the prologue and the first hundred pages and will have most of the framework.

* * *

Donella Meadows (1941–2001)

An American systems scientist, author, and lead author of *The Limits to Growth*, the 1972 report that modeled the trajectory of the global economy against its resource base and concluded that continued exponential growth would hit physical limits within the twenty-first century. She spent the rest of her life teaching people to see systems: the feedback loops, the stocks and flows, the leverage points where small interventions could change the behaviour of the whole.

Meadows anchors Passage 9. The essay Sami reads is "Leverage Points: Places to Intervene in a System," her ranked list of twelve places where you can act to change a system, from weakest to strongest. The weakest leverage points are the ones everybody works on: adjusting parameters, changing numbers. The strongest are the ones almost nobody works on: changing the paradigm, changing the goal, changing the mindset from which the system was designed. What Meadows helps Sami see is why nothing will change after the crisis. The institutions that would have to change are stuck at the weakest leverage points, because those are the only ones they know how to operate.

Reading recommendation: *Donella Meadows, *Thinking in Systems: A Primer* (Chelsea Green).* The book is short, clear, and written for someone who has never thought in systems before. The essay on leverage points is included as an appendix and can also be read for free on the Donella Meadows Project website.

* * *

Ubuntu (philosophical tradition)

Not a person but a concept, from the Nguni language family of southern Africa. The usual English rendering is *I am because we are*. A better rendering, less tidy, is that a person is a person through other persons, that the self is not a solitary fact but a thing that comes into existence inside a community of other selves, and that damage to any of them is damage to all.

Ubuntu opens and closes this book. It opens in Passage 1, where Sami reaches for it on the first night of the anchorage, in his cabin, with the Iranian coast visible in the distance. He had encountered the concept as a student and had come across it again in port cities along the east coast of Africa. He had liked it then as an idea. On night one of the anchorage, it stopped feeling like an idea and started feeling like a description of what was happening. It closes in Passage 10, where a Chewa proverb given to Sami

by a woman in Chipata, the mother of Tendai, the young Zambian bosun, arrives in a different language and says the same thing.

*Reading recommendation: *Michael Onyebuchi Eze, Intellectual History in Contemporary South Africa (Palgrave Macmillan). Eze traces the concept of Ubuntu through its South African intellectual history, including its role in the Truth and Reconciliation Commission. For a shorter introduction, Archbishop Desmond Tutu's essay "Ubuntu: On the Nature of Human Community" in his book No Future Without Forgiveness* is one page long and enough.*

* * *

Esnatch Phiri (died 1978)

Esnatch Phiri was a Zambian woman who lived in a village north of Chipata, who had fourteen children of whom nine lived to adulthood, and who made the best *ndiwo* anyone in her village remembered making. She did not write books. She did not give lectures. She sat on overturned buckets at the edges of plots and she watched her daughters work, and she said what she said so that her granddaughters would say it too.

One of the things she said was: *Kali kokha nkanyama. Aliawiri ndi anthu*. Alone, one is an animal. Two, they are people.

That sentence, carried across fifty years and eighteen hundred miles by her granddaughter, is the last sentence of this book. It is here because the philosophers above said what they said in languages that got written down, and she said what she said in a language that was spoken at a kitchen table. Neither kind of wisdom is better than the other. Both kinds travel. This book is the place where one of each kind arrived at the same time.

There is no reading recommendation for Esnatch Phiri. She did not leave a book. Her recipe is written in her granddaughter's handwriting in a notebook on a shelf in a kitchen in Chipata, and that is where her work is.

* * *

Appendix B — The Ship's Manifest

Tendai's question to Sami on the first day of the anchorage was: *What does this actually become?* Sami's answer was: *It becomes almost everything.* This appendix is

the long version of that answer.

A modern refinery produces more than six thousand distinct products from a barrel of crude oil. Only about half of each barrel becomes fuel. The rest becomes the physical substrate of almost every room a person walks into. The exercise below is a walk through a typical house. In each room, the items made from petroleum are named, and where the number is known, the dependency is given.

This is not a call for guilt. A person cannot live a petroleum-free life in 2026. The point of the walk-through is to replace the word *oil* with an accurate picture of what it is. Oil is the floor. The floor has been invisible because it has been there the whole time.

* * *

The Kitchen

Fertilizer in the food. The most important petroleum-derived product in the kitchen is the food itself. All commercial fertilizers are based on ammonia. Seventy percent of the world's ammonia is made from natural gas. The remainder is made almost entirely from coal. Four percent of the world's gas supply is spent on making ammonia. Ten to fifteen calories of fossil fuel energy go into producing one calorie of the food on the table. The United States food system is the single largest consumer of petroleum products of any industry in the country. It takes about fifteen hundred litres of oil equivalent to feed one American for one year.

The seed before the fertilizer. The fertilizer's reach extends one step further back, into the seed itself. For most of human history, farmers saved seed from their own harvest and planted it the following year. Beginning in the 1960s, in most of the world, that practice was replaced by commercial hybrid seeds that produced higher yields for a single season but did not breed true in the next. A farmer who saved seed from a hybrid crop got back a fraction of the parent yield, or nothing at all. From the 1990s onward, genetically engineered seeds extended the dependency further. The seeds are bred to pair with a specific weed killer sold by the same company. They ship coated with pesticides, held to the seed by a polymer shell. The pesticide is petroleum, typically a neonicotinoid. The polymer is petroleum. The fertilizer the seeds need to deliver their promised yield is petroleum. The weed killer they are designed to survive is petroleum. By the 1990s an estimated ninety-five percent of farmers in developed countries and forty percent of farmers in developing countries had shifted to hybrid or genetically engineered varieties. Ninety percent of the corn grown in the United States

and seventy-six percent of the soybeans are now planted with coated seeds. The seed itself is not usually counted as a petroleum product, and it is not one in the strictest chemical sense. But the package it arrives in, the coatings it carries, the fertilizer it cannot function without, and the chemistry it was designed to interlock with are all petroleum, and the farmer who plants it is buying into all of it. The farmer who once saved her own seed is now a customer of a supply chain that runs through the same straits as everything else in this appendix.

Plastic everywhere. The containers the food arrived in are polyethylene, polypropylene, or polyethylene terephthalate. The cling film is low-density polyethylene. The freezer bags, the bread bags, the cereal liners, the milk jugs, the yoghurt tubs, the juice bottles, the condiment squeeze bottles, the takeaway clamshells, the microwave trays: all petroleum.

The appliances. The insulation in the refrigerator is a rigid polyurethane foam. The gaskets on the refrigerator door are a thermoplastic elastomer. The handles on the oven, the stove, and the dishwasher are engineering plastics. The electric kettle housing is polypropylene. The dishwasher basket is a nylon-coated wire. The washing-up sponge is polyurethane foam with polyester scrubber. The washing-up liquid is a blend of petroleum-derived surfactants.

The surfaces. The countertops, if laminate, are a petrochemical resin on compressed wood. If composite stone, the binder is polyester resin. The sealant around the sink is silicone, synthesised from petrochemicals. The paint on the walls is a latex or acrylic emulsion, both petroleum-derived. The flooring, if vinyl, is polyvinyl chloride. If laminate, the backing and melamine surface are petroleum-based.

The small things. The non-stick coating on the frying pan is polytetrafluoroethylene. The silicone spatula is a polymer of silicon and oxygen but the binding plasticisers are petroleum. The rubber bands on the asparagus are synthetic rubber. The pegs on the drying rack are polystyrene. The matches in the drawer have a paraffin wax tip. Even the chewing gum in the pocket of the jacket hung on the back of the chair contains petroleum wax and synthetic rubber bases.

* * *

The Bathroom

Pharmaceuticals. Nearly all of them. Ninety-nine percent of pharmaceutical feedstocks and reagents are derived from petrochemicals. Only about three percent of petroleum production is used for pharmaceutical manufacturing, but virtually every

drug on the shelf has a petroleum fingerprint somewhere. The active pharmaceutical ingredients, the binders, the coatings, the gel caps, the blister packaging, the child-proof cap, the cotton wool inside the bottle, the label, the ink on the label. There are no viable current substitutes for petroleum inputs into most pharmaceutical products.

The medicine cabinet. Aspirin: an acetylated derivative of salicylic acid, commercially synthesised from petroleum feedstocks. Paracetamol: synthesised from phenol, a petroleum derivative. Ibuprofen: synthesised from isobutylbenzene, a petroleum derivative. Antibiotics: the molecules themselves are fermented from living organisms, but the excipients, binders, and coatings that turn them into a pill are petrochemical. Insulin: the molecule is biological, but the delivery device, the vial, the syringe, the packaging, and many of the stabilising excipients are petrochemical.

Personal care. The shampoo bottle is polyethylene terephthalate. The shampoo itself is a blend of surfactants, most of which are petroleum-derived. The conditioner contains silicones and synthetic polymers. The soap, if synthetic detergent rather than true soap, is petroleum. The toothbrush handle is polypropylene. The bristles are nylon. The toothpaste tube is a laminate with a polyethylene inner layer. The toothpaste itself contains sodium lauryl sulfate, a petroleum surfactant, and often polyethylene glycol. The razor handle is polystyrene. The disposable razor head cartridge is polycarbonate.

The towels and the robe. If polyester, petroleum. If a polyester-cotton blend, partially petroleum. If pure cotton, grown with petroleum-derived fertilizer and pesticides and shipped in petroleum-powered vessels.

The toilet. The plastic seat is high-density polyethylene. The cistern parts are ABS. The toilet paper is wood pulp, but it was bleached with petroleum-derived chemicals, wrapped in polyethylene film, and shipped on petroleum-powered trucks.

* * *

The Bedroom

The bed. The mattress, if memory foam, is polyurethane. If an innerspring, the topper and the foam layers are polyurethane. The fire-retardant barrier is a polyester blend. The pillows are polyester fibre fill. The sheets, if polyester or microfibre, are petroleum. If cotton, see above. The duvet cover is the same story.

The wardrobe. Polyester, nylon, acrylic, spandex, fleece: all petroleum. Synthetic fibres now account for roughly two-thirds of all textile production. Even the cotton

items were grown with petroleum-derived fertilizer, picked by petroleum-powered machinery, and transported on petroleum-powered ships. The buttons, zippers, and plastic snaps are petroleum. The shoes, if trainers, are a mix of synthetic rubber, polyurethane foam, nylon mesh, and thermoplastic polyurethane. The leather shoes are tanned with petroleum-derived chemicals and finished with petroleum-derived dyes.

The electronics. The phone charging on the nightstand contains semiconductors manufactured with ultra-pure gases including helium, of which Qatar supplies about a third of the world's production. The phone's housing is polycarbonate. The screen laminate is a petroleum polymer. The charging cable insulation is polyvinyl chloride. The laptop on the desk is the same story. The television in the corner is the same story. The batteries in all of them now contain synthetic graphite for the anodes, which is made from petroleum coke, a byproduct of oil refining.

The books on the shelf. The paper is wood pulp, bleached with petroleum-derived chemicals. The ink is a petroleum-derived pigment suspended in a petroleum-derived carrier. The covers are a petrochemical coating over board.

* * *

The Living Room

The furniture. The sofa cushions are polyurethane foam. The upholstery is polyester or a polyester blend unless it is wool or leather, both of which see above. The coffee table, if veneered, is petrochemical resin on compressed wood. If painted, the paint is petroleum. The rug, if synthetic, is nylon or polypropylene. The curtains, unless silk or linen, are petroleum.

The insulation. The walls contain either fibreglass bonded with phenolic resin, polyisocyanurate foam, extruded polystyrene, or spray polyurethane foam. All of these are petroleum. Even mineral wool insulation uses petroleum-derived binders.

The paint on everything. Modern wall paints are acrylic or vinyl emulsions. The binder is petroleum. The solvent, if present, is petroleum. The preservatives are petroleum. The thickeners are petroleum.

* * *

The Garage

The car. A car is approximately thirty percent petroleum products by weight, not counting the fuel. The tyres are synthetic rubber, polyester belts, and steel wire. The dashboard is polyvinyl chloride and ABS. The seats are polyester upholstery over polyurethane foam. The carpet is nylon. The paint is a multi-layer polyurethane system. The adhesives holding the car together are petroleum. The hoses are petroleum. The wiring insulation is petroleum. The battery casing is polypropylene. The windscreen is glass laminated with polyvinyl butyral, a petroleum polymer.

The electric vehicle. The battery anode is synthetic graphite, made from petroleum coke. The cathode binder is polyvinylidene fluoride, a petroleum polymer. The separator membrane is polyethylene or polypropylene. The cable insulation, the housing, the dashboard, the seats, the tyres: petroleum, exactly as with the internal combustion vehicle. The electric vehicle uses less petroleum than the petrol vehicle over its lifetime. It does not use zero.

Everything else in the garage. The garden hose. The bicycle tyres. The paint cans. The weed killer. The fertilizer. The plant pots. The extension cords. The plastic-handled tools. The rope. The tarp. The oil for the lawnmower. The lawnmower itself.

* * *

Outside the House

The road. The asphalt on the road is a petroleum product. It is the residue of crude oil after the lighter fractions have been distilled off. Every road surface in the country is petroleum.

The roof. The shingles, if asphalt, are petroleum. The underlayment is a petroleum-based felt or synthetic. The gutters, if vinyl, are petroleum. If aluminium, they were smelted using electricity, a significant fraction of which came from natural gas or coal.

The lawn. The fertilizer on the lawn is ammonia-based. The weed killer is petroleum-derived. The grass seed was treated with petroleum-derived coatings. The sprinkler system pipe is polyvinyl chloride.

The solar panels, if present. The glass is manufactured in gas-fired furnaces. The silicon wafers are processed in facilities that use petroleum-derived chemicals. The backsheet is a polymer. The frame is aluminium, see above. The electric vehicles that the solar panels charge have the dependencies named above. The solar panels are a real improvement on burning oil for electricity. They are not a separation from oil.

* * *

A Note on the Calculation

The figure of six thousand products from petroleum is from refinery engineering references. It does not mean there are six thousand different chemicals in the average household. It means that the refinery stream is splittable into six thousand distinct product grades, each with different applications. Most people encounter some hundreds of these grades in daily life, often without knowing it.

When a reader finishes walking through this appendix and looks up from the page into the room they are sitting in, the expected reaction is not alarm. The expected reaction is recognition. The floor was always there. The book has only shown it to them.

* * *

Source notes: Product derivation data from International Energy Agency, "The Future of Petrochemicals" (2018); EPCLand 2026 Engineering Guide; Industrial Info Resources, "Petrochemicals Fuel 99% of Pharmaceutical Feedstock"; Resilience.org via Hendrickson (1996); CIEL (Center for International Environmental Law); World Economic Forum, "Beyond oil: 9 commodities impacted by the Strait of Hormuz crisis" (April 2026); OilPrice.com on Qatar helium production. Seed and seed-coating data from Douglas and Tooker, "Large-Scale Deployment of Seed Treatments Has Driven Rapid Increase in Use of Neonicotinoid Insecticides and Preemptive Pest Management in U.S. Field Crops" (Environmental Science & Technology, 2015); Center for Food Safety, "Pesticide-Coated Seeds" (2024); Tooker, Douglas, and Krupke, "Neonicotinoid Seed Treatments: Limitations and Compatibility with Integrated Pest Management" (Agricultural & Environmental Letters, 2017); Food and Agriculture Organization, "The State of the World's Plant Genetic Resources for Food and Agriculture." See Appendix E for full methodology.

* * *

Appendix C — The Cascade Map

A crisis of this kind does not happen all at once. It happens in order, one sector at a time, along lines of dependency that existed before the trigger and were invisible until the trigger made them visible. What follows is the sequential failure chain of the

forty-two day closure of the Strait of Hormuz, beginning on the twenty-eighth of February 2026. Each row is a real event or a real class of events. The intent of the appendix is to let a reader see, by date, how the chain actually moved.

This is the map Sami would have drawn on a piece of paper if he had been asked, on the morning of the twenty-fourth of April, to explain to a new captain what had happened.

* * *

Day 1: The Trigger

**28 February 2026. Iran closes the Strait of Hormuz following the start of military operations. About twenty million barrels of crude oil per day, roughly twenty percent of global petroleum liquids consumption, stop moving through the strait. The Noor Salaam* drops her anchor fourteen nautical miles south of Bandar Abbas and joins the anchorage. She is one of some sixty vessels that stop in the first twenty-four hours. The count climbs through the week.*

The immediate financial effect is an oil price spike. The immediate physical effect is zero. Nobody runs out of anything on Day 1. The anchorages are full but the tanks ashore are not yet empty. The cascade has begun but it is not yet visible to a person who is not looking at a shipping tracker or a commodities terminal.

* * *

Days 4–7: Industrial Inputs Begin to Fail

**4 March.* QatarEnergy declares force majeure on LNG shipments. Twenty percent of global LNG supply is affected. Natural gas is the feedstock for ammonia. Ammonia is the feedstock for fertilizer. Ammonia is also used for industrial refrigeration, for water treatment, and as a chemical precursor across dozens of manufacturing lines. The gas that was going to move through the strait is not moving. The gas that could have substituted for it is not there.*

**4–6 March.* European natural gas spot prices rise sharply. European chemical plants, already operating on thin margins, begin to reduce output or idle entirely. Ammonia plants in Egypt, which depend on natural gas that was routed through or near the strait, begin curtailment. Indonesian and Malaysian refineries announce allocation reductions to downstream customers.*

6 March. The United States issues an emergency waiver authorising Russian oil purchases. India is one of the countries granted the waiver. India depends on imported crude for roughly sixty percent of its consumption. The waiver is framed as a temporary measure. It arrives eight days into the closure. The fact that it has to be issued this quickly is the tell.

* * *

Days 8–14: The Fragile Sectors Begin to Show

**11 March. Multiple ship attacks in the Gulf region. The Mayuree Naree, a Thai-operated tanker, catches fire. Twenty crewmen are rescued. Three are missing. Somchai, the Noor Salaam's bosun, knows the Mayuree Naree's bosun personally. The danger to crews ceases to be theoretical. Shipping insurance premiums for Gulf transit rise further. Several flag states advise their ships to avoid the region.*

Mid-March, ongoing. The pharmaceutical supply chain begins to tighten in Europe. Approximately ninety-nine percent of pharmaceutical feedstocks are petroleum-derived, and the just-in-time inventory model that pharma companies have operated under since the 1990s means that most plants carry under thirty days of feedstock. Generic drug manufacturers in India, who supply roughly forty percent of the world's generic medicines, begin to reduce output. The effect will not appear on pharmacy shelves for another four to six weeks. It is being built into the pipeline now.

* * *

Days 15–24: The Gas Layer Cascades

Mid-to-late March. European fertilizer plants, which have been running at reduced capacity for three weeks, announce extended shutdowns. The Northern Hemisphere planting season is beginning. Farmers in Germany, France, and the United Kingdom begin to source fertilizer at prices two to three times pre-closure levels. Farmers in North America do the same.

Mid-to-late March. Sub-Saharan Africa, which imports over ninety percent of its fertilizer from outside the continent, does not yet see a visible price rise because most African fertilizer arrives on a delayed schedule. The price rise will appear in May and June, in time for the African short rainy season. In East African markets, the effect will be felt first as a shortage, then as a price shock, then as a reduction in the area

planted. The full consequence for African harvests will begin to appear in October and November.

24 March. The Philippines declares an energy emergency. The country imports ninety-eight percent of its oil. Petrol queues in Manila begin to reach two blocks in length. Jun's wife photographs one of the queues and sends it to him on the ship. This is the moment when Jun's ship becomes his family's problem rather than an abstract concern.

* * *

Days 25–31: The Known Shortages Become Visible

**26 March.* Iran announces that it will allow ships from India, China, Russia, Iraq, and Pakistan to transit. The announcement is framed as a humanitarian concession. Operationally, it is a selective closure that rewards countries Iran regards as strategic allies and punishes the rest. The Noor Salaam*, Syrian-flagged, is not in the allowed category. The flag matters now in a way it did not matter two weeks earlier.

**27 March.* The United Nations secures an agreement to allow fertilizer shipments through the strait, on humanitarian grounds, for developing nations. In the same twenty-four hour window, the Mayuree Naree, the Thai tanker that burned on 11 March, grounds on Qeshm Island. Her remaining crew is detained. Somchai, who has known the Mayuree Naree's bosun since they were both apprentices, is the first person on the Noor Salaam* to hear.

**30–31 March.* The Al Salmi, a VLCC similar to the Noor Salaam and owned by a company in the same group, is hit by a drone at Dubai while discharging cargo. The ship catches fire. Three crew members die. An oil spill begins. The attack on the Al Salmi* shifts the danger calculation for every VLCC crew in the region. The strait was the danger. The strait and the ports are now the danger. There is no safe geography.

* * *

Days 32–40: The Price Structures Shift

Early April. Jet fuel reserves in Europe are now under two weeks. The 8-to-10 day European reserve figure, which was known in aviation circles before the closure, becomes public knowledge when Airlines for Europe and Willie Walsh of IATA issue coordinated statements. The United States has approximately sixty-nine days of jet

fuel reserves. Australia has twenty-eight. The asymmetry reveals which governments thought carefully about energy and which ones did not.

2 April. Iran extends the transit permission to Philippine-flagged vessels. Jun reads the news in the mess hall. His ship is not a Philippine-flagged vessel. The permission helps other Filipino seafarers on other ships. It does not help him.

7 April. Pakistan brokers a ceasefire between Iran and its regional counterparts. The blockade is announced as "technically suspended." Ships do not resume normal transit because the insurance markets do not move that fast, and because no captain with a full crew and a full cargo is willing to be the first ship shot at during a ceasefire.

9 April. Olivier Jankovec, Director General of ACI Europe, sends a letter to the European Commission documenting the scale of the airport fuel crisis. Airlines for Europe coordinates. The letter is made public within the week. Willie Walsh of IATA makes the 9/11 comparison in a public briefing. European aviation, on current trajectory, will begin cancelling long-haul flights within ten to fourteen days.

* * *

Days 41–42: The Blockade Closes

13 April. The United States Navy announces a full blockade on Iranian oil exports. This is a different category of action from the strait closure. The strait closure was a denial of transit. The blockade is a denial of export from Iran specifically. Iran responds by tightening its own closure criteria and retaliating on vessels it can reach.

15 April. The United States announces that its blockade now covers all Iranian ports.

**18 April. Iran closes the strait again. Indian-flagged ships are fired on despite having previously issued clearances. The Sanmar Herald*,* an Indian-flagged tanker that had transited the strait three days earlier under Iran's 26 March concession, is hit by gunboats on her return. Her captain is Arjun Ramachandran's former shipmate. Arjun reads the news from the hospital in Thane, where his father is on dialysis.

* * *

The Shape of the Chain

Across forty-two days, the chain moved through the following sectors in approximately this order.

First, the shipping sector itself. Ships stopped. Crews waited. Insurance markets adjusted.

Second, the gas layer. LNG shipments, then ammonia, then fertilizer, then the Northern Hemisphere planting season, then, on a delay, the African harvest.

Third, the refined products. Jet fuel first, because it was the sector with the thinnest inventory. Road diesel next. Petrol on a delay because gasoline reserves in most developed nations were larger.

Fourth, the petrochemical layer. Polymers, plastics, synthetic rubber, pharmaceuticals. This cascade is still running as the book closes. A pharmaceutical pipeline that tightens in March does not show up on a pharmacy shelf until May or June. The full consequences of the forty-two day closure for medicine availability will not be known for six to nine months after the strait reopens.

Fifth, the industrial downstream. Semiconductor manufacturing paused in some facilities as helium stockpiles were drawn down. Electric vehicle battery producers reduced output as petroleum coke and synthetic graphite supplies contracted. Solar panel manufacturers reported feedstock constraints.

Sixth, the financial layer. Commodities markets repriced. Equities adjusted. Sovereign bond spreads in fragile economies widened. The financial effects are the ones that get reported in real time. They are not the most important ones.

The most important consequences are the ones that moved through the chain slowly and will arrive in October and November and through the following year. The unplanted hectares in East Africa. The antibiotics that did not reach a Pakistani hospital. The insulin that ran short in a small town in central Asia. The letters from companies to their customers announcing an allocation cut that quietly becomes a permanent reduction.

A person reading the news in April 2026 saw the price spike and the tanker attacks. The chain itself was not visible in the news. It was visible only by date, by sector, and by consequence. That is what this appendix is for.

* * *

Data sources: QatarEnergy force majeure, 4 March 2026 (Reuters); US emergency waiver on Russian oil, 6 March 2026 (US Treasury, OFAC); Philippines energy emergency declaration, 24 March 2026 (Philippine Department of Energy); Iran selective transit concessions, 26 March and 2 April 2026 (Iranian Ministry of Petroleum announcements); Al Salmi drone attack, 30–31 March 2026 (Lloyd's List); ACI Europe Jankovec letter to European Commission, 9 April 2026 (ACI Europe

public record); US naval blockade announcements, 13 and 15 April 2026 (US Department of Defense); Sanmar Herald attack, 18 April 2026 (Indian Directorate General of Shipping). Sectoral cascade data from International Energy Agency monthly reports (March, April 2026); Civil Eats coverage of the Persian Gulf fertilizer chokepoint; World Economic Forum, "Beyond oil: 9 commodities impacted by the Strait of Hormuz crisis" (April 2026). Ship tracking data from vesselfinder.com and marinetraffic.com. See Appendix E for full methodology.

* * *

Appendix D — The Solution Scorecard

Taleb's framework is the one Sami used in Passage 8 to think about the *Noor Salaam* and the system she served. The ship was robust. The system was fragile. An antifragile system would have used the closure of the strait to become more resilient than it had been before, rather than less. Almost nothing in the global energy system is antifragile.

This appendix applies the same framework to every solution model named in the book. Each one is rated fragile, robust, or antifragile. The rating is not a moral judgment. It is a description of how the system responds to shock. A fragile system breaks. A robust system survives. An antifragile system gets stronger.

Where a model deserves credit, it is given credit. Where a model has a limitation, the limitation is named. Honest accounting is the point.

* * *

Norway: Sovereign Wealth Management

Rating: Antifragile (oil wealth specifically), Robust (broader energy security)

The Government Pension Fund Global, established in 1990 with first deposit in 1996, holds approximately two trillion dollars as of early 2026. This is the largest sovereign wealth fund in the world. It translates to over three hundred forty thousand dollars per Norwegian citizen. The fund generated a fifteen percent return in 2025, about two hundred forty-eight billion dollars in profit. The average annual return since 1998 has been six point six percent.

The reason Norway is antifragile with respect to oil wealth, rather than merely robust, is that every oil shock that destabilises a typical petro-state strengthens

Norway's position. When oil prices rise, the fund's cash flows rise. When oil prices fall, the fund buys equities at a discount. The fund's 2.8 percent annual spending cap, recently revised down from four percent, keeps the capital intact for future generations. The seventy-five percent marginal tax rate on oil companies, in place since the 1970s, captures the economic rent of the resource for the public. The Peterson Institute ranks the fund 100 out of 100 on transparency, first among sixty-four sovereign wealth funds worldwide.

The limitation: Norway's model is antifragile with respect to oil revenue. It is not antifragile with respect to oil dependency. Norway still imports most of its refined fuels. Its transport fleet, its chemical feedstocks, and its fertilizer supply are all vulnerable to the same disruptions that affect the rest of Europe. Norway solved the question of what to do with oil money. It did not solve the question of what to do when oil stops.

The harder limitation: Norway had strong democratic institutions before oil was discovered. Countries that discovered oil before building strong institutions, like Venezuela, Nigeria, Angola, and Libya, experienced the resource curse rather than Norwegian wealth. The model is therefore not directly replicable by any state still struggling to build its institutions. It is a proof of what is possible, not a template.

* * *

Denmark: Community Wind

Rating: Robust (electricity), Fragile (oil-as-material)

Denmark generates eighty-eight percent of its electricity from renewable sources, on track for one hundred percent by 2030. Wind provides fifty-eight percent of the country's electricity. More than half of Denmark's wind generation capacity is community-owned. For nearly fifteen years, the government has required that all new wind projects be at least twenty percent community-owned. Samsø Island became the first island in the world to be one hundred percent renewable-powered. Danish companies Vestas and Siemens Gamesa together hold about one-third of global wind turbine installations.

Denmark's model is robust because it survives shocks without breaking. A closure of the Strait of Hormuz does not turn off Denmark's lights. But it is not antifragile, because the model does not get stronger from the closure. A longer crisis would not accelerate Denmark's transition in any automatic way. The transition is the product of fifty years of sustained policy work, not of crisis response.

The limitation: Denmark has solved the electricity problem. It has not solved the oil-as-material problem. Transport, industrial heat, fertilizer, plastics, and pharmaceuticals still depend on fossil fuels. If the strait closes, Danish farmers still face fertilizer price shocks, Danish hospitals still face pharmaceutical supply constraints, and Danish freight still runs on diesel. Denmark has built the electricity half of the solution. The other half is where most of the oil still goes.

* * *

Iceland: Geothermal and Hydroelectric

Rating: Robust (electricity), not transferable as a model

Iceland generates one hundred percent of its electricity from renewable sources: seventy-one percent hydroelectric, the remainder mostly geothermal. It began the transition in the 1970s specifically to reduce reliance on expensive imported fossil fuels. Its location on the volcanic Mid-Atlantic Ridge provides abundant geothermal heating and electricity.

The limitation: Iceland sits on a volcanic ridge. Most countries do not. The resource base that makes the Icelandic model work cannot be replicated by a country without Iceland's geography. Iceland is a proof of concept that one hundred percent renewable electricity is possible. It is not a scalable template. The real lesson from Iceland is not the geography, which is a gift, but the decision to respond to a 1970s oil shock with permanent structural change rather than with waiting for prices to fall.

* * *

Costa Rica: Political Will Without Geographic Advantage

Rating: Robust

Costa Rica has operated on more than ninety-eight percent renewable energy for several consecutive years. In 2016, the country ran entirely on renewables for more than two hundred fifty days. Costa Rica does not sit on a volcanic ridge. Its success comes from policy choices: primarily hydroelectric, with wind, solar, and geothermal contributing, and a government commitment to environmental preservation that has survived multiple administrations.

Costa Rica matters because it disproves two common excuses. It disproves the size excuse: a small country can restructure its energy system. And it disproves the

geographic excuse: you do not need Iceland's volcanism to do it.

The limitation: Costa Rica's population is about five million. Whether the model scales to an economy the size of Germany or Japan is unproven. And like Denmark, Costa Rica has solved a large part of the electricity problem and has not solved the oil-as-material problem.

* * *

Cuba: Agroecology Under Duress

Rating: Antifragile during the Special Period, Fragile again afterwards

When the Soviet Union collapsed in the early 1990s, Cuba lost access to Soviet petrochemical imports. Fertilizer, pesticides, and oil-based machinery simply stopped arriving. The Cuban food system faced total collapse. Instead of collapsing, it was rebuilt around agroecology: cover cropping, composting, biological pest control, urban gardening, and animal traction. Cuba remains the only country to have executed a nationwide agroecological transition of this scale. Agricultural output recovered. People were fed. The proof that oil-free farming can work at national scale exists because of Cuba.

This was an antifragile response during the Special Period. The shock made the system stronger.

**The limitation: Once Venezuela began supplying subsidised oil to Cuba under Chávez, the agroecological model began to be partly displaced by a return to fossil-fuel-dependent agriculture. The transition was forced, not chosen. When the forcing stopped, the old dependencies partially returned. The open question for the rest of the world is whether a country can choose the agroecological path before* a crisis forces it, and sustain the choice when oil is cheap and available.*

* * *

China: Strategic Stockpiling

Rating: Robust (with significant caveats)

China holds an estimated 1.4 billion barrels of crude oil in strategic reserves, the largest stockpile in the world. It generates approximately fifty percent of its electricity from renewable sources, the furthest along of any major economy. It leads the world in solar and wind manufacturing capacity and in battery and electric vehicle

production. During the 2026 Hormuz crisis, China was the least affected major Asian economy. Iran reportedly allowed Chinese vessels to transit the strait while blocking most others.

The combination of strategic stockpiles, renewable generation, and diplomatic leverage made China robust during this crisis.

The limitation: China remains the world's largest LNG importer and has significant remaining fossil fuel dependency. The stockpile buys time but does not eliminate dependency. And the political model that makes this level of sustained strategic planning possible, five-year plans executed across multiple decades, is not replicable in democracies where political horizons rarely exceed one electoral cycle. China proves that strategic insulation is possible. It does not prove that it is possible everywhere.

* * *

France: Nuclear Response to 1973

Rating: Robust

The 1973 Arab oil embargo triggered France's decision to build a massive nuclear programme. Today France generates approximately seventy percent of its electricity from nuclear power, the highest proportion of any country. This is the strongest historical example of an oil shock producing a permanent structural response rather than a temporary adjustment. Most countries that responded to the 1973 embargo returned to oil dependency when prices fell. France did not.

The limitation: Nuclear has its own risks, its own waste challenges, and its own political headwinds. The French model worked in a country with strong central technocratic institutions and a political consensus that has mostly held for fifty years. Germany made the opposite decision and is now more fossil-fuel dependent than it would otherwise be. The nuclear path is a proof that permanent structural response to an oil shock is possible. Whether it is the right response depends on questions the scorecard cannot answer.

* * *

Japan: Efficiency Response to 1979

Rating: Robust (efficiency), Fragile (dependency)

The 1979 Iranian Revolution drove Japan into the most aggressive energy-efficiency push in the history of industrial nations. The efficiency gains are real and permanent. Japanese industry uses less energy per unit of output than any other major economy. Japan now holds two hundred fifty-four days of oil reserves, the strongest strategic stockpile in Asia.

The limitation: Ninety-four percent of Japan's crude imports still come from the Middle East. Efficiency reduced consumption but did not eliminate dependency. Japan is the most energy-efficient major economy in the world and is still critically exposed to a closure of the Strait of Hormuz. The lesson is that efficiency is necessary but insufficient. A fully efficient system that still depends on the same vulnerable supply chain is still a vulnerable system.

* * *

Nitricity: Decentralised Fertilizer

Rating: Antifragile (technology), Not yet deployed

Nitricity is a US-based company that has developed modular nitrogen fertilizer production systems using renewable electricity and locally available agricultural byproducts. The technology can be deployed near farming regions anywhere, breaking the Haber-Bosch process's dependency on centralised natural gas infrastructure. It is the technological proof that the fertilizer chain can be decoupled from the strait, from the pipelines, and from the ammonia plants of Qatar, Egypt, and the Persian Gulf.

The technology is antifragile because a closure of the strait that would cripple the existing fertilizer supply chain would strengthen the case for distributed fertilizer production. Every crisis increases the pressure to deploy it.

The limitation: It is not deployed at scale. The incentive structures of the current system favour the incumbents. Oil companies own stakes in fertilizer production. Board members overlap between fossil fuel companies and fertilizer companies. The technology exists. The political and economic will to deploy it does not yet. The question is not can we? but why aren't we doing it faster?

* * *

The Default Global System: Just-in-Time Delivery

Rating: Fragile

For completeness, the system against which the scorecard measures the solution models.

The default global system, which most countries, companies, and consumers operate within by omission rather than by decision, is a just-in-time delivery network built on continuous transit through a small number of chokepoints. The Strait of Hormuz is the most important of these. Twenty percent of global petroleum liquids consumption passes through the strait each day. European jet fuel reserves are eight to ten days. Most pharmaceutical plants hold under thirty days of feedstock. Sub-Saharan Africa imports over ninety percent of its fertilizer and has limited ability to store what it imports.

This system is fragile by Taleb's definition. It breaks under stress. The *Noor Salaam's* forty-two day anchorage demonstrated this in a way that nothing had demonstrated it since 1973. The fragility is not an accident or an oversight. It is the result of decades of decisions that rewarded efficiency over resilience, and that treated the chokepoints as permanent features of geography rather than as deliberate choices that could be unwound.

* * *

The Composite Picture

No single model in the scorecard solves the whole problem. But the models together solve almost every component of it.

Norway solves the question of managing oil wealth. Denmark, Iceland, Costa Rica, and France solve large parts of the electricity question, each in a different way. Cuba proves that agroecology can replace fossil-fuel-dependent agriculture at national scale. China demonstrates that strategic stockpiling and renewable investment can insulate a nation against oil shocks, if the political will exists. Japan shows that dramatic efficiency gains are possible but insufficient on their own. Nitricity proves that the fertilizer chain can be decoupled from the strait through technology that already exists.

A country that wanted to construct a fully antifragile energy posture would take from each model the parts that suit its circumstances, rather than adopting any single model wholesale. The scorecard exists to make that composition possible.

What the scorecard cannot do is produce the political will to act on any of it. That is the question this book is about, and the answer, so far, from the historical record of the last three oil shocks, is that nations act after the damage, not before it. Until that

pattern breaks, every crisis will continue to look like the last one.

* * *

Framework: Nassim Nicholas Taleb, Antifragile: Things That Gain from Disorder (Random House, 2012). Norway data: NBIM annual reports; Norwegian Government Pension Fund Global; Peterson Institute Transparency Scoreboard; EditorialGE 2026; Yale case study. Denmark data: Climate Council; Construction21; WRI; Samsø Energy Academy. Iceland data: Icelandic National Energy Authority; Climate Council. Costa Rica data: World Economic Forum; Climate Council. Cuba data: UN FAO; various academic studies of the Special Period. China data: Soufan Center; Time; IEA. France nuclear data: EDF; IEA Country Profile France. Japan data: Agency for Natural Resources and Energy (ANRE); IEA Country Profile Japan. Nitricity data: Nitricity company publications; agricultural technology trade coverage. Default system fragility data: International Energy Agency; Civil Eats; ACI Europe; WEF. See Appendix E for full methodology.

* * *

Appendix E — Sources and Methodology

This appendix describes how the book was researched, written, and verified. Transparency about method matters more in 2026 than it did five years ago, because the tools used to produce this book are new enough that the reader is owed an honest account of what they are looking at.

The short version is this. The characters are fictional. The events are real. The writing is the author's. The research was extensive, and it was accelerated by artificial intelligence in the same way that a writer a generation ago was accelerated by the arrival of online search. The method is described below in enough detail to let a reader evaluate it, and with enough discretion to respect the investment that went into assembling the underlying material.

* * *

What Is Factual in This Book

The closure of the Strait of Hormuz beginning on the twenty-eighth of February 2026 is real. The cascading sector failures described across the passages are real. The

specific dated events, including the QatarEnergy force majeure on the fourth of March, the United States emergency waiver on the sixth of March, the Iranian selective transit concessions on the twenty-sixth of March and the second of April, the attack on the *Al Salmi* at Dubai on the thirtieth of March, the Jankovec letter to the European Commission on the ninth of April, the United States naval blockade on the thirteenth of April, and the *Sanmar Herald* incident on the eighteenth of April, are drawn from public reporting at the time those events occurred.

The statistical and structural facts in the book are also real. The six thousand products from a refinery barrel. The ninety-nine percent of pharmaceutical feedstocks that are petroleum-derived. The eight to ten day European jet fuel reserve figure and the twenty-eight day Australian figure and the sixty-nine day United States figure. Norway's two trillion dollar sovereign wealth fund and its 2.8 percent spending cap. Japan's two hundred fifty-four day oil reserve. China's 1.4 billion barrel stockpile. Denmark's eighty-eight percent renewable electricity share. Cuba's Special Period agroecological transition. The Nitricity technology. The Qatar helium supply. The Haber-Bosch process. All of these are documented in publicly available institutional reports, peer-reviewed literature, and journalistic coverage.

The philosophers are real. The texts they are associated with are real. The specific passages attributed to them in the book are drawn from their actual works, paraphrased to fit Sami's voice and the rhythm of the narrative, but not invented.

The Chewa proverb *Kali kokha nkanyama, aliawiri ndi anthu* is a genuine proverb of the Chewa people of Eastern Zambia and Malawi, documented in multiple Chewa cultural sources. *Chikuni chimodzi sichipsetsa mphika* is also a genuine Chewa proverb, documented in the same tradition.

The geographic, maritime, and industrial details have been checked. Where a detail could not be verified to the author's satisfaction, the book errs toward the general rather than inventing a specific. Kanyanta as a named village, for instance, was replaced in late editing with the verified general form *a village north of Chipata*, because the author could not confirm that a specific village of that name existed, and preferred a true generality to a fabricated specificity.

* * *

What Is Invented

Sami Kassar and the crew of the *Noor Salaam* are fictional characters. So is the ship. So is the specific anchoring at fourteen nautical miles south of Bandar Abbas. So are

the conversations between the characters, including the Evening Questions, the exchanges in the engine room, the satellite phone calls after the crisis, and the call with Tendai's mother in Chipata.

These characters are invented in order to carry the facts. A book that presented the same data as a policy paper or a news summary would not be read by the people who most need to understand what the data mean. A book that attaches the data to human lives, and asks the reader to follow a captain and a chief engineer and a young Zambian bosun and a Bangladeshi cook through forty-two days of waiting, is a book that lets the data arrive in a form that can actually be absorbed.

This is the oldest technique in non-fiction writing. It is used by John McPhee, by Michael Lewis, by Tracy Kidder, and by many other writers who have told true stories about complex systems by anchoring them in specific people, some real and some composite. The method is declared here so that no reader mistakes the fiction of the characters for a fiction of the data.

Esnatch Phiri is invented as a named grandmother. Her proverbs are not invented. Her name, her dates, and her life details are fiction placed as a vehicle for real Chewa wisdom, because the author believed the proverb deserved to arrive in the book through a specific person rather than as an anonymous aphorism. A reader who wants to verify the proverbs themselves will find them documented in Chewa cultural and linguistic sources. Esnatch Phiri stands for the women whose names were not written down but whose words were.

* * *

How the Book Was Made

This book was written in 2026. The method that produced it is new enough to be worth describing in detail.

The first phase was research. The author used artificial intelligence, primarily large language models, to accelerate the gathering of publicly available information across maritime logistics, petrochemical supply chains, the 2026 geopolitical record, the history of oil shocks, the comparative energy policies of the named countries, the technical specifications of VLCC tankers, the cultural and linguistic traditions of the nationalities represented in the crew, and the philosophical traditions named across the passages. The AI was used as a research accelerator, not as a source in itself. Every factual claim in the book was checked against original institutional, journalistic, or academic sources. Where the AI produced an assertion that could not be verified

against a primary source, the assertion was removed.

The second phase was sifting. The volume of material gathered in the first phase was several orders of magnitude larger than what appears in the book. The author read through the material, identified the through-lines, decided which facts mattered and which did not, and decided which facts would land through the characters and which would land through direct exposition in the appendices. This phase was entirely human work. The selection of what to tell, in what order, through whose eyes, and with what emotional weight, is the author's alone.

The third phase was brainstorming. Once the through-lines were identified, the author used AI as a collaborator for exploring how to structure the book, how to pace the chapters, how to have the characters speak in distinct voices, and how to hand the reader difficult technical material in language that could be absorbed. The AI functioned here as a thinking partner. The conversational back-and-forth that shaped the skeleton, the character arcs, and the structural decisions was a human-AI collaboration in the sense the word has come to mean in 2026.

The fourth phase was drafting. The author wrote the book. Sections were drafted by the author, revised by the author, audited against a list of language patterns the author wanted to avoid, and reworked through multiple passes. AI was used during drafting as an assistant, in the same way an editor or a research assistant would have been used in an earlier era. The author read every sentence that appears in this book, and the author decided whether each sentence stayed or went. Nothing was generated and inserted without review. Many things were generated and rejected.

The fifth phase was fact-checking. Every date, every statistic, every attribution, every name of a real person, every name of a real institution, every geographic detail, every technical claim about ships or refineries or fertilizers or pharmaceuticals was checked against publicly available primary sources. Where an initial draft contained a fact that could not be verified, the fact was removed rather than softened. This is the phase where words like *ntolo*, which was an invented food term initially drafted into the book, were replaced with verified terms like *ndiwo*. It is the phase where unverified village names were replaced with verified general descriptions. The fact-checking pass was the longest phase of the project by duration.

The sixth and final phase was language-pattern auditing. The author maintains a list of phrases, cadences, and rhetorical devices that characterise AI-generated prose in 2026 and that the author wants no part of. Every passage in the book was audited against this list. Instances of contemplation tics, of the *it is not X, it is Y* rhetorical construction, of excessive rhetorical *particulars*, of words like *profound* and *deeply*

and *fundamentally* used as filler, of em-dashes used to simulate thoughtful pauses, of any sentence structure that signalled *this is an AI pretending to be a human*, were rewritten. The book is clean of these patterns to the best of the author's ability. A reader who finds a residual instance is invited to flag it. The work of removing these patterns is continuous.

* * *

Why the Method Is Declared

The author believes that honest disclosure of how a book like this is made is more useful than concealment.

The alternative, a book that pretends it was produced by a single writer in a quiet room with nothing but a typewriter and a stack of newspapers, would be dishonest about the tools actually used in 2026. The tools are new. They change what is possible. A book about the 2026 Hormuz crisis, written with the depth of research and the breadth of reference this book attempts, was almost certainly not going to be publishable within the same year as the crisis without the acceleration that AI research assistance provides. A book that arrived two years after the events would not have the urgency that a book arriving in the same year has. Speed matters. The tools that enable speed should be acknowledged, not hidden.

The method described above is a collaboration between a human writer and a set of new tools, in the same way that every piece of writing produced in the last thirty years has been a collaboration between a writer and a word processor, an internet search engine, and a digital archive. The tools have changed. The writer is still required. The research, the selection, the structural decisions, the character voices, the fact-checking, the editing, the rewriting, and the final responsibility for every word are all the writer's. The tools assist. They do not replace.

A reader who is worried that artificial intelligence is going to replace human writers is invited to consider this book as evidence that the replacement frame is the wrong one. The collaboration frame is the right one. What a human writer can do with these tools is more than what a human writer could do without them. What the tools can do without a human writer is much less than what is advertised. The production that results from the collaboration is something neither party could produce alone. That is the honest description.

The author hopes that declaring the method in this way is useful to other writers who are figuring out how to use these tools well, and useful to readers who are trying

to understand what kind of object a book like this actually is. It is the author's book. It was made with new tools. Both of those things are true at once. The book exists because they are both true.

* * *

What the Appendix Will Not Disclose

The specific combination of research prompts, internal documents, conversation transcripts, and drafting scaffolds that produced the book is not published here. That combination is the author's craft. A plumber is not required to publish the sequence of tools and techniques they use to fix a pipe. A lawyer is not required to publish the research methodology they used to prepare a brief. A novelist is not required to publish the outline and the notebooks and the discarded drafts that produced a finished book. The declaration above is the honest description of the method in kind. The specific implementation is the author's trade.

Readers who want to verify facts in the book will find the public sources cited in the appendix source notes for each section. Readers who want to see the method in action are reading the book, which is the artifact the method produced.

* * *

Acknowledgment

Research assistance: Claude (Anthropic), across multiple sessions between late 2025 and April 2026. The AI was used as a research accelerator, a brainstorming partner, and a drafting assistant, under the author's direction and with the author's review of every line that appears in the book. All factual content was verified against publicly available primary sources. All editorial judgments, character voices, structural decisions, and final text are the author's.

Pumulo Sikaneta April 2026

* * *

End of book.

Dirty Wisdom: An Old Playbook for a World That Thinks It's New

Pumulo Sikaneta. April 2026.

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